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NORWAY

1997

SPECIAL FEATURES

LABOUR MARKET

EDUCATION AND TRAINING

OECD ECONOMIC SURVEYS

1996-1997

NORWAY

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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Table of contents

Assessment and recommendations 1

I. Recent developments and short-term prospects 9

Reaching capacity limits 9
Continued buoyant oil activity 12
Strong non-oil exports 13
Sustained household demand 17
Robust business investment 20
Falling unemployment 20
Still subdued inflation 24
Rising current account surplus 27
Short-term outlook 27

II. Economic policies 31

Overview 31
The fiscal stance 32
Monetary policy 41
Structural reform 49

III. Implementing the OECD Jobs Strategy 56

Introduction 56
Labour market and employment performance 57
Policy requirements 65
Recent policy initiatives 91
Assessment and scope for further action 93

IV. Education and training 97
Why human capital is essential for Norway 98
The education system in the light of past and planned reforms 108
The contribution of continuing education and training 123
Looking ahead 128

Notes 132

Annex
Calendar of main economic events 142

Statistical annex and structural indicators 147

Boxes

1. The Norwegian salmon farming industry 16
2. The Government Petroleum Fund 37
3. The OECD Jobs Strategy: synopsis of recommendations for Norway 66

Tables

Text

1. Demand and output 10
2. World salmon production 16
3. Household appropriation account 19
4. Labour market development 22
5. Prices, wages and costs 25
6. Short-term projections 29
7. Budget balances: projections and outcomes 33
8. Money and credit 48
9. Bank operating results 53
10. Concentration in the banking sector 54
11. Employment developments 58
12. Employment protection legislation in Europe 75
13. Income replacement for people at working age 79

14. Net replacement rates 81
15. Active Labour Market Programmes for Ordinary Unemployed 84
16. Comparative levels of labour productivity 100
17. Rates of return to formal education in the Nordic countries 101
18. Labour market situation by level of education 104
19. Labour market situation of individuals six months after completing education, by level and field of study 105
20. Employment by occupation 108
21. Financial support for students 111
22. Resource inputs in education 115
23. Graduation in upper-secondary and university education 116
24. University degrees by disciplines 117
25. Adult education and training programmes 124

Statistical annex and structural indicators

A. Selected background statistics 148
B. Supply and use of resources 149
C. Gross domestic product by origin 150
D. Gross domestic product by origin (volume) 151
E. General government income and expenditure 152
F. Labour market 153
G. Balance of payments 154
H. Foreign trade by area 155
I. Prices and wages 156
J. Money and credit 157
K. Production and employment structures 158
L. Productivity and investment structure 159
M. Labour market indicators 160
N. Public sector, general government income and expenditure structures 161

Figures

Text

1. Growth performance 11
2. Oil and gas production 13

3. Export performance and cost competitiveness 14
4. Exports by commodity group 15
5. Household spending 18
6. Household financial position 19
7. Mainland business investment 21
8. The labour market 23
9. Wage and price developments 26
10. External position 28
11. State budget balance 35
12. State pension expenditure and petroleum revenues 40
13. Exchange rate behaviour and official reserves 43
14. Interest rate developments 45
15. Money and credit 47
16. Features of the agricultural sector 50
17. Agricultural trade and support 51
18. Rationalisation in the banking sector 55
19. Labour market indicators 59
20. Key features of unemployment 61
21. Duration of unemployment, 1995 62
22. The equilibrium rate of unemployment 64
23. Development of earnings distribution 72
24. Part-time work 73
25. Recipients of labour market income support 76
26. Government spending and social transfers 77
27. Unemployment benefits 80
28. Tax wedges 83
29. Government expenditure on education and training 98
30. Education and labour market performance 102
31. Employment and unemployment by occupation 107
32. Reading frequency and literacy 113
33. Teacher compensation 114
34. The trade-off between school and work 120
35. Growth in student numbers 121
36. Expenditure on job-related training and education in the private sector 125

BASIC STATISTICS OF NORWAY

THE LAND

Area (1 000 sq.km)	324
Agricultural area (1 000 sq. km)	9
Productive forests (1 000 sq.km)	65

Major cities (1.1.95)	
Oslo	482 555
Bergen	221 645

THE PEOPLE

Population, (31.12.1994)	4 347 695
Number of inhabitants per sq. km	13
Net natural increase (average 1990-94)	14 939
Per 1 000 inhabitants (average 1990-94)	2.9

Civilian employment, 1994	2 003 000
of which:	
Industry (%)	23.4
Agriculture, forestry and fishing (%)	5.3
Other activities (%)	71.3

PRODUCTION

Gross domestic product, 1995 (NKr million)	935 927
GDP per head (1995, US$)	33 971

Gross fixed capital investment (1995):	
Percentage of GDP	20.4
Per head, US$	6 917

THE GOVERNMENT

Public consumption in 1995 (percentage of GDP)	20.3
General government current and capital expenditure in 1995 (percentage of GDP)	47.1
General government revenue in 1995 (percentage of GDP)	50.2

Composition of Parliament (number of seats):	
Labour Party	67
Conservative Party	28
Christian Democratic Party	13
Centre (Agrarian Party)	32
Progress Party	10
Social Left Party	13
Others	2
Total	165

Last general elections: 1993

Next general elections: 1997

FOREIGN TRADE

Exports of goods and services (average 1990-95, as a per cent of GDP)	39.3
of which:	
Gross freight and oil drilling (1990-95)	5.8
Main exports in 1995 (percentage of total commodity exports):	
Forestry products	4.8
Base metals and products thereof	12.0
Fish and fish products	7.2
Machinery, apparatus and transport equipment (excluding ships)	9.1

Imports of goods and services (average 1990-95, as a per cent of GDP)	32.3
Main imports in 1995 (percentage of total commodity imports):	
Ships	3.0
Machinery, apparatus and transport equipment (excluding ships)	33.9
Raw meterials (non-edible), including fuel and chemicals	13.3
Base metals and products thereof	10.6

THE CURRENCY

Monetary unit: Krone

Currency units per US$, average of daily figures:	
Year 1996	6.46
December 1996	6.47

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Norway by the Economic and Development Review Committee on 13 January 1997.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 3 February 1997.

•

The previous Survey of Norway was issued in August 1995.

Assessment and recommendations

The upswing is entering its fourth year

Following a strong rebound in the second half of 1993, the Norwegian economy has expanded rapidly, with overall real GDP growth averaging 4½ per cent in the years 1994 to 1996. The growth impetus was provided by robust demand conditions, both abroad and at home. Soaring petroleum exports and a pickup of sales of non-oil products (such as fish food) contributed to a sharp widening of the current account surplus. Private consumption was boosted by easier monetary conditions and substantial gains in real earnings. Investment on both the mainland and the Norwegian continental shelf also surged, reflecting high corporate profits and capacity utilisation rates. Excluding petroleum output, growth of mainland GDP averaged almost 3½ per cent in the past three years, leading to substantial job creation. This has resulted both in making the employment level one of the highest among OECD countries and in reducing the unemployment rate to a seasonally adjusted 4.5 per cent by the end of 1996 – 1.5 percentage points below its 1993 peak.

But mainland growth is approaching capacity limits...

With mainland output growth having exceeded its estimated potential for three successive years, the output gap has turned positive, which suggests that the economy is now operating close to capacity. Indeed, there are signs that the availability of labour is progressively becoming a constraint on economic activity. After a sharp cyclical rebound,

labour supply is at a record high, which, given the decline in long-term unemployment, seems to indicate that labour reserves may soon be exhausted. The number of unfilled vacancies is on an upward trend, with employers in several industries reporting difficulties in hiring suitably skilled workers. In particular, tight labour supply conditions are being felt in some parts of the manufacturing industry (such as metal products) as well as in the health sector. The unemployment rate may thus have fallen below its ''equilibrium'' level, and upward pressure on wages is likely to be building up.

... and despite continued low inflation...

Wage growth has, in fact, already accelerated from around 3 per cent per annum over the first half of the 1990s to over 4 per cent in 1996, prompted by the strong profitability of resource-based industries following the global hike in commodity prices beginning in 1993. In real terms, earnings rose 3 per cent in 1996 – the highest since the mid-1980s boom – as consumer price inflation continued to decline, helped *inter alia* by reductions in indirect taxes (notably a sharp decline in auto taxes) and weak import prices in the wake of the slowdown in the European economy. Although the downward trend in inflation was reversed in the autumn of 1996 by a sharp rise in hydro-electricity charges, domestic price increases averaged a comfortable 1.3 per cent for the year as a whole – one of the best performances among European economies.

... the economy risks overheating

The Secretariat projects overall economic growth to lose some momentum, slowing to 3 per cent in 1997 and 2.3 per cent in 1998, based on an expected stabilisation of petroleum production and capital formation (mainland and offshore) at a high level, the maturation of the consumer durable goods cycle and continued public expenditure restraint. By contrast, with growth in the European econ-

omy recovering, non-oil exports should increase further. Accordingly, the mainland economy is projected to expand by 2.5 per cent in 1997 and by about 2 per cent in 1998, with the unemployment rate declining towards 3.5 per cent. Wage inflation is expected to drift up, reaching 5 per cent by 1998, while price increases should stay subdued in the 2 to 3 per cent range. Such a wage and price scenario calls for continued moderation in pay settlements. Otherwise, there is a risk of substantially higher inflation, the more so since the booming real estate market could contribute to boosting domestic demand further.

As a result, the overall policy strategy is coming under strain

In such circumstances, a continuation of the current macroeconomic policy framework – with monetary policy focusing primarily on exchange rate stability against European currencies, in order to provide a ''level playing field'' for wage bargaining and international competition, while fiscal and income policies are geared towards inflation control – may become problematic. This framework, which proved successful in recent years when there was still considerable slack in the economy, is coming under strain now that the risk of inflationary tensions is rising. There are three main reasons for this. *First*, the conduct of monetary policy is increasingly complicated by upward pressure on the krone generated by the widening current account surplus. *Second*, there is concern that the soaring petroleum revenues may trigger claims for additional public expenditure, in particular if oil prices turn out to be higher than envisaged. *Third*, with the tightening of labour market conditions, the incomes policy objective of wage moderation may be more difficult to achieve in the period ahead.

A restrictive monetary policy is therefore needed...

Consequently, a restrictive monetary policy is now necessary. In order to contain speculative capital inflows, central bank deposit rates, which constitute the floor for money market rates, have been reduced by some 150 basis points

since March 1996, from a level of 4.75 per cent prevailing at the beginning of the year. As a result, the short-term interest rate differential against Germany has almost been eliminated. These developments have contributed to higher asset prices, both in the real estate and stock markets. With, as noted, the economy running near potential, such a course, if maintained, risks providing undue impetus to domestic credit and private demand, with resultant overheating. Therefore, whatever room exists for tightening monetary conditions, within the present framework of maintaining a stable exchange rate over the medium term, needs to be used to forestall an acceleration of inflation.

... together with a continued tight fiscal stance...

Tight fiscal policy can help take pressure off monetary policy by reducing the potential conflict between exchange-rate stability and inflation control. Following deficits in the early 1990s, the general government account has recorded widening surpluses, increasing from 0.3 per cent of GDP in 1994 to 3.5 per cent in 1995 and 6.3 per cent in 1996. Although the bulk of this improvement resulted from rising petroleum proceeds and the strong expansion of the mainland economy, reduced growth in public expenditure also played a role. The Budget for 1997 envisages a continued move towards fiscal restriction, of the order of ¾ per cent of GDP, which – under present oil price assumptions – implies maintaining the general government surplus at around 6½ per cent of GDP. With budget surpluses officially projected to continue during the rest of the decade, Norway's fiscal outlook thus compares favourably with those of most other OECD economies – the more so since the public sector enjoys a comfortable net asset position. Even so, the particular circumstances of Norway seem to call for continued efforts to tighten fiscal policy as a means of restraining possible real appreciation of the currency and associated crowding out of the tradeables sector.

... and an accumulation of foreign assets by the government

The current policy is to invest the State surpluses in the government's ''Petroleum Fund'' which was established in 1991 to preserve a portion of the wealth generated by the oil and gas resources for future generations. As operated by the Bank of Norway, under government guidelines, the Fund's investment currently takes the form of foreign currency denominated deposits and bonds. Thus, by absorbing part of the currency inflows associated with the government 's oil and gas proceeds, the Fund also helps counter the upward pressure on the krone's exchange rate stemming from the strong external position. Since the Fund's assets might create an incentive for additional government spending, it would seem advisable to explore ways by which these assets could be made available for the financing of future government liabilities, such as pensions. As discussed in detail in the 1995 Survey of Norway, this would have the advantage of providing now for the financing of the increase in welfare costs associated with an ageing population that is expected to come at a time when oil revenues are projected to be declining.

Incomes policy also has an important role to play...

By promoting wage moderation, incomes policy could continue to support monetary and fiscal measures in maintaining low inflation. Since the early 1990s, the ''Solidarity Alternative'' agreement between the government and the social partners has been used to this effect, with the view to preserving the international cost competitiveness of the mainland industry. The agreement is based on the exchange rate remaining stable, this being intended to avoid job losses associated with a currency appreciation or an erosion of real wages resulting from a depreciation. As the fiscal concessions made by the government to facilitate this agreement have been limited, the present incomes policy has proved to be quite successful so far even if there has been some increase in wage drift over the past year.

... but flexible relative wages are necessary

However, while this system appears to have delivered moderate wage settlements at the macro level, it has also tended to prevent market forces from being reflected in the structure of relative wages. Indeed, by supporting more rapid growth of lower pay scales, wages for some young and inexperienced workers have tended to become so high as to restrict their employment opportunities. As a result, many of these people have been induced to enrol in education or to participate in labour market programmes. Moreover, the rigid earnings structure has contributed to entrenching skill mismatches. The government should therefore encourage the social partners to explore the scope for enhancing wage flexibility. A more active competition policy in key industries would help in this regard, as this would reduce the risk of high wage increases in such industries spilling over to the rest of the economy.

... as well as measures to stimulate labour supply

In addition, attempts by the government to increase the size and quality of the labour force, in order to boost the growth potential of the economy, should be continued and even reinforced. In this regard, raising the effective retirement age while avoiding reductions in standard working hours would be beneficial. A number of salient features of the social transfer system would also need to be corrected. Norway has one of the most generous social transfer regimes in the OECD area, providing income replacement to nearly 15 per cent of the working-age population through the National Insurance Scheme. Almost half of this support is for disability pensions, while the remainder provides unemployment benefits and various other forms of temporary income maintenance. Together with active labour market programmes and sharp increases in public employment (the level of which currently reaches over one-third of total employment), such a policy has helped the country to avoid the secular rise in ‘‘open’’ unemployment recorded in most

other European OECD economies. Despite a number of recent reforms, the social transfer system and the high benefit levels it provides continue to distort work incentives, effectively creating a wage floor and potentially reducing labour supply. High marginal tax rates exacerbate such drawbacks, again affecting mostly low skill and young workers. In practice, the high cost of the system has tended to raise the overall level of taxation in the economy as well as wage costs per worker. Streamlining the system and curbing public sector employment growth would therefore help improve the functioning of the labour market, and, in turn, create room for a more efficient allocation of resources.

... and improved education and training

As discussed in Chapter IV of the Survey, the Norwegian economy could also benefit from better education and training to meet changing skill requirements. The 1994 reform in that area ensured all teenagers a place in the upper-secondary school system (including apprenticeship training for those who have opted for vocational education) and was designed to relieve pressure on the system from the massive student inflows following the recession in 1988-92. At the same time, participation in higher education increased substantially, boosting the overall cost of the system. However, there is a need for continued efforts to develop the vocational stream of upper-secondary education and a better targeting of higher education on labour market needs. The effectiveness of such policies would be strengthened by establishing an incentive structure geared towards encouraging human capital investment consistent with emerging employment opportunities. Combined with enhanced wage flexibility, this would contribute to reducing mismatches between the supply and demand of labour and, as a result, help improve productivity.

At the same time, structural reform should proceed in other areas

There is scope for structural initiatives in other areas as well. The presence of monopoly rents in some sectors of the economy indicates that market entry of new suppliers is restricted. A more active competition policy would have the advantage of lowering such barriers, in particular for small and medium-sized enterprises. Such firms could also benefit from government initiatives to set up co-operation networks, most importantly in the field of research and development. With a view to lowering cost levels through lower food prices, support to the agriculture sector should be reduced, both in terms of direct subsidies and border protection. Moreover, further rationalisation in the banking industry is needed in order to promote a leaner and more efficient financial system. This would call for continuing the process of privatisation of the commercial banks, which appear to have overcome the crisis of the early 1990s. In the meantime, it is important that the government exercise its role as owner to assure that the banks are managed efficiently and keep earning the required rate of return on their capital. There is also a need for further targeting of the activities of the traditional state banks whose role should be scaled down and limited to providing funding for socially profitable but commercially risky investments. All such structural adjustments would pave the way for a smooth transition to an economy less dependent on petroleum activities.

I. Recent developments and short-term prospects

Reaching capacity limits

The upturn from the 1993 trough is now entering its fourth year, with mainland GDP growth accelerating from 2¾ per cent in 1995 to 3½ per cent in 1996 (Table 1). Economic expansion is thus approaching the momentum of the mid-1980s boom, and for the fifth year in a row is outpacing growth in the European Union (Figure 1). Mainland economic activity has become more consumption driven, as fixed investment has been levelling off, despite a pick up in capital formation on the Norwegian continental shelf (NCS) in the first half of 1996. Moreover, the net foreign trade contribution to mainland growth turned positive in 1996, as traditional (non-petroleum) exports were boosted by strong demand in Asia and Eastern Europe. As has been regularly the case in the past decade, overall GDP (mainland and offshore activities) growth has seen an even more pronounced rise, accelerating from 3¼ per cent in 1995 to over 5 per cent in 1996, on account of the continuing surge in petroleum exports.

After several years of robust economic growth, the output gap in 1996 turned positive for the first time since 1988 (Figure 1, Panel B). The emerging capacity constraint is primarily explained by tightening labour market conditions, whereas the capacity utilisation rate (structures and equipment in use as share of total) has actually declined, due to high growth rates of business investment in recent years. Indeed, the unemployment rate fell below 4½ per cent in late 1996, and is expected to decline further in 1997. The current rate of unemployment is now well below the Secretariat's estimate of the non-accelerating wage inflation rate of unemployment (NAWRU). The emerging labour scarcity is illustrated also by employers' reporting growing difficulties in recruiting suitable labour and by the rapid decline in long-term unemployment. After a jump to 4¼ per cent in

Table 1. **Demand and output**

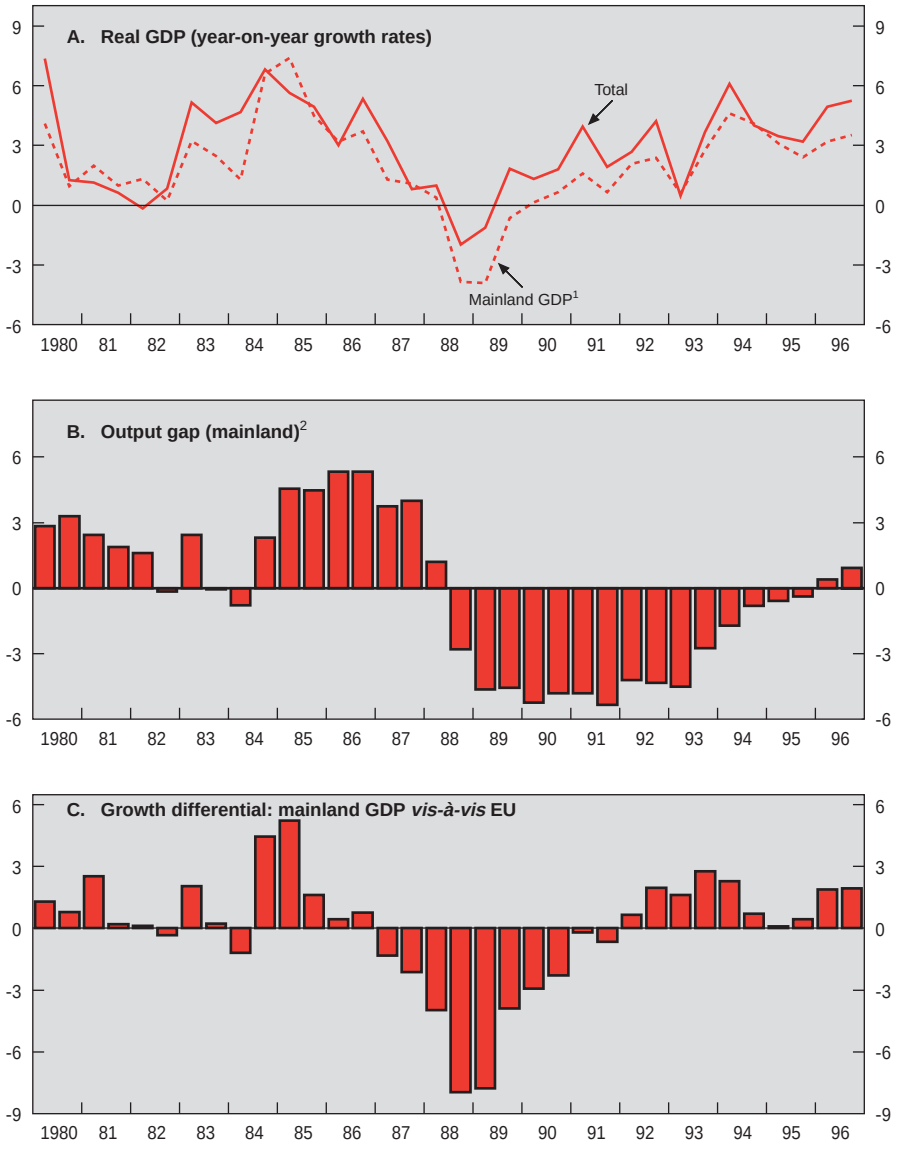
Percentage changes from previous period, seasonally adjusted annual rates, volumes

	1993 current prices NKr billion	1994	1995	1996[1]	1995 1st half	1995 2nd half	1996 1st half	1996[1] 2nd half
Private consumption	411.6	4.1	2.6	4.0	1.0	4.9	4.3	2.7
Government consumption	179.6	0.7	0.2	2.0	–0.3	1.3	2.7	1.3
Gross fixed capital formation	168.2	6.9	4.5	6.5	9.6	1.0	0.9	24.4
of which:								
Public sector	27.3	1.6	–0.5	2.4	–1.0	–5.4	10.7	–5.2
Petroleum	57.2	–7.3	–13.1	9.2	–17.1	24.0	–17.7	65.1
Residential	17.4	34.9	12.7	8.7	12.6	–6.6	–5.9	63.9
Other private	66.3	14.0	16.1	6.6	31.2	–5.1	11.1	10.2
Stockbuilding[2]	9.6	0.5	1.5	0.0	0.7	2.1	–2.6	3.1
Total domestic demand	769.0	4.4	3.7	4.0	3.3	5.4	0.2	10.5
Exports of goods and services	316.0	8.2	3.8	7.6	–0.9	7.8	6.7	9.3
of which:								
Traditional goods	112.4	13.1	4.1	12.9	–2.1	2.5	19.1	11.6
Petroleum	104.0	11.6	8.4	8.5	6.2	18.5	15.3	–11.9
Ships and platforms	11.8	–12.0	5.2	..	–40.5	33.2	–64.2	..
Services	87.8	0.6	–3.0	..	–1.8	–1.4	–3.4	..
Imports of goods and services	261.7	6.9	5.1	5.0	6.5	5.3	–0.5	16.4
Foreign balance[2]	54.3	0.9	–0.1	1.4	–2.4	1.3	2.8	–1.4
GDP	823.6	5.0	3.3	5.1	0.6	6.3	3.9	6.4
of which: Mainland	695.3	4.3	2.7	3.4	0.1	4.7	1.7	5.5

1. OECD Secretariat estimates.
2. Contribution to GDP growth.

Source: Statistics Norway and OECD Secretariat.

Figure 1. **GROWTH PERFORMANCE**



1. Mainland GDP excludes North Sea production, oil platforms and ships.
2. The difference between actual and trend mainland GDP.

Source: OECD, *Quarterly National Accounts;* Secretariat estimates.

1996, a further acceleration of wage increases is projected for 1997, which could undermine the profits and international competitiveness of mainland industry and potentially contribute to stronger domestic demand and economic overheating.

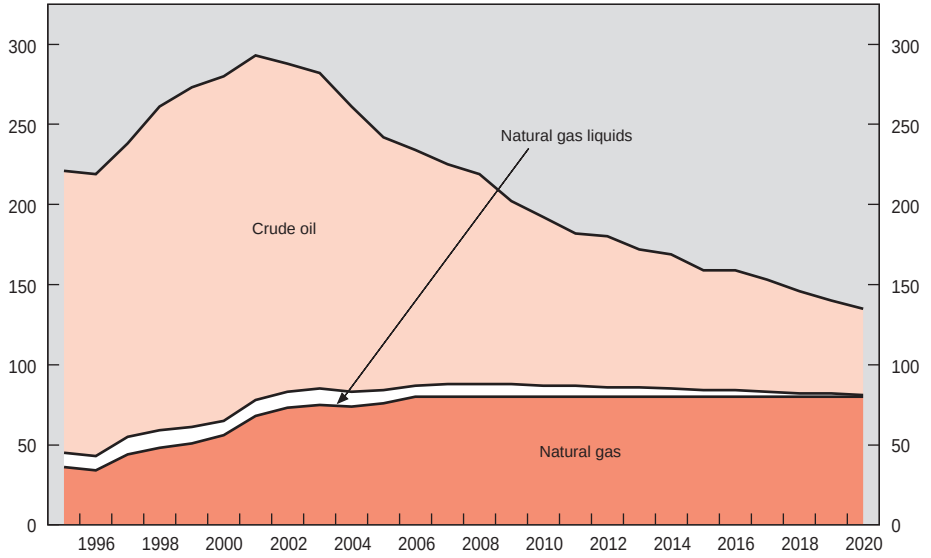
The following paragraphs examine in more detail the main features of the present upswing, starting with a review of offshore and non-oil export developments. This is followed by an analysis of mainland demand, the labour market, inflation and the external position. The chapter concludes with the economic projections for 1997-98.

Continued buoyant oil activity

The petroleum sector has continued to expand strongly in 1996, with oil and gas exports growing at around 10 per cent for the third successive year, due to large investments in new production facilities on the NCS. Moreover, while no major new fields have been discovered since the 1980s, the *recovery factor* (the amount of oil and gas that can be produced in proportion to the total estimated stock) has increased tremendously, based on innovations in exploration and extraction technologies. As a result, Norway has become the world's second largest net exporter of crude oil and natural gas liquids (NGLs), after Saudi Arabia. Petroleum production now represents one-sixth of overall GDP, one-third of total exports and a quarter of total fixed investment. As discussed in the 1995 *OECD Survey of Norway*, the increases in the recovery factor have resulted in the government's long-term projections of petroleum production consistently erring on the downside, with the 1996 output level one-third higher than officially projected in the 1994-97 Long-Term Programme released in 1993.

While subject to many uncertainties, the most recent official projections for the long run suggest that total oil and gas production will peak at around 292 million standard cubic metres of oil equivalent (Sm^3o.e.) in the year 2001, before gradually falling to 135 million Sm^3o.e. by 2020 (Figure 2). The projected decline after the turn of the century is attributable to the maturing of the major oil fields discovered in the 1970s and early 1980s. As a result, future output has to rely progressively on smaller (and more expensive) fields and on the still substantial natural gas resources. As discussed in detail in the 1995 *Economic Survey*, both future gas production and the development of small oil fields are subject to

Figure 2. **OIL AND GAS PRODUCTION**
Million standard cubic meters of oil equivalent



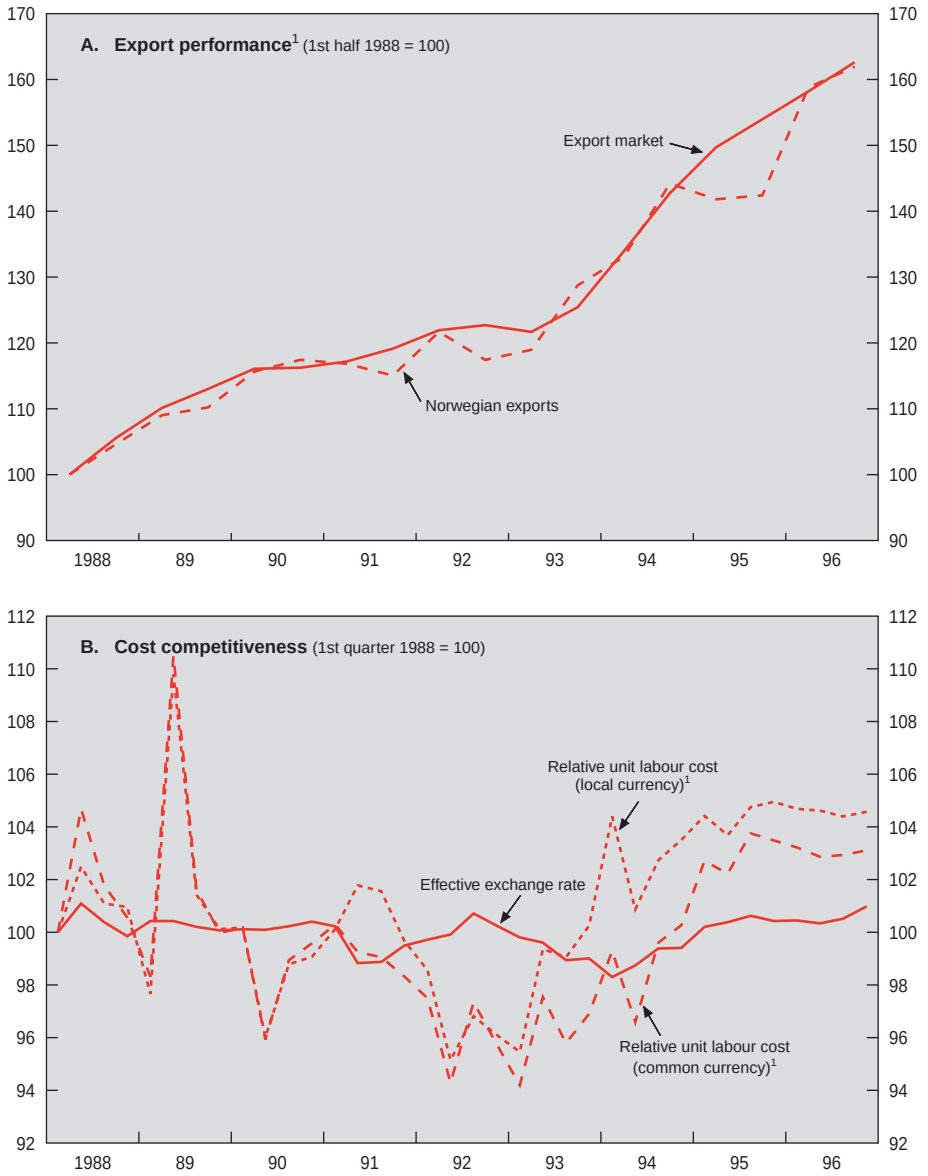
Source: Ministry of Industry and Energy.

major uncertainties related to developments in the global energy market – including the price of crude oil and the future supply of natural gas by competing producers present on the European market, Russia and Algeria in particular.[1]

Strong non-oil exports

Petroleum exports aside, ‘‘traditional’’ merchandise exports in Norway have increased by over 30 per cent in real terms in the 1994-96 period, notwithstanding a short-lived downturn in 1995 due to weak demand in Europe (Table 1 and Figure 3, Panel A). Such remarkable outcomes were achieved despite a significant rise in relative unit labour costs (as measured in a common currency)

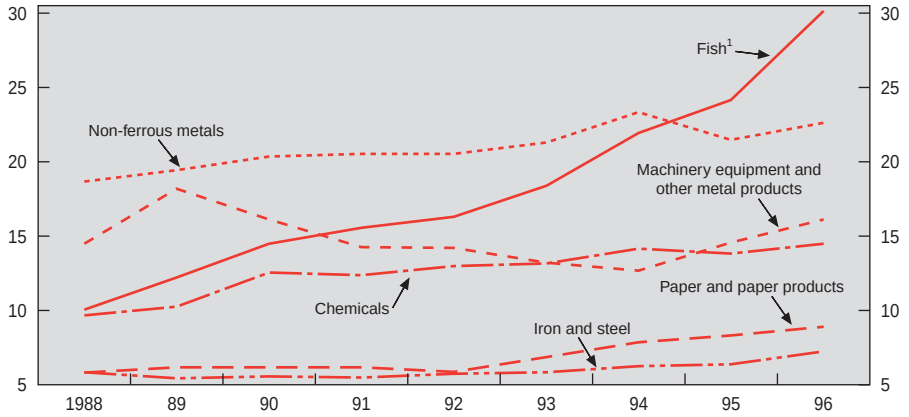
Figure 3. **EXPORT PERFORMANCE AND COST COMPETITIVENESS**



1. Manufacturing.
Source: OECD Secretariat.

Figure 4. **EXPORTS BY COMMODITY GROUP**

NKr billion, 1988 prices



1. Including processed fish and feed.
Source: Statistics Norway.

between 1993 and 1995 (Figure 3, Panel B), as exports have shifted to strongly growing markets in the United States, the United Kingdom and the Nordic countries, as well as to new markets in eastern Europe and south-east Asia.[2] These trends appear to have continued in 1996, with exports to eastern Europe and south-east Asia being particularly buoyant.

Most impetus for growth in traditional exports has been provided by the exports of fish, which have virtually tripled (in real terms) since 1988 (Figure 4). This reflects a sharp rebound in fish catches from their 1990 trough and soaring sales of farmed salmon to Europe and Asia (see Box 1 for details). The exports of paper and paper products, as well as of iron and steel, non-ferrous metals (predominantly aluminium) and chemicals have also expanded significantly during the 1990s. The latter two product categories were, however, adversely affected by the stock adjustments related to the temporary slack in Europe in 1995. By contrast, the protracted decline in exports of machinery and equipment was reversed in that year.

Box 1. **The Norwegian salmon farming industry**

With the introduction of new technologies, the share of Norway in the world supply of salmon has increased significantly since the early 1980s. The global market share of Norwegian farmed salmon has indeed risen from 3 per cent in 1985 to 17 per cent in 1995, while the world production of salmon doubled during this period (Table 2). Norway has thus become the world's largest producer of farmed salmon, before Chile.

Table 2. **World salmon production**

	× 1 000 tonnes			Per cent share of total		
	1985	1990	1995	1985	1990	1995
Total	868	1 091	1 479	100	100	100
of which: Farmed salmon	48	275	535	6	25	36
of which:						
Norway	**30**	**146**	**249**	**3**	**13**	**17**
United Kingdom	7	32	72	1	3	5
Ireland	1	6	12	0	1	1
Chile	1	23	106	0	2	7
Canada	1	19	41	0	2	3

Source: FAO.

As the tariff structure in the European Union discourages the importation of processed fish products, the Norwegian salmon industry has specialised in the production and exportation of fresh and frozen salmon. Norway is now the largest single supplier of these products to the EU area[1] and a growing exporter to South-East Asia. Norway's main European competitors in these markets are the United Kingdom[2] and Ireland. Following the United States government decision, in 1991, to introduce a 23.8 per cent anti-dumping duty and a 2.27 per cent countervailing duty on Norwegian salmon exports, Norway has seen its share in the US market declining rapidly to the benefit of Chile which has become the dominant exporter to North America.

Since the early 1980s the Norwegian fishery authorities have attempted to contain the growth in the production of farmed salmon in order to sustain price levels, product quality and environmental conditions. Its primary instrument has been the granting of site licences, with a 12 000 m^3 limit on the water volume enclosed per site. In 1990 and 1991 the Fish Farmers' Sales Organisation (FOS) froze the excess supply of fresh salmon in an attempt to prevent a further decline in prices. However, the profits of salmon farmers plummeted and many suppliers went bankrupt. This prompted the authorities to liberalise the ownership regulations in the industry in order to bring in new capital, and to enhance

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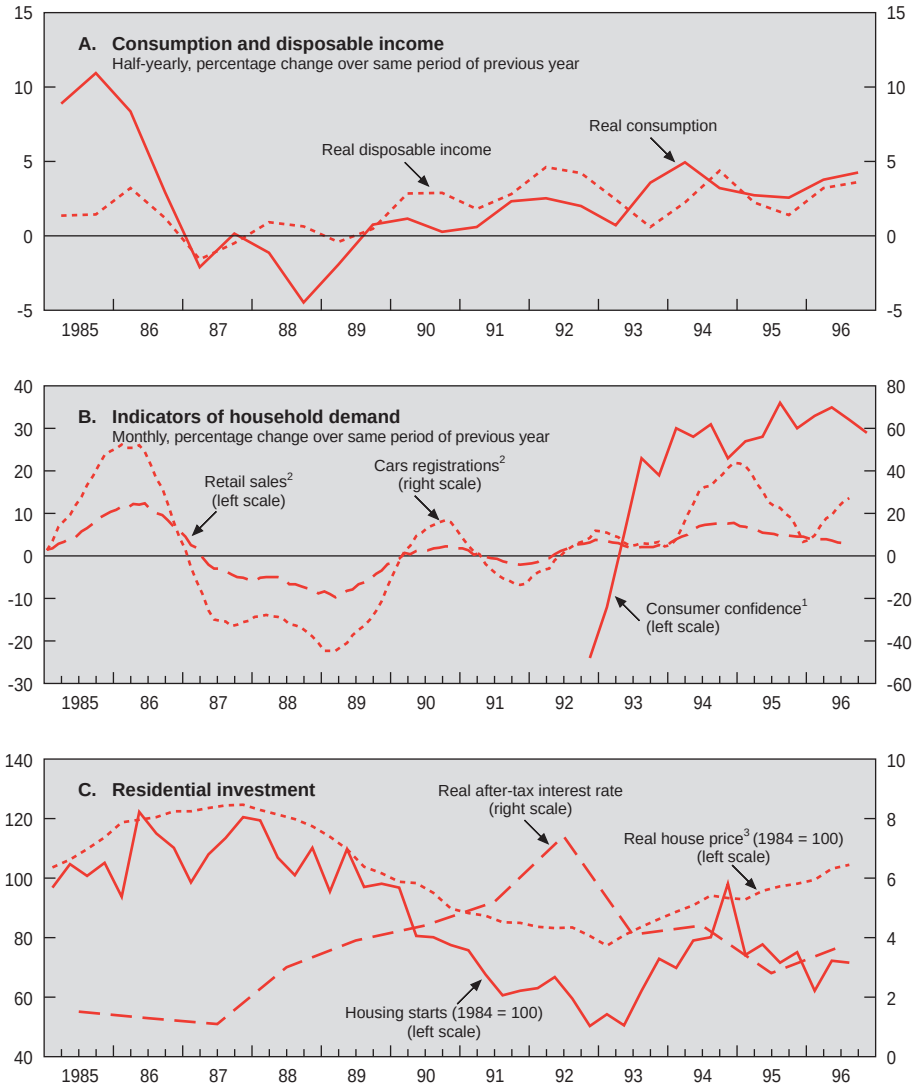
the efficiency and international competitiveness of the sector through a better allocation of resources and the achievement of scale economies. As a result, many young entrepreneurs and investors were attracted to the industry, thereby raising the availability of modern production technology and marketing know-how. This has sparked a wave of innovations, with the breeding time of farmed salmon reduced by around two-thirds. At the same time, production losses from diseases have become smaller and input prices lower (of smolts and feeds in particular). Reflecting these developments, the real unit cost of farmed salmon (in constant 1995 prices) has declined from NKr 41 per kilo in 1990 to NKr 19 in 1995 and, despite the continued decline in market prices from around NKr 36 to NKr 30 per kilo (1995 prices) in the same period, gross margins have turned positive.[3]

1. Within that area, Denmark has become the main location for transit trade of Norwegian salmon and a leading producer of smoked salmon.
2. In 1996, following complaints by the organisations representing the salmon farming industry in Scotland and Shetland, the European Commission started anti-dumping and anti-subsidy investigations against the Norwegian salmon farming industry. Awaiting the final verdict, the Commission decided not to take immediate action.
3. A decision by the European Commission in December 1995 introduced a minimum price for salmon products for the first half of 1996 of 3.4 ECU per kilo (NKr 27.5). The price was again liberalised following a decision by the Norwegian authorities to introduce rationing of feed through ''stops'' and quotas in order to limit the supply of salmon products.

Sustained household demand

The trend rise in real household consumption since early 1993 resumed in 1996 – after losing momentum in the previous year – based on a sharp increase in real wage earnings and a decline in the household savings ratio (Figure 5, Panel A and Table 3). Growth in consumer demand has been most pronounced for private cars. In fact, following some weakening in 1995, car purchases rebounded strongly in response to a significant cut in auto taxes and an increase in the deposit refund for scrapped vehicles at the beginning of the year (Figure 5, Panel B). Not surprisingly, consumer sentiment is currently upbeat, reflecting also the downward trend in real (after-tax) interest rates and the unabated improvement in household wealth since 1993 (Figure 5, Panel C; Figure 6). Households' financial consolidation, which marked the 1988-1992 recession, is clearly over, with the ratio of household net financial assets to disposable incomes having returned to the pre-recession level of almost 40 per cent.

Figure 5. **HOUSEHOLD SPENDING**



1. The indicator reflects consumer perception of economic developments for the past 12 and next 12 months and whether present conditions are conducive to larger purchases.
2. Percentage change over 12 months, 12-month moving average.
3. Second hand market.

Source: Statistics Norway; Ministry of Finance; OECD, *National Accounts, Main Economic Indicators.*

Table 3. **Household appropriation account**

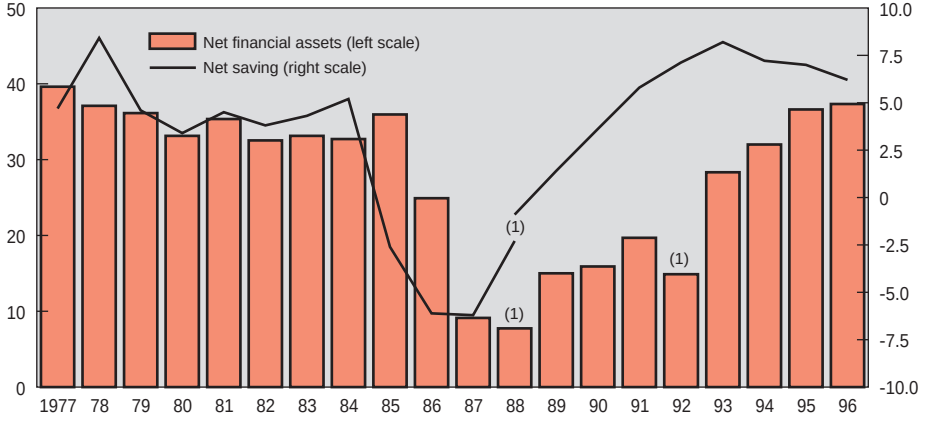
Percentage change from previous year

	1992	1993	1994	1995	1996[1]
Real disposable income	**3.5**	**3.4**	**2.5**	**2.3**	**3.1**
Contribution from:					
Wages	0.6	1.2	2.5	2.4	4.0
Profit income	–0.6	0.6	0.4	–0.5	0.0
Net transfers	1.4	0.7	0.1	0.7	0.7
Net interest receipts	0.1	0.8	1.3	0.7	0.3
Other income	1.5	1.1	–0.1	0.4	–0.2
Direct taxes	0.6	–1.1	–1.6	–1.4	–1.8
Real consumption expenditure	**2.2**	**2.2**	**4.1**	**2.6**	**4.0**
Saving ratio, level in per cent	**7.1**	**8.2**	**7.2**	**7.0**	**6.2**

1. Estimates.
Source: Statistics Norway.

Figure 6. **HOUSEHOLD FINANCIAL POSITION**

Per cent of disposable income



1. Break in series due to revision on the National Accounts.
Source: Bank of Norway; Ministry of Finance.

In line with the marked improvement in household financial wealth, the real estate market has been recovering strongly in recent years, with real house prices increasing by a total of 15 per cent in 1994 and 1995, and between 10 and 20 per cent[3] in 1996 (Figure 5, Panel C). Nevertheless, residential investment has been surprisingly weak since mid-1995, due to: *i)* limited availability of construction sites and rising land prices in urban areas; *ii)* capacity constraints in the construction sector associated with strong building activity in the non-residential sector (including a new airport near Oslo); and *iii)* demographic factors, including reduced demand for new housing by young adults, reflecting increased participation in education (see also Chapter IV). However, housing starts rebounded in the second half of 1996, indicating that residential investment is on the verge of a significant recovery.

Robust business investment

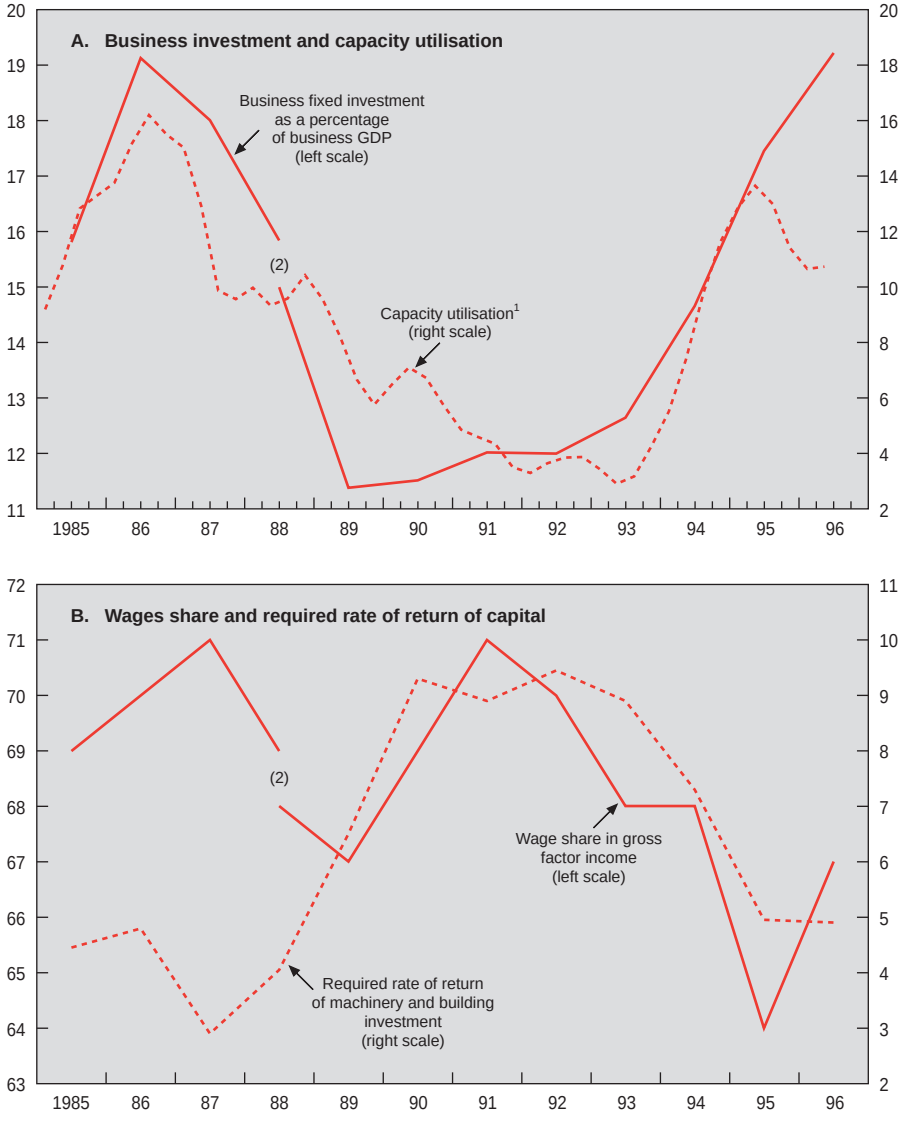
With the ratio of mainland business investment to mainland GDP having returned to its 19 per cent peak of 1986, growth in mainland business investment finally tapered off from the double-digit rates recorded in 1994 and 1995, to reach an estimated 6½ per cent in 1996 (Figure 7, Panel A, Table 1). A number of factors have contributed in this regard, most prominently the investment induced decline in capacity utilisation, especially in the manufacturing sector. The sharp increase in manufacturing profits achieved in recent years was reversed in 1996, due to an acceleration of wage growth and a downward adjustment in international commodity prices. The conditions sustaining mainland business investment at high levels have, however, remained relatively favourable, with both the wage share in value-added and capital costs at historical lows (Figure 7, Panel B).

Falling unemployment

Employment growth has been on an upward trend since 1993, from 1½ per cent in 1994 to 2¼ per cent in 1995 and 3 per cent in the first three quarters of 1996 (Table 4 and Figure 8, Panel A), with job creation gradually shifting from the public to the private sector. At the same time, employment in full-time equivalents grew around 1 per cent, suggesting that the share of part-time labour

Figure 7. **MAINLAND BUSINESS INVESTMENT**

Per cent



1. Percentage of manufacturing firms reporting capacity shortages.
2. Break in series due to National Accounts revision.

Source: Bank of Norway; Statistics Norway.

Table 4. **Labour market developments**

Percentage changes, Mainland

	1990 1 000 persons	1991	1992	1993	1994	1995	1996 first three quarters [1]
Mainland GDP		1.1	2.2	2.8	4.3	2.7	3.4
Labour productivity		2.1	2.5	2.6	3.1	0.6	0.6
Employment [2]	2 030	–1.0	–0.3	0.0	1.5	2.2	2.8
id. full-time equivalents [3]	..	–0.3	0.4	0.1	1.0	2.0	..
Labour force	2 142	–0.7	0.2	0.1	0.9	1.6	1.0
Unemployment [2, 4]	113	5.5	5.9	6.0	5.4	4.9	4.5 [5]
of which: Males	66	5.9	6.5	6.6	6.0	5.2	..
Females	46	5.0	5.1	5.2	4.7	4.6	..
Teenagers (< 20 years)	18	17.4	16.7	18.1	17.1	16.1	19.3 [6]
Young adults (20-< 25 years)	24	11.0	12.4	12.0	10.9	10.2	10.3 [6]
More than 12 months unemployed [7]	19	21	26	29	29	27	16 [6]
Participation in active labour market programmes	40	2.1	2.3	2.7	2.6	2.0	1.9 [6]
Id. including special measures	51	2.7	2.9	3.4	3.4	2.9	2.7 [6]

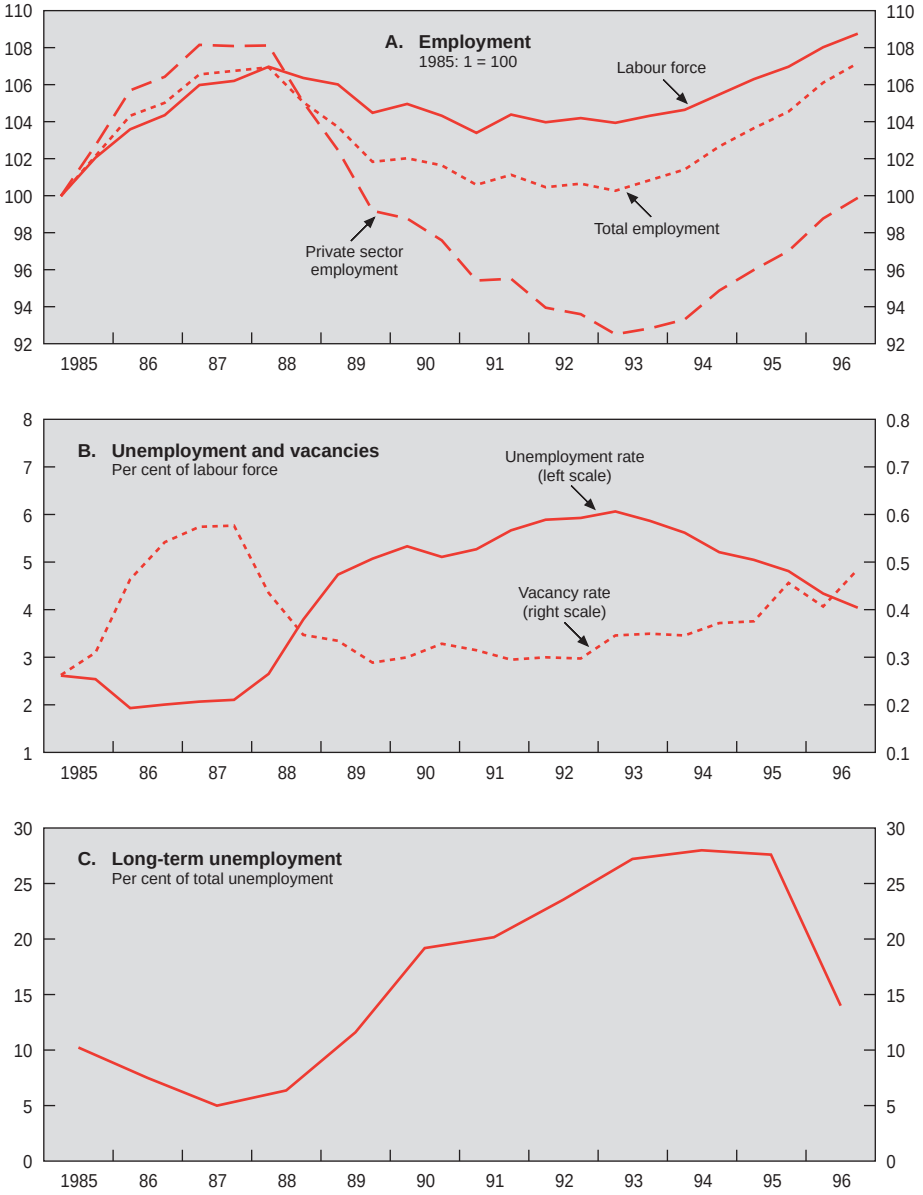
1. Growth rates relative to the same period in the previous year.
2. Labour force survey.
3. National accounts figures.
4. Per cent of the labour force, level.
5. Seasonally-adjusted.
6. First half of the year.
7. Incidence.

Source: Statistics Norway.

has increased substantially. The increase in the labour supply has been boosted since 1994, mainly by the entry of students and discouraged workers to the labour market and the in-flow of workers from the other Nordic countries (notably to the construction and health sectors). These developments explain the modest decline in the unemployment rate, from the 6 per cent peak in 1993 to 4.5 per cent in the third quarter of 1996.[4] Including participation in active labour market programmes, the adjusted unemployment rate declined from the 8 per cent peak in 1993 to 6½ per cent in the first half of 1996. The decline in unemployment has involved all categories of workers (by gender and age), except teenagers.[5]

Taken together, a number of factors point to labour market conditions having been tightening to a point where they could act as a drag on further

Figure 8. **THE LABOUR MARKET**



Source: OECD, *Quarterly National Accounts; Main Economic Indicators.*

increases in production. In particular, the number of unfilled vacancies is on an upward trend, with many employers reporting difficulties in recruiting suitably skilled labour (Figure 8, Panel B). Moreover, the incidence of long-term unemployment (more than twelve months) has fallen sharply from the 29 per cent peak in 1994 to 16 per cent in the first half of 1996, suggesting that existing labour reserves are being exhausted. The tighter situation in the labour market – in particular in the metal products industry and health care – is compounded by an emerging limit to further increases in labour supply, with the labour force participation rate having almost reached its 1987 record level of 80 per cent.

Still subdued inflation

After accelerating by 1 percentage point in 1995, consumer price inflation subsided to 1¼ per cent in 1996 (Table 5). The moderate inflation outcome was supported by the waning effect of a 1 percentage point increase in the VAT in 1995, the substantial reduction in new car registration charges at the beginning of 1996, and a decline in import prices associated with a strong krone and weak international commodity markets. The downward trend in consumer price inflation was reversed over the summer, however, due to a drought-related hike in hydro-electricity prices and the elimination of VAT reductions on dairy and meat products from 1 July (Figure 9, Panel A). The subdued consumer price inflation for the whole of 1996 is nonetheless well below that of Norway's main trading partners.

By contrast, the increase in the overall GDP deflator accelerated from 3 per cent in 1995 to almost 3½ per cent in 1996 (Figure 9, Panel B). This development reflected the continued improvement in the terms of trade since 1993, which was supported primarily by rising energy prices. The spot price of crude oil (Brent Blend) registered a 20 per cent increase in 1996 – up from an average $17 per barrel in 1995 to $20.30 – due to colder weather conditions and the delay of Iraqi oil entering the world market in the autumn. Other commodity prices, however, have weakened since mid-1995, due to the sluggish European economy and the collapse of the copper trade market, with commodity prices (as measured by the HWWA commodity price index excluding energy) falling by 12 per cent in the twelve months to July 1996. For 1996 as a whole, the decline in commodity

Table 5. **Prices, wages and costs**

Percentage change from previous year

	1992	1993	1994	1995	1996[1]
Consumer price index	**2.3**	**2.3**	**1.4**	**2.4**	**1.3**
Hourly earnings per employee	**2.7**	**3.5**	**3.2**	**4.1**	**4.2**
of which: Mainland	2.7	3.3	3.1	4.2	4.2
Manufacturing and mining	2.8	2.8	2.9	4.4	4.0
Private services	2.9	3.0	3.1	4.1	4.2
Public sector	2.8	3.4	3.3	4.0	4.2
Relative unit labour cost					
local currency	–4.4	2.0	4.1	3.4	–0.2
common currency	–3.5	0.0	2.6	5.8	–0.2

1. Estimate.

Source: Bank of Norway, OECD Secretariat.

prices is estimated at 7 per cent. As a result, the increase in the mainland GDP deflator diminished.

Since 1990, wage costs have provided virtually no inflationary impetus, as they hovered in a range of 3 to 3½ per cent per year, and were largely matched by increases in labour productivity (Table 4). This situation changed from 1995, when labour productivity levelled off, followed by an acceleration of wage inflation in 1996, to 4¼ per cent (and somewhat more in manufacturing). The pay increase was due to wage drift prompted by the payment of substantial profit related bonuses in several industries – following the boom in commodity prices in 1994 and early 1995 – and higher wage settlements in the spring of 1996. With inflation declining, the growth in real wages reached 3 per cent in 1996, the highest level in two decades. Price competitiveness – as gauged by the stabilisation of relative unit labour costs (in a common currency) – has not been affected due to a narrowing of profit margins in 1996, in particular in metal industries. Notwithstanding the weaker profit developments, a number of factors indicate that wage inflation pressure is building up, including increased strike activity, and the apparent weaker influence of the main federation of labour unions over local wage settlements. As a result, wage moderation by the central labour market organisations – in the framework of the so-called ‘‘Solidarity Alternative’’ (see Chapter III) – which proved successful in recent years, may now be at risk.[6]

Figure 9. **WAGE AND PRICE DEVELOPMENTS**

Per cent change over corresponding period of previous year

1. Manufacturing.

Source: OECD, *National Accounts.*

Rising current account surplus

With export developments buoyant, the current account surplus has soared, from around 3 per cent of GDP in the early 1990s, to an estimated 6½ per cent of GDP in 1996 (Figure 10, Panel A). This was underpinned by a widening surplus on the oil and gas balance, from the already comfortable 12 per cent of GDP in the 1993-95 period to almost 14 per cent in 1996, and a pronounced contraction in the ''traditional'' trade deficit from 7 per cent of GDP in 1995 to 5½ per cent in 1996. A further decline in the deficit on the investment-income account, owing to the gradual elimination of net foreign debt, also contributed in this regard (Figure 10, Panel B). As a result, Norway has become an international creditor since 1995.

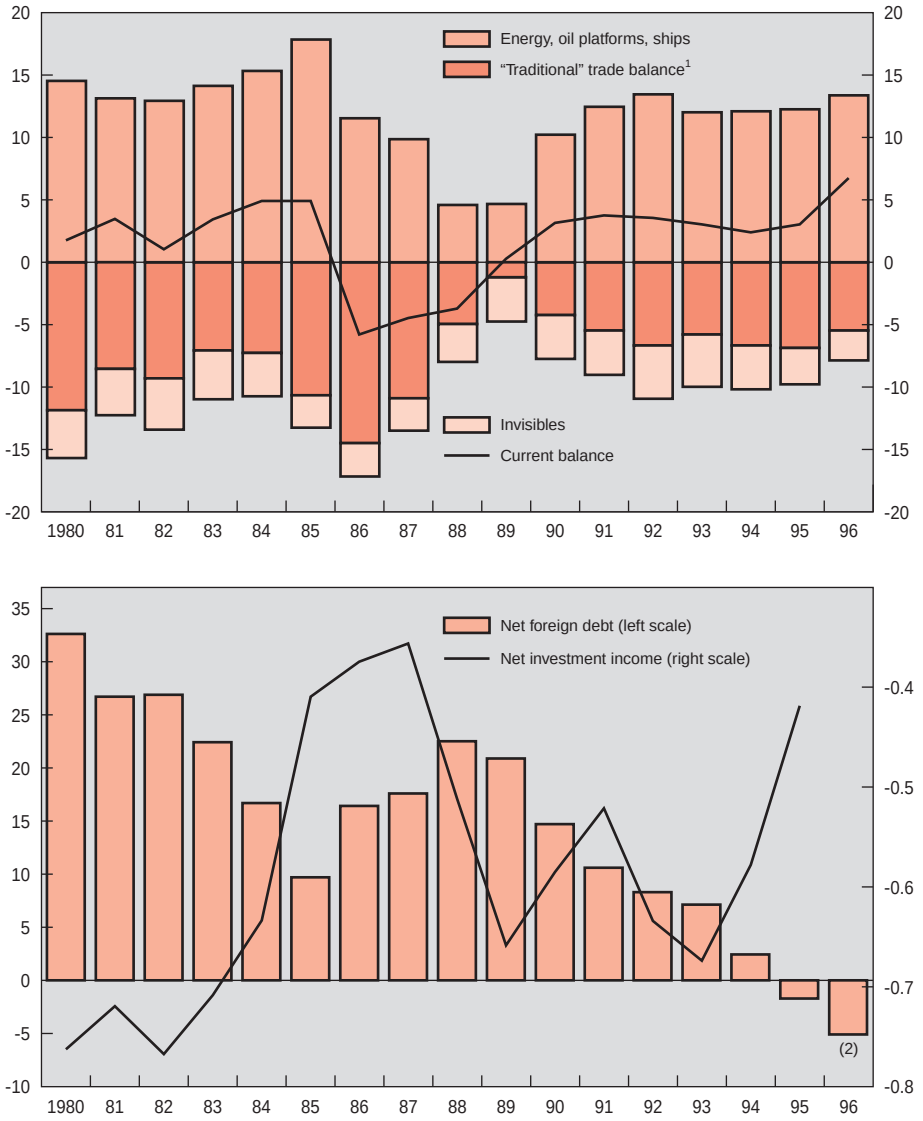
Short-term outlook

The Secretariat projects overall GDP growth to slow down from 5 per cent in 1996 to 3 per cent in 1997 and 2.3 in 1998, based on the expected levelling off in petroleum production and a moderation of domestic demand (Table 6). In particular, capital formation (mainland and offshore) should stabilise at a high level, while the durable consumer goods cycle matures – following the jump in car purchases at the beginning of 1996 – and public expenditure restraint continues. By contrast, with economic activity in Europe picking up, the demand for traditional exports should remain fairly strong. Accordingly, the mainland economy is projected to expand by 2.5 per cent in 1997 and by 2.1 per cent in 1998. Despite the continued cyclical rebound in labour supply, this should suffice to reduce the unemployment rate further from 4.3 per cent in 1996 to 3.6 per cent by the end of 1998. With labour market conditions thus tightening, wage growth is projected to accelerate to over 5 per cent and consumer price inflation, after attaining 1¼ per cent in 1996, could creep up to around 2½ per cent in 1997 and broadly remain at that level in 1998. Despite an anticipated correction of the recent oil price hike, the current account surplus is expected to remain above 6 per cent of GDP.

These projections, which are consistent with those published in *Economic Outlook, No. 60* (December 1996), are based upon the following assumptions:

Figure 10. **EXTERNAL POSITION**

As a percentage of GDP



1. Excluding exports and imports of petroleum, oil rigs and ships.
2. First half-year only.

Source: Statistics Norway.

Table 6. **Short-term projections**[1]

Percentage change, constant prices

	NKr billion in 1993	1995	1996	1997	1998
Private consumption	411.6	2.6	4.0	3.2	2.6
Government consumption	179.6	0.2	2.0	0.7	1.9
Gross fixed investment[2]	168.2	4.5	6.5	3.7	2.4
of which: Oil sector[2]	57.2	–13.1	9.2	3.8	–1.2
Non-oil business sector	66.3	16.1	6.9	3.9	3.3
Residential construction	17.4	12.7	8.7	6.6	6.2
Public sector	27.3	–0.5	2.4	1.6	1.3
Stockbuilding[3]	9.6	1.1	0.0	0.0	0.0
Total domestic demand	769.0	3.7	4.0	2.7	2.3
Exports of goods and services	316.0	3.8	7.6	5.7	3.2
of which: Energy exports	104.1	8.4	8.5	2.7	1.7
Imports of goods and services	261.7	5.1	5.0	5.4	3.6
Foreign balance[4]	54.3	–0.1	1.4	0.5	0.1
GDP	823.6	3.3	5.1	3.0	2.3
of which: Mainland	695.3	2.7	3.4	2.5	2.1
Memorandum items:					
GDP deflator	–	3.1	3.4	1.6	2.4
Private consumption deflator	–	2.4	1.3	2.5	2.4
Employment	2 036.7[5]	2.1	2.4	1.3	1.1
Unemployment rate	6.0	4.9	4.2	3.9	3.6
Private compensation per employee	–	3.5	4.2	4.3	5.3

1. Projections published in the OECD *Economic Outlook 60*, December 1996.
2. Includes platforms under construction.
3. Contribution to GDP growth, excluding platforms under construction.
4. Contribution to GDP growth.
5. × 1 000 persons, National accounts figures.

Source: OECD Secretariat.

– The predicted growth pattern in the OECD area should lead Norwegian manufacturing export markets to grow by 6½ and 7 per cent, respectively, in 1997 and 1998;
– The average OECD oil-import price is projected to decline from the November 1996 level of $20 per barrel, to approximately $18 in 1997 and 1998, with supply and demand conditions stabilising;
– A modest tightening of monetary conditions in Europe in the second half of 1997 is assumed to be followed by Norwegian authorities to pre-empt inflationary pressures and keep the exchange rate *vis-à-vis* the ECU roughly unchanged at its November 1996 level. However, long-term

interest rates are expected to increase slightly from their current level, and somewhat more than German rates, due to a modest pick up in inflation;

– The ‘‘underlying’’ growth of central government expenditure (excluding unemployment benefits and corrected for changes in accounting practices) is assumed to be restrained to ¼ per cent in real terms, and income taxation to be slightly reduced, consistent with the 1997 Budget proposal (see Chapter II).

Since the projections were finalised in November 1996, the oil price (in US dollars) and the krone exchange rate *vis-à-vis* the ecu have moved up substantially, while the Bank of Norway cut its official interest rates by 75 basis points in January 1997 to counter the upward pressure on the currency. Moreover, the official projection for ‘‘underlying’’ central government expenditure growth has been revised upward to 1 per cent in 1997, in line with the National Budget adopted in December 1996 (see Chapter II). Given these developments, domestic demand and economic growth may turn out to be somewhat stronger than suggested by the current projections. This points to a risk of economic overheating. The tightness of the current labour market situation, in particular, with rising numbers of unfilled vacancies in certain sectors, may lead to higher than expected wage and price inflation. Domestic demand may be boosted by wealth effects associated with a potential re-emergence of speculation in the real estate market, more robust employment and earnings developments, and higher than expected public expenditure associated with rising public budget surpluses.

II. Economic policies

Overview

With activity having strongly recovered from the protracted recession of the 1988-92 period, the economic policy setting in Norway is geared towards creating the conditions for sustained economic growth while avoiding a re-emergence of inflation. *Monetary policy* is tied to the commitment of a stable value of the krone against European currencies, at the level prevailing after the floating of the exchange rate in December 1992. Exchange rate stability is deemed to: *i)* provide a credible low-inflation anchor, thereby contributing to low interest rates; *ii)* preserve a ''level playing field'' for the social partners with a view to making wage negotiations focus on real rather than nominal increases; and *iii)* create a stable economic environment for the mainland tradeables industry, in order to help reduce the economy's petroleum-dependency. *Fiscal policy*, after having been extensively used to stimulate the economy during the recession, currently aims at budget consolidation in anticipation of the decline in State petroleum revenues and the increase in welfare expenditure associated with the ageing of the population. Tight fiscal policy, however, is also considered essential on conjunctural grounds. At the same time, *incomes policy* attempts to moderate the wage settlements between employers and labour unions in order to preserve the international competitiveness of Norwegian exports along the lines of the ''Solidarity Alternative'' agreement adopted by the government and the social partners in 1993.[7] This agreement is based on a stable exchange rate, without which imported inflation (in case of a depreciation) or losses in competitiveness (if the krone appreciates) risk undermining the wage settlements. Finally, *structural policies* focus essentially on the harmonisation of trade regulations and competition laws with the EU's *Acquis Communautaire* in the framework of the EEA-agreement (in force since early-1994)[8] and, more recently, the implementation of the WTO-agreement.

The pursuit of this strategy may prove quite demanding in the current environment of strong economic growth, booming petroleum activity, large current account surpluses and tighter labour market conditions. With the objective of keeping a stable exchange rate as a binding constraint on the active use of monetary policy for stabilisation purposes, the authorities are essentially relying on restrictive fiscal policies as a means to choke off excess demand and prevent the economy from overheating. Although this approach has been successful so far, whether this will continue to be the case is highly uncertain, the more so since the government's soaring petroleum revenues could complicate the task of the fiscal authorities in implementing a budget austerity programme. As a result, the authorities may face the dilemma of either giving priority to exchange rate stability, with the risk of undermining inflation control, or raising interest rates to keep a rein on inflation expectations but with the risk of feeding the upward pressure on the krone.

The present chapter reviews the progress in implementing and maintaining this strategy. It starts with a discussion of fiscal policy developments, with an emphasis on possible approaches for the management of the government's petroleum wealth in the long run. It then examines monetary management and financial market behaviour in the last eighteen months or so. This is followed by a brief appraisal of structural reform in the fields of agriculture and financial services. Structural initiatives in other areas, as well as incomes policy, are discussed in Chapter III dealing with the implementation of the OECD *Jobs Strategy*.

The fiscal stance

Widening budget surpluses in 1995 and 1996

After having been in deficit in 1992 and 1993, the general government account has been showing widening surpluses, ranging from 0.3 per cent of GDP in 1994 to 3.5 per cent in 1995 and an estimated 6.3 per cent in 1996 (Table 7). The bulk of these surpluses originate at the State level, due to the surge in oil-related revenues, which more than doubled between 1994 and 1996. But even excluding such revenues, the State balance also posted a significant improvement,

Table 7. **Budget balances: projections and outcomes**

	1994	1995		1996		1997	
	Outcome	Proposed[1]	Outcome	Proposed[1]	Estimate[2]	Proposed[1]	Approved[3]
Central government (NKr billion)							
Revenues	**344.5**	**349.6**	**372.4**	**373.8**	**402.2**	**419.3**	**430.0**
of which:							
Mainland taxes	217.4	254.2	263.7	273.5	287.4	317.7	319.2
Petroleum revenues	44.0	47.0	50.2	48.8	60.4	61.1	69.1
Expenditure	**373.4**	**370.5**	**368.3**	**363.2**	**356.4**	**378.4**	**375.4**
of which:							
Petroleum expenditure[4]	18.4	15.3	11.7	–0.5	–10.8	–4.0	–8.6
Balance	**–28.9**	**–20.9**	**4.1**	**10.6**	**45.8**	**40.9**	**54.6**
of which:							
''Net cash flow'' from petroleum activities	25.6	31.7	38.5	49.3	71.3	65.1	77.7
Balance excluding petroleum activities	–54.5	–52.6	–34.4	–38.7	–25.5	–24.2	–23.1
Petroleum Fund financial investment	**0.0**	**0.0**	**2.0**	**10.6**	**45.8**	**43.3**	**57.4**
of which:							
Transfer from the Central Government	0.0	0.0	2.0	10.6	45.7	40.9	54.6
Return on financial assets	0.0	0.0	0.0	0.0	0.1	2.4	2.8
Net lending[5]	**–0.3**	**–4.9**	**29.6**	**18.7**	**64.5**	**57.9**	**70.4**
General government (per cent of GDP)							
Net lending	**0.3**	**–0.5**	**3.5**	**1.9**	**6.3**	**5.1**	**6.4**
of which:							
Central government	0.0	–0.7	3.2	1.9	6.4	5.5	6.6
Local government	0.3	0.3	0.3	0.0	–0.1	–0.4	–0.2
Memorandum items:							
Mainland GDP volume growth	4.3	2.7	2.7	2.6	3.4	2.5	2.5
Oil price (kroner per barrel)	111	115	108	105	125	115	125
Oil and gas production[6]	180	163	193	188	222	236	236

1. October of the preceding year.
2. November 1996.
3. December 1996; memorandum items November 1996.
4. Net of operating profits of the State Direct Financial Interest.
5. Accrual basis, including lending operations and surpluses on social security accounts and special funds.
6. Million standard cubic meters of oil equivalent.

Source: Ministry of Finance.

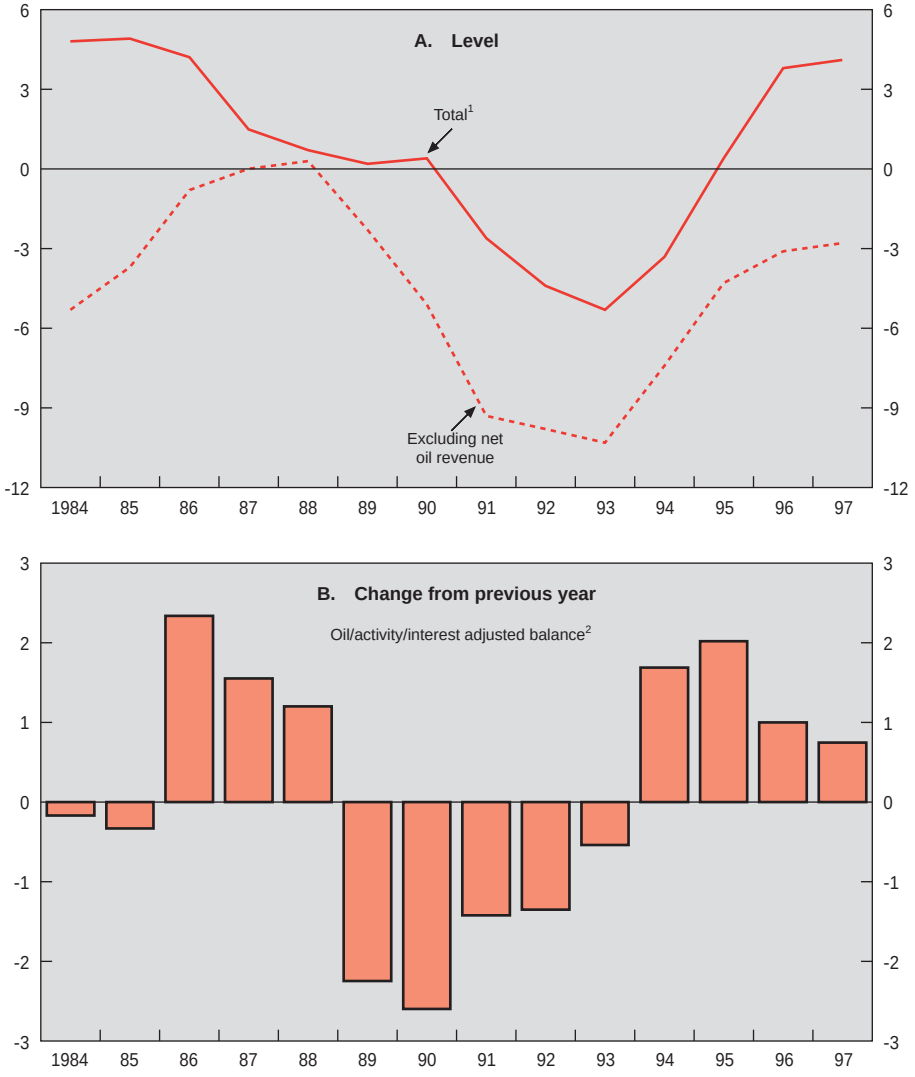
with the non-oil deficit declining from 7.4 per cent of mainland GDP in 1994 to 4.3 per cent in 1995 and to 3.1 per cent in 1996 (Figure 11, Panel A). Although most of this improvement resulted from the strong growth of the mainland economy, fiscal consolidation played a role in this regard. In fact, since the recovery started in 1993 the authorities have aimed at restraining real growth in central government ''underlying'' expenditure (excluding unemployment benefits and one-off spending items and corrected for changes in accounting methods) to less than 1 per cent per year. As a result, the stance of fiscal policy has moved markedly towards restriction, as indicated by the State's non-oil cyclically-adjusted primary deficit (net of debt interest payments) – which is estimated by the authorities to have decreased by 2 and 0.9 percentage points in 1995 and 1996, respectively (Figure 11, Panel B).

Although such favourable fiscal outcomes in part reflect policy actions, the extent of the improvements has largely come as a surprise to the authorities (Table 7). Indeed, the results for general government net lending have exceeded the initial budget estimates by as much as 4 per cent of GDP in 1995 and 4.5 per cent in 1996. In both years, roughly half of the difference was due to better-than-projected non-oil tax revenues, resulting from the greater strength of the mainland economy since 1994, with the central government's mainland tax revenues outstripping earlier budget estimates by NKr 9 billion in 1995 and NKr 14 billion in 1996. Local governments' tax revenues portrayed a similar development – although their net lending position deteriorated as their expenditure rose strongly at the same time. Moreover, as indicated in Chapter I, official projections of oil and gas production underlying the budget have been exceeded by around 15 per cent in both 1995 and 1996. As a result, the State's net cash flow from petroleum activities in 1995 was NKr 7 billion higher than the initial projections and, given the substantial unexpected increase in the oil price, three times that amount in 1996.

Given such favourable fiscal outcomes, the government has started to transfer sums into the foreign-currency denominated *Government Petroleum Fund*, in order to build up foreign financial assets as a means to preserve the public wealth emanating from the oil and gas sector – an official objective which was reiterated in the Revised National Budget released in May 1995. The first allocation of NKr 2 billion took place in May 1996 when the fiscal accounts for 1995 were

Figure 11. **STATE BUDGET BALANCE**

As a percentage of mainland GDP



Note: Data for 1996 are estimates; for 1997, projections.

1. As a percentage of total GDP.
2. Adjusted for cyclical developments, petroleum income, foreign interest payments, capital transactions and transfers from the Bank of Norway.

Source: Ministry of Finance.

finalised. For the fiscal year 1996 over NKr 45 billion – an amount equivalent to the estimated central government surplus, and four times that announced in the 1996 Budget – was transferred into the Fund at the end of the year. The Fund, which is managed by the Central Bank in the same way as the ordinary currency reserves, consists of foreign currency-denominated deposits and bonds (see Box 2 for further details). According to current plans, the Fund should contain assets equivalent to NKr 270 billion, or roughly 30 per cent of GDP, by the end of the year 2000. According to new guidelines for monetary policy formulated in the 1995 Revised Budget, and which have been reiterated in subsequent budget documents, the Bank of Norway is committed to purchase foreign currency in the exchange market corresponding to the sums allocated to the Petroleum Fund. As discussed further below, this has so far not conflicted with the objective of exchange rate stability. In fact, upward pressure on the value of the krone has led the bank to purchase even larger amounts of foreign currency than required by Petroleum Fund transactions.

Consolidating the surplus in 1997

The Approved Budget for 1997 foreshadows continued fiscal restraint, aimed at maintaining the general government surplus at around 6½ per cent of GDP in spite of some slowdown in mainland output growth and a levelling-off of petroleum revenues.[9] The extent of further fiscal tightening, as gauged by the increase in the State's non-oil cyclically adjusted primary balance, is officially estimated at ¾ per cent of mainland GDP (Figure 11, Panel B). Unadjusted for the impact of the business cycle and interest payments, the non-oil State deficit would reach 2.8 per cent of mainland GDP. As has been the case in previous budgets, fiscal consolidation measures are concentrated on expenditure restraint. As a result, the increase in the central government's ''underlying'' expenditures is projected to reach 1 per cent in real terms.[10] Areas where expenditure should grow strongly include health care (due to the construction of new hospitals in the Oslo and Bergen areas) and education (through the introduction of special grants to municipalities in view of the extension of the compulsory school age by one year – see Chapter IV).

Box 2. **The Government Petroleum Fund**

In the 1995 Revised National Budget, the government re-emphasised the future role of the Petroleum Fund, reiterating its wish to build up financial assets as a means to preserve wealth emanating from the petroleum sector. The Fund should invest in foreign currency-denominated assets in order to: *i)* absorb the currency inflows associated with the government's petroleum receipts from abroad (and hence contribute to containing the upward pressure on the Krone exchange rate); and *ii)* to diversify away from the risk of a combined fall in oil prices and market values of domestic assets.

While the Petroleum Fund was formally established in 1991, the first allocation of NKr 2 billion took place in May 1996 (accruing retroactively to the 1995 final accounts), followed by a second allocation of over NKr 45 billion at the end of the year. By the year 2000 the Fund should contain around NKr 270 billion, including the accrual of returns on the capital. The capital is valued against its historical cost or its market value, which ever is the lowest.

Pursuant to §7 of the Act of 22 June 1990 No. 36, the Government Petroleum Fund:

- Is managed by the Bank of Norway on behalf of the Ministry of Finance, involving also the reporting on the management and keeping the accounts for the Fund in accordance with the criteria laid down by the Ministry of Finance.
- Is placed as Krone deposits in a Treasury account with the Bank of Norway. The Bank should invest these resources separately in its own name in assets denominated in foreign currencies. The value of the Treasury account is set equal to the value of the corresponding portfolio of foreign financial assets, including the book return.
- Should achieve the highest possible return within the following limits: *i)* bonds, bills, notes and equivalent instruments must be issued by sovereign nations, enterprises with government guarantees or highly-rated international organisations; *ii)* investments may also be made in securities where the authorities are considered to back the lending programme, but for which there are no formal guarantees; *iii)* it is not permitted to invest in shares or other forms of equity capital, or in securities issued by borrowers other than those approved by the Central Bank Governor for the investment of foreign exchange reserves; *iv)* Bank deposits shall not exceed a value of 25 per cent of the Fund's market value. Such deposits can have up to six months' maturity; and *v)* the Fund's resources can only be invested in highly liquid instruments and in banks approved by the Central Bank Governor.
- Should be invested in foreign currencies, with the currency shares reflecting the geographical composition of Norwegian imports and an interest-rate risk limited to a maximum of 5 per cent of capital loss associated with a 1 percentage point increase in yields.

Taxes, on the other hand, will be slightly reduced, although the impact on the economy is likely to be rather small. The tax threshold, below which no tax is paid, has been raised by somewhat more than the estimated nominal earnings growth of 3½ per cent, while the special tax rebate for small children has been increased – although this has been partly compensated by the freezing of the standard tax deduction for dependent children at its 1996 level. Moreover, State revenues from custom duties are projected to decline consequent to the WTO agreement, although this is to some extent offset by an increase in excise duties on tobacco, diesel and packaging. A noteworthy initiative has been a shift in income taxation from municipalities to the State, which explains part of the increase in State non-oil tax revenues projected for 1997 (Table 7). This measure has been implemented in order to prevent higher municipal tax revenues, due to stronger economic activity, from further boosting local spending. The associated increased emphasis on block grants from the State for the financing of local governments has reduced the need for ''perequation'' transfers (inter-municipal transfers to offset regional differences in economic strength) through the off-budget ''Tax Equalisation Fund'', which will therefore be phased out.[11]

Although the oil price is assumed to remain at its 1996 level of NKr 125 per barrel in 1997, the State's net cash flow from petroleum activities should rise further, reaching NKr 78 billion for the whole year. As a result, overall State net lending is projected to attain 6.6 per cent of GDP – 0.2 percentage points higher than in 1996. A widening deficit at the local level, however, is expected to lead to a stabilisation in general government net lending at around 6½ per cent in 1997. The State budget surplus is planned to be invested in the Petroleum Fund, which, as a result, is expected to have accumulated assets worth 11 per cent of GDP, or NKr 100 billion, by the end of 1997. The current policy is thus to let the accumulation of assets prevail over debt redemption – in fact keeping the amount of outstanding debt fixed at roughly 30 per cent of GDP. Accordingly, the central government borrowing requirement in 1997 is projected to be practically nil (less than ½ per cent of GDP), as it has been the case since 1994.

In light of the experience with previous episodes of rising budget surpluses, in particular in the early 1980s, there is a distinct risk that such a comfortable fiscal situation may trigger political claims for additional government expenditure. This risk would be heightened if oil prices turn out to be higher than projected in 1997 (as a rule of thumb, a NKr 1 increase in the annual average oil

price raises the overall State surplus by NKr 1 billion). Given the already tight conditions in product and labour markets, further government spending would make it very difficult to avoid an overheating of the economy, in particular if growth in household expenditure were also to accelerate due to the perceived increases in public financial wealth. The section below discusses possible strategies to reduce such risks in the future.

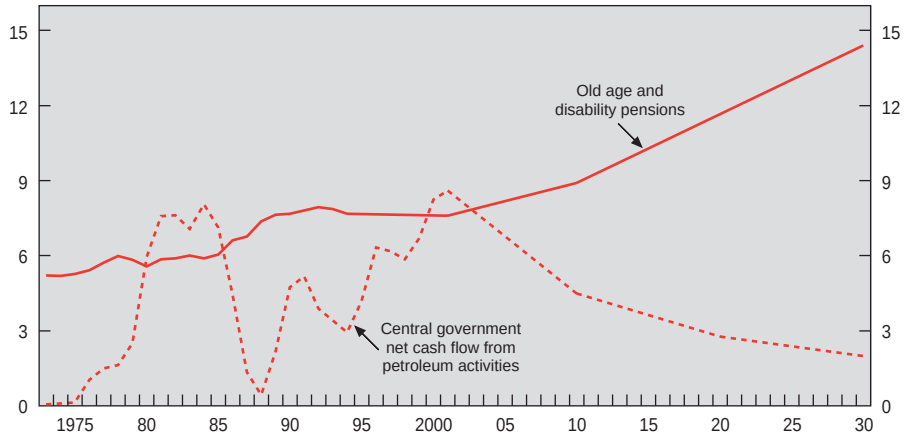
Managing the petroleum wealth in the long run

Together with the favourable budget outcomes, the recent review of pension rules, following the government's White Paper on the future of the social security system,[12] has eased concerns about the sustainability of fiscal policy in the long term. Indeed, an update of Norway's generational accounts[13] – estimating net transfers of wealth between generations through taxes and public expenditure programmes (assuming unchanged individual entitlements to pensions, social transfers and government services such as health care and education) – suggests that major inter-generational imbalances are unlikely to emerge, despite the ageing of the population. In this regard, Norway is in an unique position compared with other OECD countries, where the financing of future pension entitlements would require major tax increases in the future.[14] The main reason for this is the petroleum wealth, which allows the accumulation of substantial financial assets by the Norwegian Government well before public pension expenditure is expected to surge in the second and third decades of the next century (Figure 12). According to official projections, a sharp increase in pension expenditure, ranging from 8 per cent of GDP in the period 1995-2010 to 15 per cent in 2030, will be preceded by a doubling of the State's net cash flow from petroleum activities from 4 per cent of GDP in 1995 to 8 per cent in 2000.

While generational accounts are useful as a means to illustrate the impact of changes in policies on the inter-generational distribution of wealth, they are very sensitive to the underlying assumptions, such as the rate of interest used in the calculations. In particular, a small but sustained fall in the real interest rate from its present level of around 4 per cent would make it more difficult for Norway to keep public finances balanced in the long run, due to a decline in the return on the government's financial assets, including those accumulating in the Petroleum Fund.[15] Moreover, long-run projections for entitlement programmes are also highly dependent upon the cyclical position of the economy in the given base

Figure 12. **STATE PENSION EXPENDITURE AND PETROLEUM REVENUES**

As a percentage of GDP



Source: Ministry of Finance.

year on which the generational accounting model is calibrated (usually the latest year for which a budget is available). For example, if the economy appears to have been at a cyclical peak in the base year, benefit claims (unemployment, disability, early retirement, social welfare, etc.) may increase in the subsequent downturn, thus accentuating the rise in social expenditure emerging from the ageing of the population.

Such uncertainties call for a mechanism to prevent undue increases in government spending as these would jeopardise the accumulation of financial assets in the Petroleum Fund and thereby the financing of future expenditure. At present, transfers to the Petroleum Fund are decided as part of the normal budget procedure and, therefore, are subject to approval by the parliament on an annual basis. As a result, the parliament is entitled to reduce the allocations to the Petroleum Fund if needed to finance current expenditure. A possible way to avoid this could be to legally earmark a portion of the financial assets accumulating in the Petroleum Fund – *e.g.* to finance public pension obligations. A drawback of this approach may be that such earmarked assets could then no

longer serve as a fiscal buffer against adverse economic shocks – which is currently one of the official aims of the Petroleum Fund. However, it is unlikely that the whole Fund would be needed for that purpose, notably if the projection of 30 per cent of GDP in the year 2000 is realised.

As an alternative strategy, the government could aim at reducing the need for public financial asset formation by shifting the responsibility for retirement compensation partly to the private sector. For example, as recently proposed by the Norwegian Banker's Association, the government could legally require large firms and/or branch organisations to set up funded occupational pension schemes to cover earnings-related pension entitlements. According to this proposal, the start-up capital for such schemes would be provided by the government and financed by a liquidation of part of the Petroleum Fund.[16] Next, the government would give tax credits to workers contributing to such schemes. Proceeding this way could have a number of advantages. First, the reduction in income taxation would narrow the ''tax wedge'', which should help improve the functioning of the labour market. Second, the relationship between pension contributions and benefits would become more transparent, and possibly reduce the incentives for early retirement (including retirement through the disability pension scheme).[17] Third, the creation of pension funds could help strengthen the local stock market, thus widening the scope for equity financing of Norwegian firms and possibly serving as a lever for corporate growth. However, such an approach would also make it more difficult for the government to ensure that the national saving surplus be largely invested in financial assets abroad, which is one of the official objectives of the Petroleum Fund. Indeed, solvency requirements would normally lead occupational pension schemes to invest mainly in domestic assets, given their future obligations in national currency.[18] A recently installed official commission is expected to report on this overall approach and possible alternatives by July 1998.

Monetary policy

The policy formulation

After the decision by the Norwegian monetary authorities to let the krone float in response to the turmoil in European exchange markets in late 1992,

monetary policy has been geared towards maintaining a stable exchange rate *vis-à-vis* countries pursuing low-inflation objectives. Such an orientation was formalised by the adoption of a policy framework in the May 1994 Revised National Budget, stipulating that the Bank of Norway should focus on ''the stability in the value of the krone as measured against European currencies'' at the level prevailing after the floating, which was about 3 to 5 per cent below the pre-floating ECU parity. The 1996 National Budget set out additional guidelines, which legally commit the Bank of Norway to purchase foreign currency ''at a scale that over time corresponds to the net allocation of capital to the Petroleum Fund'', although ''in a situation with a strong krone and a sharp cyclical upturn, it may be appropriate for the Bank of Norway to undertake net purchases of foreign currency exceeding the Petroleum Fund allocations''.[19] It is within these constraints that monetary policy and other policy instruments (fiscal policy, incomes policy) need to operate with a view to keeping inflation under control.

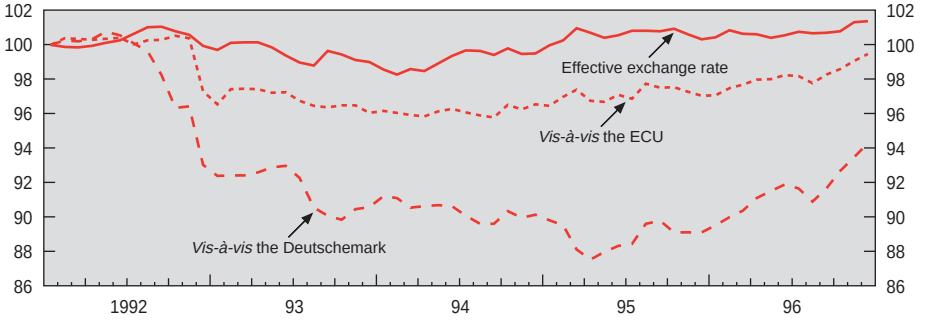
As the inflation record since the currency depreciation in 1992 has been favourable, financial markets have shown confidence in the ability of the monetary authorities to maintain price stability while achieving the stable exchange rate objective. This has been reflected in the behaviour of long term interest rates, which have gradually declined to a low level by Norwegian historical standards. However, due to the rising external surplus, the exchange rate has been subject to upward pressure in 1996. This has required the Bank of Norway to purchase large of amounts of foreign currency and has prevented a tightening of monetary policy which would have been expected given the strength of the economic upswing. In fact, as discussed below, the authorities may face the dilemma that avoiding an appreciation of the currency would call for a relatively easy stance of monetary policy at a time when activity is strong and the economy is continuing its upturn. The monetary policy framework would be subject to further strains if, as argued in the previous chapter, inflation risks materialised as a result of tight labour market conditions and budget discipline was weakened by the rising oil revenues.

Exchange and interest rate developments

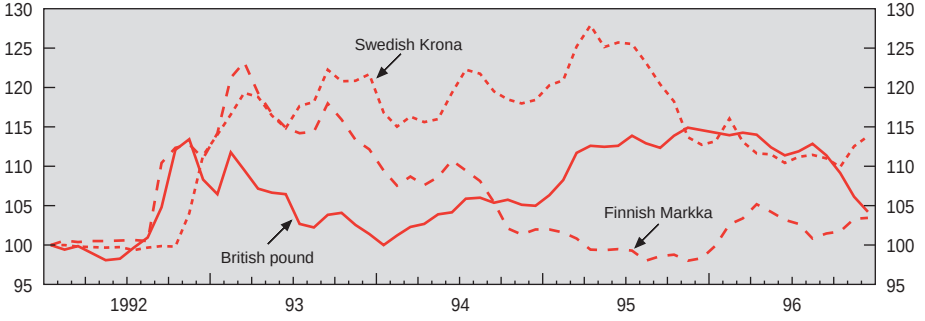
Since the start of the economic recovery in 1993, the 4 per cent depreciation against the ECU in the aftermath of the floating of the currency in December 1992 has been fully reversed (Figure 13, Panel A). Moreover, with the

Figure 13. **EXCHANGE RATE BEHAVIOUR AND OFFICIAL RESERVES**

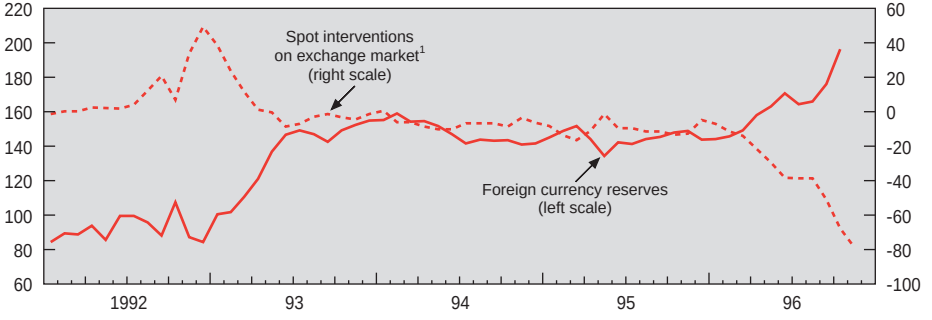
A. Exchange rate *vis-à-vis* the ECU, the Deutschemark and effective exchange rate
January 1992 = 100



B. Exchange rate *vis-à-vis* some other floating European currencies
January 1992 = 100



C. Official foreign currency reserves and spot interventions
Billion NKr

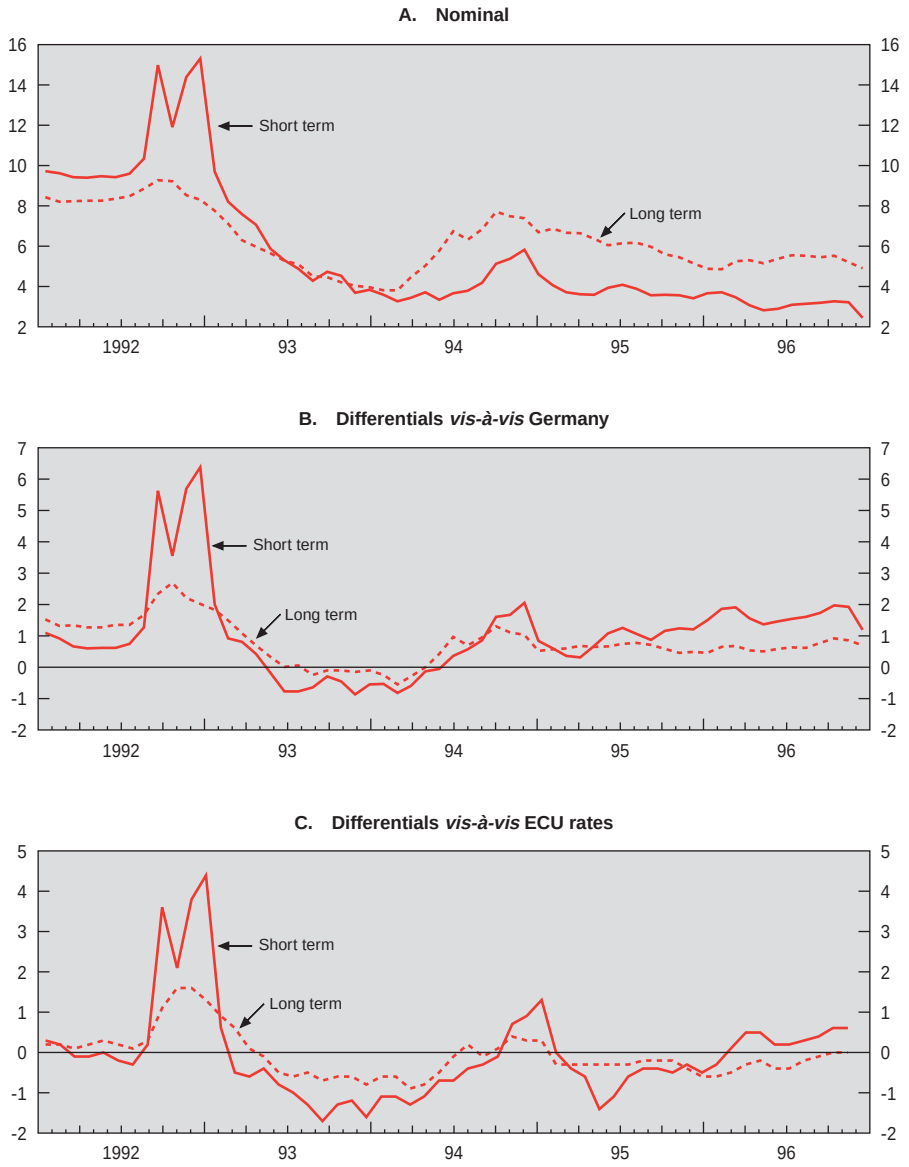


1. Net accumulated sales of currency to banks from January 1992 onwards.
Source: Bank of Norway; Statistics Norway.

depreciation after the floating almost negligible in wider trade-weighted terms – due to the simultaneous steep fall in the value of the Swedish krone and the Finnish markka (Figure 13, Panel B) – the effective exchange rate currently exceeds its pre-floating level. Since the beginning of 1996, large spot interventions by the Bank of Norway have been needed to contain upward pressure on the krone resulting from the sharp widening of the current account (Figure 13, Panel C). Foreign currency purchases by the Bank of Norway between January and December 1996 totalled NKr 92 billion,[20] raising official reserves to a record high of NKr 215 billion – more than twice the amount prior to the floating of the exchange rate in December 1992. In order to sterilise the liquidity generated in the banking system by such foreign exchange interventions, the Ministry of Finance issued short paper in the same proportion, in the form of Treasury Bills with a one year maturity. This facility for open market policy was created in the summer of 1993, but was used on a large scale only from early 1996.[21]

In order to offset the upward pressure on the exchange rate, the stance of monetary policy has been progressively eased since late 1996. In 1995 and most of 1996, the Bank of Norway succeeded in maintaining the three-month money market rate around 5 to 5½ per cent, notwithstanding a small cut in the official deposit and overnight lending rates by 0.25 per cent to 4.5 and 6.5 per cent, respectively, on 8 March 1996 (Figure 14, Panel A). As a result, the short-term interest differential with Germany gradually widened to reach almost 200 basis points by October 1996 (Figure 14, Panel B). Similarly, the differential against the official trade-weighted ECU rate turned positive in late 1995, and increased to nearly 100 basis points in October 1996 (Figure 14, Panel C). Since then, however, speculative capital inflows in anticipation of a possible appreciation of the krone have forced the monetary authorities to lower official interest rates, with the deposit and overnight lending rates cut by 125 basis points in December 1996 and January 1997 to 3¼ and 5¼, respectively. This resulted in a sharp decline in the three-month rate to some 3½ per cent last January, practically on a par with the equivalent German rate. While such monetary easing, combined with the temporary suspension of intervention in the foreign exchange market, appears to have succeeded in containing the capital inflows, there is a risk that exchange rate pressure will re-emerge once incoming indicators point to a further strengthening of Norway's economic fundamentals.

Figure 14. **INTEREST RATE DEVELOPMENTS**
Per cent



Source: Bank of Norway; OECD, *Main Economic Indicators.*

Long-term rates have tended to follow the movements in money market rates quite closely since late-1994. The ten-year bond rate mostly fluctuated between 6½ and 7 per cent until the fall of 1996, thus maintaining a substantial positive yield gap of around 150 basis points. The interest rate differential varied around 50 basis points against equivalent German and zero against equivalent ECU rates (Figure 14). Since November, however, the ten-year rate sharply declined, to below 6 per cent, as bond prices were boosted amid speculation by foreign investors on an appreciation of the currency and/or further easing of monetary policy. As a result, the long-term differential against Germany had virtually disappeared at the time of writing, indicating that the markets have fundamentally changed their risk appraisal of Norwegian financial assets.

Money and credit growth

After having hovered around 6 per cent in 1995, the rate of growth of the broad money aggregate M2 (narrow money plus time deposits) has been quite irregular in 1996, slowing down to 3.7 per cent on a twelve-month basis in November following an acceleration to the 7 to 9 per cent range in the first half of the year (Figure 15, Panel A). At this stage it is unknown why broad money supply has abated recently, although it may be a temporary phenomenon related to the pickup in housing investment in the second half of the year. On average, the year-on-year growth rate of M2 attained 5.5 per cent in the eleven months until November 1996. While a shift from sight to time deposits has reduced the expansion of the narrow money aggregate M1 in the course of 1995, it has converged to the growth rate for M2 in 1996.

Such developments in monetary aggregates have resulted from a recovery in domestic credit from the trough in the early 1990s, while money creation from foreign currency inflows (excluding those induced by tax payments by oil companies, which normally accrue to the government and hence do not affect domestic liquidity) has been largely sterilised (Table 8). The marked improvement in banks' balance sheets (see below) suggests that lending by commercial and savings banks is no longer bound by supply constraints, notwithstanding a gradual decline in lending margins (Figure 15, Panel B). Indeed, such lending has continued to grow strongly in 1995 and 1996, spurred by lower lending rates, booming private investment and rising credit demand by households to finance purchases of consumer durables (Figure 15, Panel C). By contrast, loans by the

Figure 15. **MONEY AND CREDIT**

1. Cash, demand deposits and unused credit held by the public; each observation represents a 3-month right adjusted moving average.
2. M1 plus time deposits.

Source: Bank of Norway; OECD, *Main Economic Indicators*.

Table 8. **Money and credit**

End of year, percentage change from previous year

	Level end 1990 NKr billion	1991	1992	1993	1994	1995	1996[1]
Total money stock (M2)	**453.4**	**10.8**	**7.5**	**0.6**	**6.7**	**5.3**	**6.4**
contribution from:							
Foreign assets, excluding oil taxes	..	–1.3	–0.5	0.2	–1.9	–5.3	1.5
Domestic credit, net[2]	..	–2.2	–4.5	–5.4	3.1	6.5	6.4
Other domestic sources, net	..	14.3	12.5	5.7	5.5	4.1	–1.5
Memorandum item:							
Payment of oil taxes	..	7.1	5.1	4.9	4.5	4.8	2.7
Total domestic credit	**849.7**	**–1.6**	**–1.4**	**–0.5**	**3.7**	**5.3**	**6.2**
of which:							
Commercial and savings banks	395.0	–2.2	–1.5	3.3	8.3	10.2	9.6
State banks	158.7	8.2	6.5	–2.4	–1.9	–0.4	–1.6
Mortgage institutions	128.7	–19.1	–20.3	–17.8	–4.6	0.5	–0.4
Life insurance companies and pension funds	65.5	13.6	8.8	–7.7	–7.3	–16.5	–5.5
Bond and certificate market	60.4	5.2	–2.7	14.3	4.3	6.7	5.8
Other	41.4	–13.0	–2.1	–0.4	19.0	20.4	–4.3

1. September.
2. Commercial, savings and State banks.

Source: Bank of Norway.

traditional State banks continued to fall as their favourable lending rate differentials relative to commercial and savings banks have virtually disappeared. Moreover, the recent decline in residential investment (see Chapter I) and the associated weakening in demand for mortgage loans has had a negative effect on credit by the State's Housing Bank and the commercial mortgage institutions. With commercial and savings bank lending accounting for almost half of the total domestic credit stock, overall credit growth doubled from 3 per cent in 1994 to around 6 per cent in the last two years.

Assessment

The experience in late 1996 and early 1997 suggests that the present monetary policy set-up, based on the commitment to exchange rate stability, may become difficult to sustain as upward pressure on the krone may increase with the rising current account surplus. Already, monetary policy has had to be eased in order to ward off speculative capital inflows, a move which, if continued, risks

providing undue impetus to domestic credit and private demand, with resultant economic overheating. Therefore, whatever room exists within the present framework of maintaining a stable exchange rate in the medium run should be used to reverse the recent relaxation of monetary policy. At the same time, the current tight stance of fiscal policy would need to be maintained, or even reinforced, to ease the strain on monetary policy by reducing the potential conflict between exchange-rate stability and inflation control.

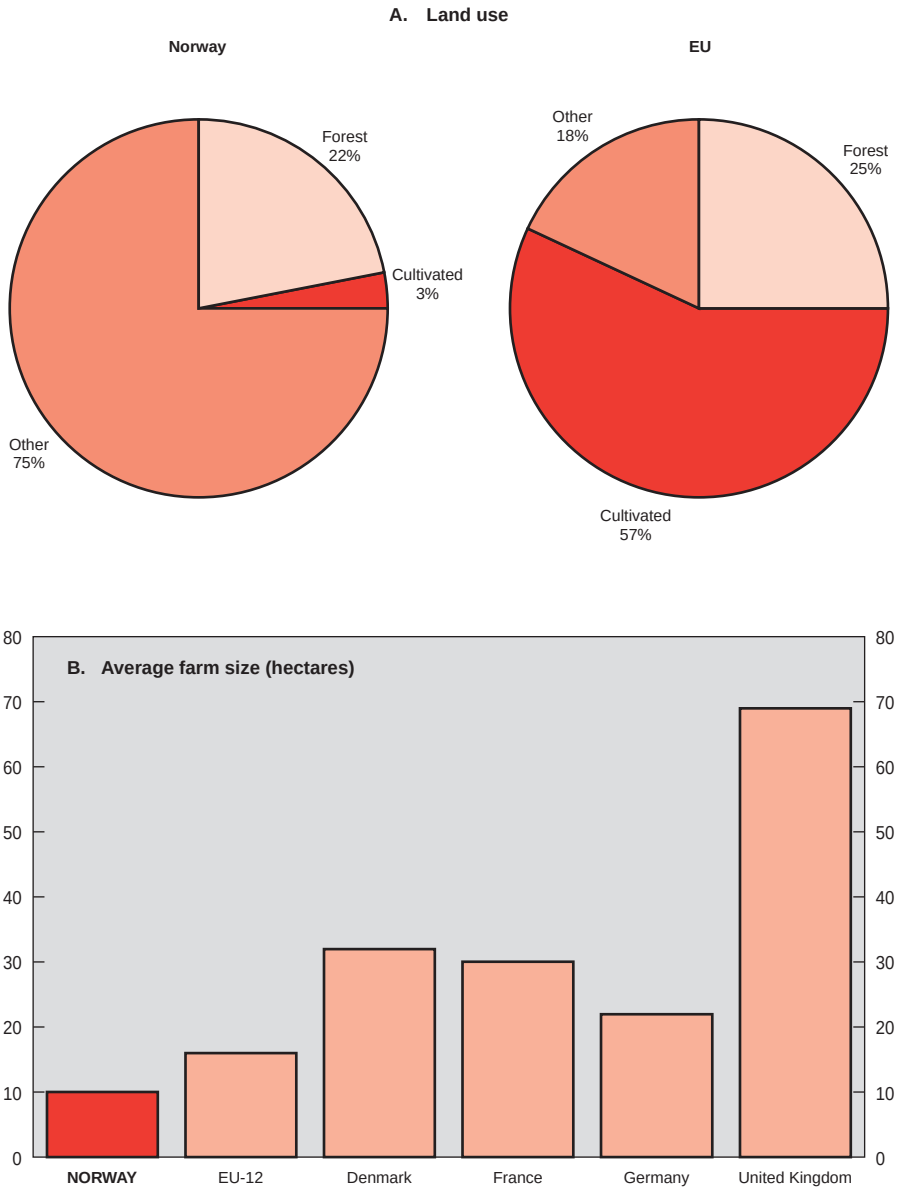
Structural reform

Since the adoption of a wide range of structural measures implied by the 1994 EEA-agreement, only a limited number of new initiatives have been taken in the last eighteen months or so. The main areas where structural policy actions have been significant are: *i) agriculture*, where a programme of measures aimed at reducing foreign trade protection has been initiated consequent to the WTO agreement; and *ii) the financial industry*, with further steps towards privatising the commercial banking sector after the government bail-outs during the crisis in the early 1990s. The sections below briefly review the reform process in these areas and identify scope for further action.

Agricultural support

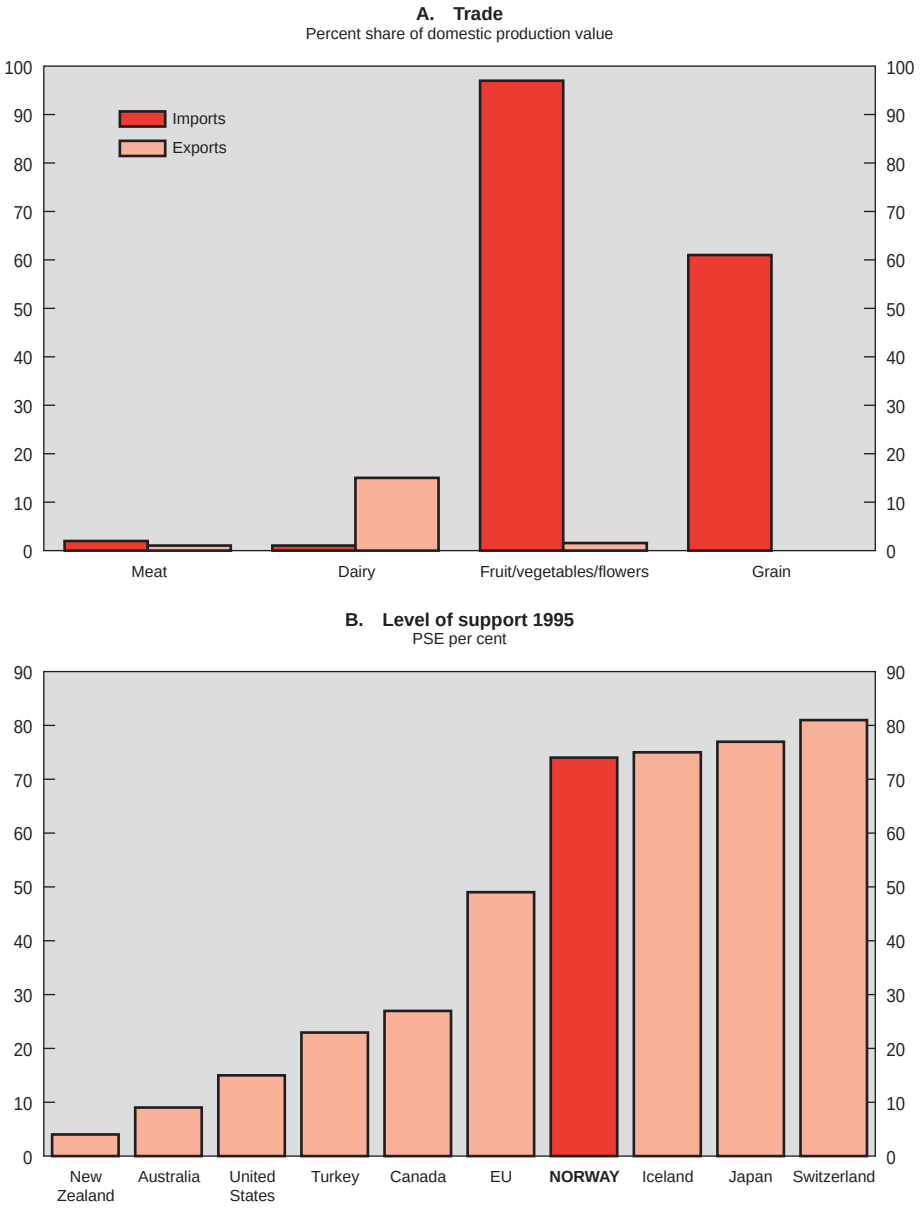
Norwegian agriculture stands out in Europe for its small scale farming, with an average farm size of ten hectares – considerably less than the average in the EU countries – occupying only 3 per cent of the total land mass (Figure 16). International trade in agricultural products in Norway is limited mainly to imports of fruit, vegetable and grain, and the export of one-fifth of the dairy production (Figure 17, Panel A). The virtual absence of agricultural exports – which amount to 1 per cent of total exports – is related to high relative prices. Moreover, the small size of agricultural imports – only about 6 per cent of total imports – is due to the significant government support, including border protection, market regulation and direct support, with the effective tariff rate in some cases exceeding 300 per cent. The producer subsidy equivalent (PSE) – including both direct and indirect forms of support – amounted to 74 per cent of the total production value in 1995, almost twice the PSE in the EU area, and exceeded only by Japan, Switzerland and Iceland.[22] More specifically, the total transfers (associated with

Figure 16. **FEATURES OF THE AGRICULTURAL SECTOR**



Source: Statistics Norway.

Figure 17. **AGRICULTURAL TRADE AND SUPPORT**



Source: Statistics Norway.

agricultural policies) per full-time farmer equivalent amounted to $43 500, almost three times the OECD average. In terms of transfers per hectare of cultivated land, support in Norway is thirteen times the average in the OECD area. All of this indicates that the agriculture sector in Norway is relatively inefficient by international standards.

Consequent to the agriculture policy reform of 1993,[23] the annual Agricultural Agreements – between the farmers union and the government – have led since to reductions in administered prices for all products (by almost 4 per cent in total) and cuts in direct budgetary support (3 per cent).[24] At the same time, the composition of budgetary support has shifted away from production-linked assistance to general income transfers. In addition, the Uruguay Round agreement has prompted the government to implement several trade measures. The state monopoly over cereal imports was abolished. Quantitative import restrictions were replaced by tariffs and remaining minimum access import quotas were allocated through auctioning of import licenses – the quota rents thereby accruing directly to the government. Moreover, the entire agreed tariff reduction of 36 per cent, which originally was to be spread over a six-year period, has been implemented at once with effect from 1 July 1995. In order to offset the adverse income effects of these reforms, the agro-food industry received investment grants and was allowed some duty free importation of raw materials. Despite the above reforms, Norway has not yet achieved a market orientation of agricultural production and the percentage PSE has remained at the same high level, indicating a continued burden on consumers and tax payers.[25] Although the progressive introduction of a tariff-based import regime has rendered border restrictions more visible, the very high tariff levels are highly punitive, virtually excluding competing imports.

Financial services

Having overcome the banking crisis which occurred in the early 1990s, the financial industry in Norway is now facing the challenge of more intense competition from elsewhere in the Scandinavian region and Europe, due in part to the opening-up of the local market for foreign bank branches consequent to the EEA agreement. Indeed, while banks have become profitable again and have had satisfactory BIS capital adequacy ratios in recent years (Table 9), the improvement in banks' operating results has been tapering off – despite further reductions in provisions against loan losses[26] – amid increasing competition within the

Table 9. **Banks' operating results**

Per cent of average total assets

	Average 1981-90	1991	1992	1993	1994	1995	1996[1]
Commercial banks							
Net interest revenue	3.0	2.5	2.8	3.1	2.9	2.5	2.2
Other operating revenues	1.3	0.9	1.1	1.5	1.0	1.1	1.2
Operating costs[2]	3.1	3.3	2.9	2.6	2.5	2.5	2.1
Operating results before losses	1.3	0.0	1.0	2.0	1.3	1.1	1.3
Loan losses	1.1	4.3	2.3	1.4	0.1	–0.3	–0.1
Operating results after loan losses	0.3	–4.3	–1.3	0.6	1.2	1.4	1.4
Memorandum item:							
Capital adequacy ratio[3]	..	7.1	8.6	12.0	12.4	11.9	11.2
Savings banks							
Net interest revenue	4.2	3.8	4.3	4.7	4.1	3.8	3.3
Other operating revenues	0.6	0.6	0.8	1.5	0.5	1.0	1.1
Operating costs[2]	3.3	3.5	3.3	3.0	2.9	3.0	2.8
Operating results before losses	1.5	0.9	1.9	3.2	1.7	1.8	1.5
Loan losses	0.9	2.1	1.8	1.2	0.4	0.2	0.0
Operating results after loan losses	0.6	–1.2	0.0	2.0	1.3	1.6	1.6
Memorandum item:							
Capital adequacy ratio[3]	..	7.8	10.6	14.4	14.9	14.2	13.2

1. First half.
2. Including write-downs and losses on sales of fixed assets.
3. BIS standards.

Source: Bank of Norway.

banking industry. Moreover, despite concerns of remaining overcapacity in the provision of financial services and low interest margins, the process of rationalisation of the banking industry appears to have slowed down, with the number of bank branches and employees practically stable since 1993 and merger activity subdued as concentration in the industry is already high (Table 10 and Figure 18).[27] On the other hand, a recently-agreed merger between the postal and banks' payment systems could yield substantial cost savings.

In addition to a review of the deposit insurance scheme, which was recognised to be a factor inducing bank managers to take excessive risk, as it allowed full bail-out of depositors in case of insolvency,[28] the government has continued the partial re-privatisation of Norway's three main commercial banks – which have been in state ownership since the bail-out in the early 1990s. In October 1995, the State sold its entire 96 per cent stake in the share capital of

Table 10. **Concentration in the banking sector**

Per cent shares, 1995 figures

	Listed capital	Assets	Pre-tax profits	*Memorandum items:* State share of capital (per cent)	*Memorandum items:* Return on assets (per cent)
Den norske Bank	30.2	27.4	29.9	52	1.67
Kredit kassen	20.9	20.6	31.3	51	2.32
Union Bank of Norway	15.1	17.4	15.8	0	1.38
Postbank	9.3	11.2	2.3	100	0.31
Eksportfinans	6.1	6.3	2.3	0	0.58
Fokus Bank	5.4	5.3	4.8	0	1.39
Regional savings banks:					
Nord Norge	3.6	3.6	4.1	0	1.74
Rogaland	3.4	3.5	3.6	0	1.62
Midt-Norge	3.0	3.0	3.7	0	1.89
Hedmark	2.9	1.6	2.1	0	1.94

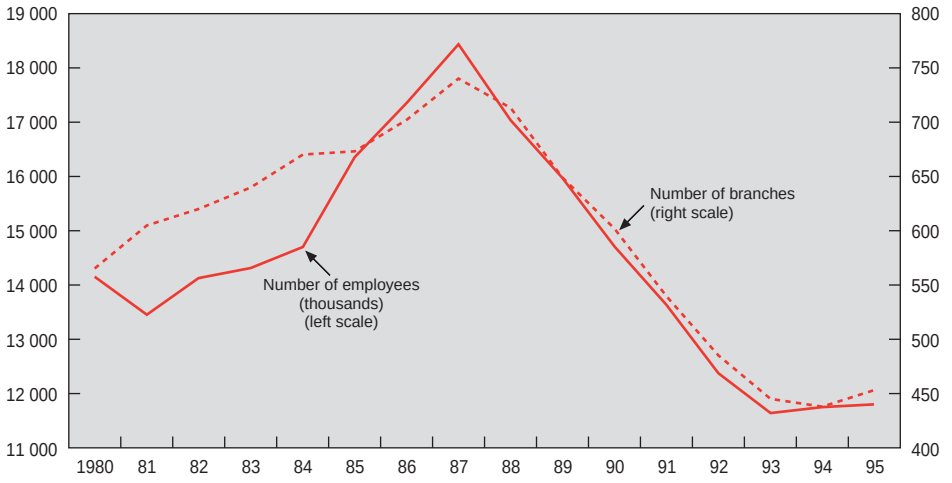
Source: *The Banker*, July 1996.

Fokus Bank, the smallest of the three commercial banks in government ownership. This initiative was followed in June 1996 by the sale of 19.8 per cent of the shares in Den norske Bank (DnB), the largest of the three, bringing the State's stake in DnB to 52 per cent. As a result, the State now holds slightly over 50 per cent in two out of the three top domestic commercial banks. The government reiterated in early 1996 an earlier decision to maintain its shareholdings of at least 50 per cent in DnB and Kreditkassen, Norway's second commercial bank, over the present term of the parliament – *e.g.* until the end of 1997.

The State thus remains strongly involved in the banking industry. As argued in the 1996 *Economic Survey* the rationale for such a policy is not obvious as, despite the stated policy that banks should function as commercial financial institutions, there is a risk of distortions in credit markets due to government interference. It would thus seem desirable for the long-run efficiency of these markets that the government exercise its role as owner to assure that the banks are managed efficiently and keep earning the required rate of return on their capital.

The role of the ''traditional'' State banks, in particular the Housing Bank and the Regional Development Fund (SND), also would need to become subject

Figure 18. **RATIONALISATION IN THE BANKING SECTOR**



Source: Statistics Norway; Bank of Norway.

to closer scrutiny. The Housing Bank, which provides mortgage loans to private households, mainly finances owner occupied accommodation in the low-priced segment of the housing market. Such financing may reach 80 per cent of the market value of the real estate, subject to a maximum amount per household, at lending rates equal to the interest rate on government bonds plus a ½ percentage point mark-up – disregarding credit rating and risk pricing of the borrowers. Similarly, SND combines banking activities with subsidy provision to small and medium-sized enterprises, and is engaged in risk lending at favourable rates. To the extent that SND provides credits to high-risk business which is typically beyond the scope of commercial banking, these practices can be very useful. However, there is also a risk of SND capturing market segments which are normally covered by commercial banks. In sum, there appears to be ample scope for better targeting of lending by the State banks, with the emphasis shifting to the servicing of clients who are in genuine need of government support, while leaving other business to the commercial banking sector.

III. Implementing the OECD Jobs Strategy

Introduction

Norway has succeeded in avoiding a shift to persistent high unemployment which has been experienced by most other European OECD countries since the late 1970s. The unemployment rate rarely exceeded 2 per cent between 1962 and 1987, and although the increasingly oil-dependent economy was hit by consecutive declines in oil prices and the US dollar during the second half of the 1980s, it rose only to about 3 per cent in 1988. It was not until asset prices and the banking system collapsed at the end of the decade and the international recession unfolded at the beginning of the 1990s, that the unemployment rate rose to 6 per cent in 1993 – a level still well below that seen elsewhere in Europe. The unemployment rate declined steadily during the pronounced recovery in economic activity in 1994 and 1995, reaching 4.3 per cent in December 1995, again in contrast with developments in Europe generally. It remained at that level throughout 1996 – despite sustained strong output growth – reflecting a flexible labour supply response to increased labour demand.

Such favourable outcomes reflect, in part, the increasing oil wealth, which enabled Norway to pursue macroeconomic stabilisation policies from the 1970s (including a counter-cyclical monetary policy) and sustain sharp growth in public sector employment, social security and labour market programmes. However, while contributing to low and relatively stable unemployment, these policies have tended to squeeze business-sector employment by distorting wage formation in this sector. In fact, in the fifteen years leading up to 1995, 176 000 jobs were created in the public sector, while 59 000 jobs were lost in the business sector. This asymmetry was especially pronounced during the 1988-1992 recession, when 57 000 jobs were created in the public sector, while 172 000 jobs were shed in the private sector. Following the recovery that emerged in late 1993, boosted

by monetary easing and fiscal stimulus, the orientation of macroeconomic policy shifted towards long-run price stability and full employment based on fiscal consolidation, a stable exchange rate and wage moderation. This approach was successful as private employment recovered and inflation continued to decline. However, as discussed in Chapter I, there is now growing concern that economic activity is expanding at a rate where capacity limits will soon be reached and further reductions in the unemployment rate may result in a re-emergence of wage inflation. Hence, in order to sustain non-inflationary output and employment growth, an emphasis should be placed on structural reform that serves to increase labour supply further and the efficiency of its use, not least by increasing work incentives, enhancing relative wage flexibility and improving product market competition.

The OECD Jobs Study recognised that most Member countries face a challenge to improve the adaptability of both their labour and product markets in order to enhance long-term output and employment prospects. As a result, it set out a programme of action to tackle it, including measures to increase the knowledge base and innovative capacity of the economies. This chapter is one in a series being published in OECD Economic Surveys dealing with the Jobs Study recommendations on a national basis. The remainder of the chapter is organised as follows. The first section presents an overview of the Norwegian labour market and employment performance (see also Chapter I). The policy requirements for Norway are discussed in the second section while a review of recent policy actions and an assessment of the scope for further action are given in the last two sections.

Labour market and employment performance

Employment growth was mostly positive between 1980 and 1987, averaging around 1½ per cent annually (Table 11). In 1988, however, the situation suddenly reversed, with employment declining on average 1¼ per cent every year until 1992. While public sector employment consistently grew over 2¼ per cent annually from 1980, private sector employment, which rose by around 1 per cent on average in the early to mid-1980s, fell by 2¾ per cent annually from 1988 to 1993. Total employment growth resumed in 1994 with private sector employment growth exceeding earlier levels, aided by improved competitiveness (due to

Table 11. **Employment developments**

Per cent change

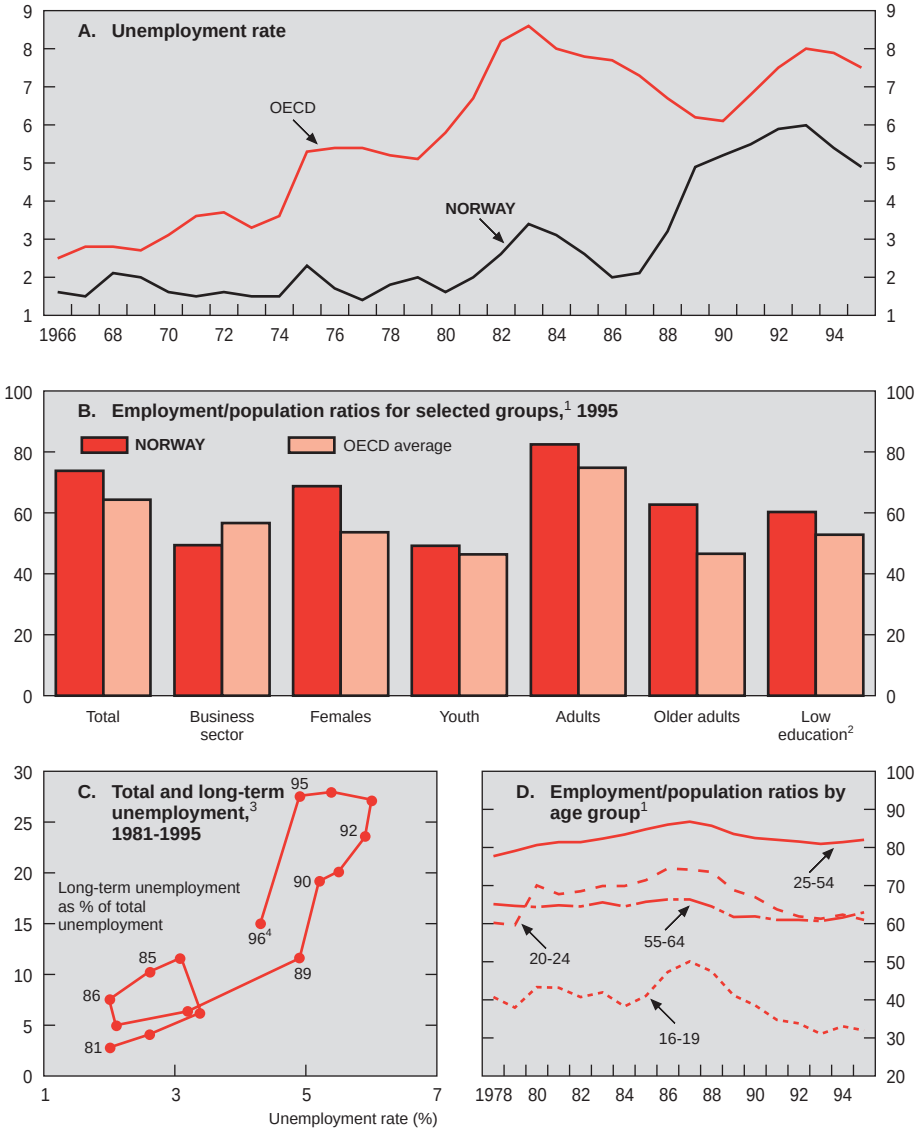
	Level 1 000 persons		Period average		1993	1994	1995
	1981	1995	1981-87	1988-92			
Agriculture, fisheries and forestry	159	105	–2.1	–4.9	–3.1	–2.1	–1.2
Oil, gas and mining	16	26	4.5	0.2	4.0	0.4	0.8
Manufacturing [1]	397	321	–1.0	–3.5	1.6	3.0	2.4
Construction	143	106	3.2	–8.3	–9.5	3.8	6.7
Private services	775	873	2.4	–1.1	–0.7	0.6	2.6
Total Private	1490	1431	1.1	–2.8	–0.8	1.2	2.7
Public	471	647	2.2	2.5	2.7	1.3	1.0
Total	1961	2079	1.4	–1.3	0.3	1.2	2.1

1. Includes gas and electricity.
Source: National Accounts.

greater progress in reducing inflation than in most trading partner countries) and strong profitability of export manufacturers of raw-materials-based goods. All of the net employment creation between 1981 and 1995 has occurred in the public sector, with a large part involving the absorption of the increase in female labour supply over the period.[29] In contrast, private sector employment fell during the same period, with increases in private services and offshore employment being overwhelmed by declines in private manufacturing and agricultural employment.[30] As a result, Norway has, after Sweden, the second highest share of public sector employment in total employment in the OECD area – 31 per cent in 1995[31] – and almost double the average of OECD Europe.[32] At the same time, the ratio of business employment to total population is considerably lower than the OECD average (Figure 19, Panel B).

The strong upswing since 1993, and the resulting gains in employment, have boosted labour force participation rates to their pre-recession peaks and to among the highest levels in the OECD area. Indeed, the labour force has demonstrated a significant responsiveness to employment conditions, with the participation rate rising steadily from about 70 per cent in the 1970s to just under 80 per cent in the 1980s, largely fuelled by increasing participation of women.[33] Consistent with earlier cyclical patterns, labour force participation during the recession declined to 76 per cent at the 1993 trough, as many women withdrew from the labour market, while many young people stayed longer in school (see Chapter IV). As

Figure 19. **LABOUR MARKET INDICATORS**



1. Defined as the percentage of each working-age population group.
2. Completed less than upper secondary education, 1994.
3. Long-term unemployment is defined as individuals looking for work for one year or more.
4. Estimate.

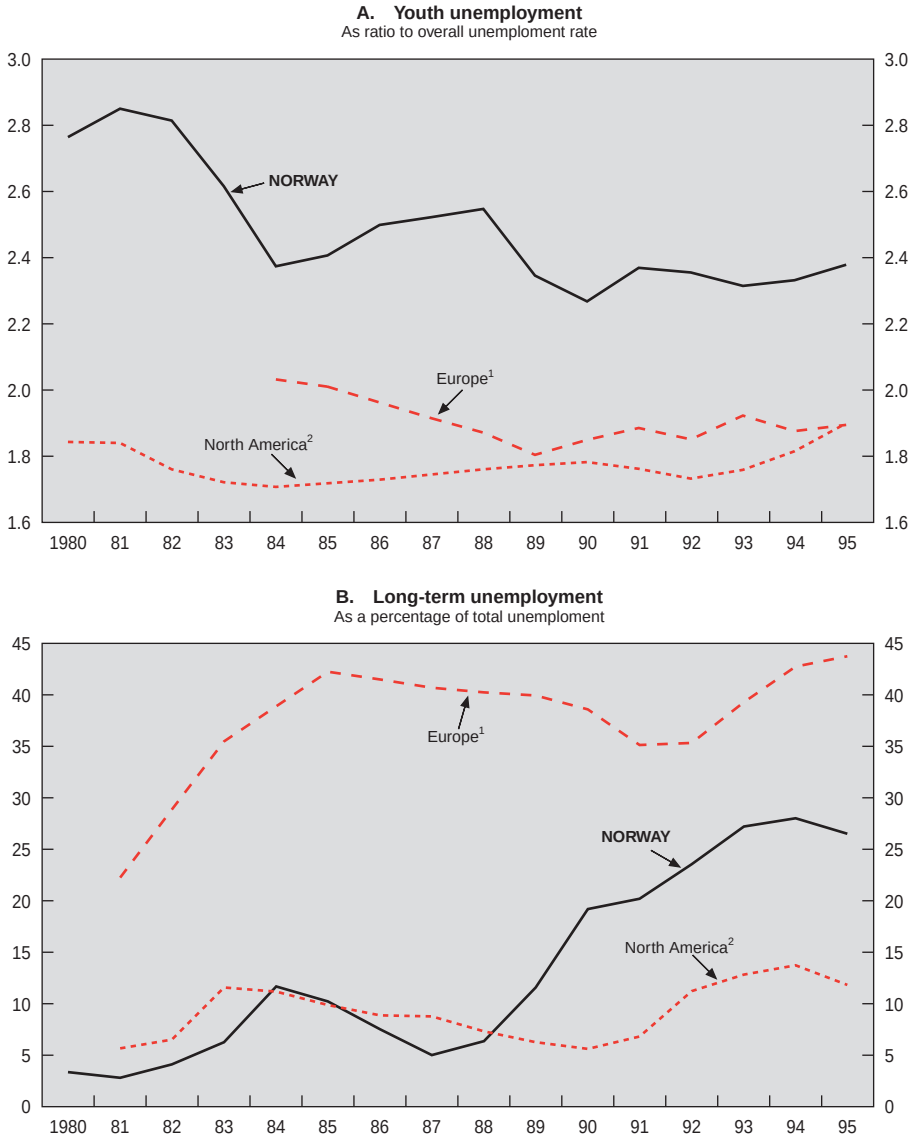
Source: OECD Secretariat.

soon as the recovery built up steam, however, the labour force participation rate began to rise, approaching 79 per cent in 1996 – only 1 percentage point below the pre-recession peak. Additionally, the increasing share of women in the labour force,[34] who work fewer hours on average than men, due to their greater preference for part-time work, has translated into a decline in average hours worked. This trend has been accentuated by a secular decline in average hours worked by both men and women: women worked 28 per cent fewer hours in 1994 than in 1962, and men 22 per cent. For the economy as a whole, average hours worked per person fell by 26 per cent over the same period, so that the amount of time Norwegians currently spend at work per year is lower than in most other OECD countries.

The responsiveness of the labour supply served to make unemployment fall or rise much less than employment conditions would otherwise have implied.[35] Prior to the mid-1980s, the unemployment rate trended higher, although remaining below 2 per cent (Figure 19, Panel A). From 1988, it began to rise, reaching 6 per cent by 1993, tempered by the decline in the labour force. Although the unemployment rate had declined to 4.3 per cent by late 1995, it has hovered around that level for most of 1996. During the recession, around 5 per cent of the working-age population (WAP) were not gainfully employed, while at the same time they were not registered as unemployed, due to their participation in active labour market programmes (ALMPs – around 1½ per cent of WAP)[36] or other social insurance programmes (*e.g.* disability pensions and rehabilitation benefits for ''non-medical'' reasons, or early retirement pensions – almost 3½ per cent of WAP). The number of inactive persons receiving income replacement – including also persons registered as unemployed – was estimated to exceed 11 per cent of the WAP in 1993, and to have declined to only around 10 per cent in 1996.

The incidence of long-term unemployment (more than twelve months) rose sharply during and in the aftermath of the recession, from the 5 to 10 per cent range prevailing during most of the 1980s, to a peak of 25 per cent in 1994, before declining modestly to 23 per cent in 1995 (Figure 20, Panel B). The upward trend in long-term unemployment appears to reflect a ''selection process'' – normal in an episode of rising unemployment – whereby the stock of less employable persons gradually expands[37] (Figure 19, Panel C). Among these are many young workers and workers with low educational attainments. In 1996, the incidence of long-term unemployment has declined further, reflecting

Figure 20. **KEY FEATURES OF UNEMPLOYMENT**



1. Finland, France, Germany, Ireland, Italy, Netherlands, Norway, Spain, Sweden, United Kingdom.
2. Canada, United States.

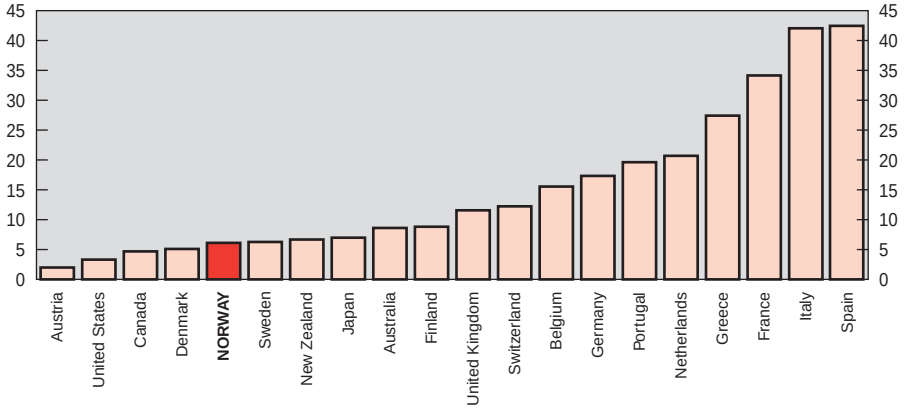
Source: OECD Secretariat.

tightening conditions in the labour market. The number of long-term unemployed, however, has tended to be understated on account of active labour market policies, which to some extent artificially interrupt unemployment spells. The fact that the (measured implicit) average duration of unemployment in Norway is low by international standards needs to be seen in this context (Figure 21).[38] The presence of many immigrants in the Oslo region (15 per cent of the labour force), has contributed to raise the incidence of long-term unemployment there above the national average.[39]

As in other OECD countries, unemployment is particularly pronounced for young workers (Figure 20, Panel A), despite a guarantee of a position in a training programme for youths under the age of 20 without a job or a place in the education system. The rate of unemployment among youngsters (16 to 19 years of age) was 17.1 per cent in 1994 and that of young adults (20 to 24) 10.9 per cent, significantly higher than the 4.5 per cent unemployment rate of prime-age workers (25 to 54). Moreover, youth unemployment appears to be particularly concentrated among those with low educational attainment, reflecting the loss of

Figure 21. **DURATION OF UNEMPLOYMENT, 1995**[1]

Months



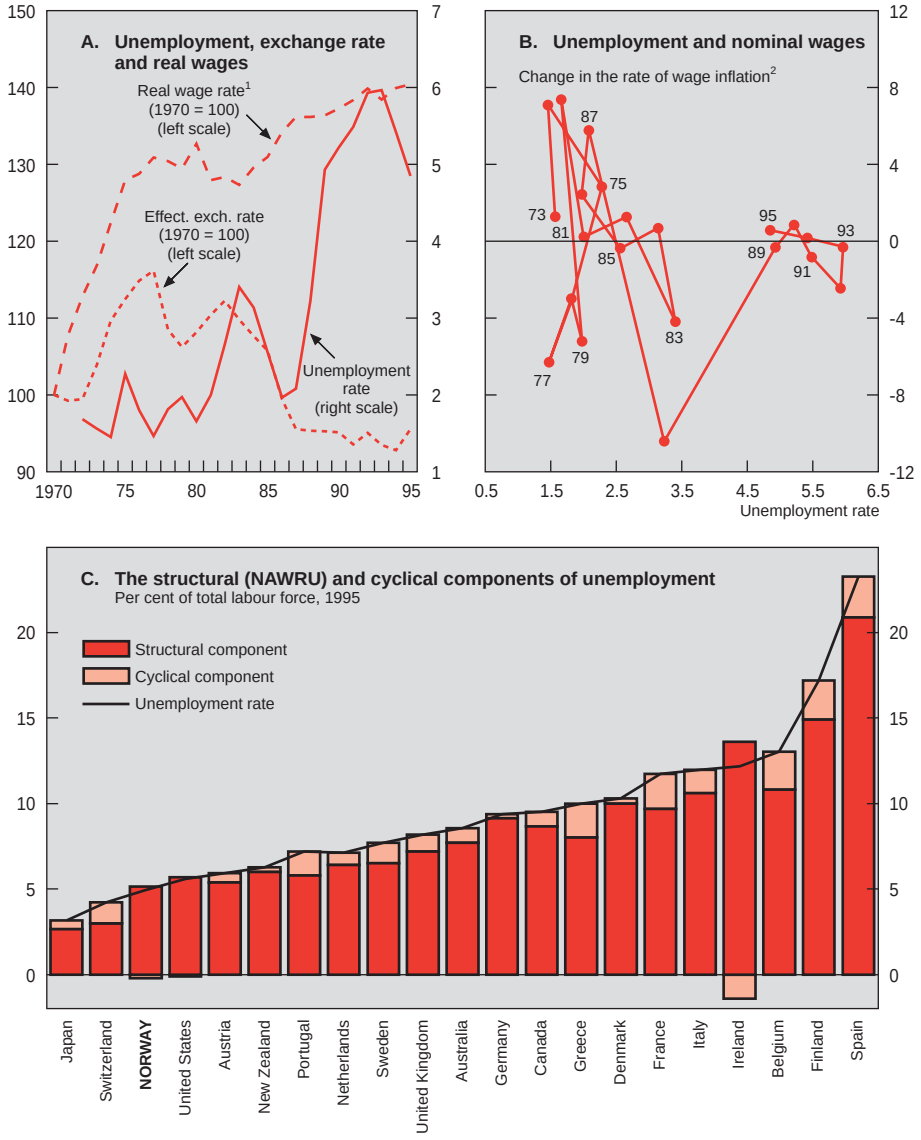
1. Calculated as the unemployment rate for the working-age population divided by the flow into unemployment.
Source: OECD Secretariat.

low-skill jobs in the recession and the increasing full-time education enrolment of young persons. The re-entry into the labour market of graduates of higher education as the economy rebounded has served to brake the decline in the unemployment rate of high-skilled workers in recent years.

In past decades, the Norwegian economy has exhibited a certain degree of real wage flexibility. In the mid-1970s and early 1980s, when unemployment tended to increase, real wages declined slightly, as a depreciation of the exchange rate – and the associated rise in the price inflation – was only partly offset by higher nominal wages (Figure 22, Panel A). In the 1988-92 recession, real wages tended to rise modestly, with the exchange rate remaining broadly stable and price inflation declining. Apparently, therefore, a downward correction of real wages during the recession has tended to be difficult to achieve while disinflation was in progress. Conversely, real wages have remained broadly stable since the recovery started in 1993. This is primarily due to the concerted effort by the labour union movement, the employers organisations and the government to establish a ''Solidarity Alternative'', which has as its aim wage restraint in order to reach full employment and price stability.[40] The achievement of this goal has been predicated on the government keeping the exchange rate stable, in order to secure real wages, as well as to prevent an appreciation of the exchange rate which would reduce the competitiveness of the mainland industry. The Solidarity Alternative is also seen as a means to maintain credibility in financial markets, thus ensuring low long-term interest rates. In this framework, workers can still achieve real wage gains based on productivity increases. In the event, real wage growth accelerated by almost 2 percentage points, to 3 per cent, in 1996, reflecting tighter labour market conditions, especially in construction and manufacturing.[41] In fact, as nominal wage increases also accelerated (Figure 22, Panel B), the unemployment rate in 1996 fell below the Secretariat's NAWRU (non-accelerating wage inflation rate of unemployment) estimate of 4.8 per cent (Figure 22, Panel C).

The question arises how Norway has succeeded in maintaining such internationally low rates of equilibrium unemployment. In other countries which have maintained a low structural unemployment rate – such as Japan and the United States – flexible labour market practices with respect to wage setting and hours worked (essentially a market price adjustment) were successfully used to this end.[42] In Norway, on the other hand, changes in the real wage have played a less

Figure 22. **THE EQUILIBRIUM RATE OF UNEMPLOYMENT**



1. Total compensation per employee, deflated by the private consumption deflator.
2. Average hourly earnings in manufacturing.

Source: OECD, *Main Economic Indicators,* Secretariat estimates.

pronounced role in keeping structural unemployment low.[43] Rather, the petroleum wealth has to some extent eased budget and current-account constraints facing the economy, enabling the authorities to increase public employment and reduce labour supply by increasing the number of persons on various public income support measures or in subsidised education programmes. In essence, the authorities adjusted labour demand and supply to avoid high unemployment ever becoming established and entrenched. In the absence of these measures, it is likely that unemployment would have shifted to a permanently higher level, as has been the case in most other European countries.[44]

The ability of macroeconomic policies to provide further non-inflationary employment gains is now close to being exhausted.[45] As discussed in Chapter II, the role of fiscal and monetary stimulus in reducing unemployment – although it proved useful during the recession – has since 1994 become less of a viable option. Moreover, with labour market conditions rapidly tightening, the scope for incomes policy has become more challenging. In the early 1990s, wage restraint was achieved through active co-operation of the government and the social partners, whereby, in some years, the government offered budgetary concessions to the central labour market organisations, in exchange for their commitment to secure wage restraint at the individual union and employer level. In the absence of such fiscal concessions in recent wage rounds, wage restraint has been eroding, with wage growth picking up markedly in 1996. Further wage increases in the 1997 wage round are likely to undermine the competitiveness of mainland export production and, with time, also employment levels. Structural reforms in both labour and product markets are, therefore, needed to sustain low unemployment in the future.

Policy requirements

The main challenge facing the Norwegian labour market at present is the creation of additional jobs without accelerating wage growth. This would require continued wage restraint and a reconsideration of policies that reduce the labour supply available for the private sector, *e.g.* the extensive use of labour market training measures, the relatively generous income replacement system, and the trend growth of public employment. From a longer-term point of view, the uncertainties attached to future proceeds from the petroleum sector would require

Box 3. **The OECD Jobs Strategy: synopsis of recommendations for Norway**

The *OECD Jobs Study* set out a strategy based on nine recommendations for improving the ability of OECD economies to cope with structural change, through enhancing the ability to adjust and adapt, and increasing the capacity to innovate and be creative. The nine distinct policy areas covered included the macroeconomic policy framework and the creation and diffusion of technological know-how. With respect to labour and product market flexibility it identified the need for initiatives in the following areas: working-time flexibility, the entrepreneurial climate, wage and labour cost flexibility, employment security provisions, active labour market policies, labour force skills and competencies, and unemployment and related benefit systems.

While the Norwegian labour market performance is among the best in the OECD area, both in terms of employment and unemployment, there is scope for further improvement. In this context, Norway is in the process of implementing reforms in some of the above-mentioned areas. Following from the detailed review of the Norwegian labour market and employment performance and the assessment of policy requirements in this chapter, the following priority areas have been identified for policy action:

Increase employment flexibility

Consider more liberal employment protection legislation, which would relax restrictions on employment, in particular the use of fixed-term contracts. This would improve the matching of skills to potential jobs and contribute to reduce youth unemployment and prepare the ground for full-time employment opportunities.

Reform unemployment and related benefit systems

Continue reforms of unemployment and related benefit entitlements to enhance work incentives by *i)* closely monitoring the impact of the 1997 reform of benefit duration rules and if necessary reducing the effective duration for which an individual can receive unemployment benefits, *ii)* limiting disincentives arising from the benefit and tax system for low-income workers, perhaps by reducing benefit withdrawal rates and/or reducing unemployment benefit replacement ratios as duration increases, *iii)* tightening eligibility rules and implement measures to ensure the stricter enforcement of existing rules, *iv)* decreasing unemployment benefit entitlements for part-time unemployed persons, *v)* introducing a waiting period for unemployment benefits, and *vi)* further limiting the inflow of workers into the disability pension scheme for reasons which are not strictly medical (at present, an estimated one-third of disability pensioners are receiving these benefits for non-medical reasons).

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Review active labour market programmes

Improve targeting for active labour market programmes, in order to make these programmes more effective and reduce their size given the tight labour market conditions. Conduct evaluations of active labour market programmes in order to identify those which raise human capital in a cost effective way. Such ALMPs should be continued and targeted more closely on low-skilled workers, especially among the young. Consider to liberalise job placement services, currently monopolised by the Public Employment Service.

Increase wage and labour cost flexibility

Explore ways to introduce more flexibility of relative earnings into the wage formation system so that it better reflects local labour market conditions and to improve the correspondence between wages and productivity. In particular, wages for youth and low-skilled workers should be aligned more closely with their experience and skill levels.

Enhance product market competition

Remove regulations that restrict competition in sectors dominated by state enterprises or with insufficient competition among private firms. Improved competition will strengthen the ability of the economy to adapt to future shocks and ensure the prospects for sustainable growth, and bring benefits to consumers in the form of reduced prices and to taxpayers through lower taxes.

Reduce direct support for agriculture, among the highest in the OECD area.

Enhance technology diffusion

Improve research links between the public and private sectors and further evaluate and rationalise the various technology diffusion programmes. Small and medium-sized firms could benefit from government initiatives to set up co-operation networks, especially in the field of research and development.

Promote entrepreneurship

Encourage the formation of SME clusters and networks in order for small firms to compete more effectively with large and established firms in the global market place. The government could also realise economies of scale in dealing with the SMEs as groups rather than individually.

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Improve labour force skills and competencies

Continue the efforts to develop the vocational stream of upper-secondary education and improve the targeting of higher education on labour market needs. More generally, human capital investment consistent with emerging employment opportunities would need to be encouraged.

more market-based flexibility of product and labour markets to facilitate adjustment to unforeseen swings in offshore revenues. In particular, the capacity of the labour market to provide a greater adaptability in relative wages could be improved, especially for young workers and those with low skill levels. Greater flexibility in the wage-setting system, accompanied by changes in the tax and benefit system, could reduce the dis-incentives for many unemployed persons to seek employment. At the same time, the education system should be geared toward upgrading and broadening the skill base of the labour force to maintain an equitable income distribution and to reduce mismatches in the labour market. Finally, as elsewhere in Europe, eliminating over-regulation of product markets, removing obstacles to the development of small and medium-sized enterprises (SMEs), fostering entrepreneurial skills and competencies should speed up employment creation in the private sector. Such an integrated approach should help reduce unemployment further while at the same time containing wage pressures on a durable basis. The paragraphs below provide some background analysis to the specific Jobs Study recommendations for Norway. A synopsis of these recommendations is presented in Box 3.

Enhancing the ability to adjust and adapt

The wage formation system

The institutional framework for wage formation may have an important bearing on the success of a country in adapting real and relative wages to changes in labour demand and supply conditions. The dominant characteristic of the Norwegian wage bargaining system is the high degree of centralisation. Another prominent feature is the relatively active role of the government in influencing labour market outcomes. For decades, incomes policy has played a major role in

overall policy making. To facilitate its implementation, a number of institutions have been created, which has tended to strengthen the centralisation of wage determination.

Workers are organised into many separate unions, most of which are affiliated with one of the three major union confederations, the largest of these being the Norwegian Federation of Trade Unions (LO). The public sector has the highest proportion of employees organised in unions, followed by the financial sector, manufacturing and transport. Other private service sectors, such as wholesale and retail trade, have comparatively low union membership. Although membership levels have increased slightly since the 1960s, the proportion of unionised workers belonging to LO has been on a declining trend – accounting for 57 per cent of total unionised workers in 1996 – reflecting both competition between rival organisations and the sharp reduction in the employment share of manufacturing. As LO still covers the majority of workers in the exposed sectors, it plays a crucial role in wage formation.[46] On the employers' side, there are two major organisations, one for the private sector – the Confederation of Norwegian Business and Industry (NHO) – the other for the public sector. Although the number of workers covered by agreements between LO and NHO has fallen since the 1960s, such settlements still play a key role as pace setters in bargaining rounds and are also extended to non-union workers in all establishments affiliated with the NHO. Moreover, LO/NHO agreements are frequently taken over by non-member firms where the workers may or may not be unionised. The effective coverage of central agreements is now close to 75 per cent of all workers.[47] The traditional pace-setting role of LO and NHO agreements has for a long time made the exposed sector the wage leader in the economy. Central agreements typically include all non-wage aspects and the overall wage increase in the first year, while in the second year a further adjustment in wage rates is negotiated.[48]

Norwegian governments have traditionally tried to influence bargaining outcomes at the central level. The most drastic form of government intervention has been to rule by decree, either freezing wages or limiting wage increases, as was the case in the late 1970s and 1980s when the international competitiveness of Norwegian producers was being eroded by relatively high wage inflation. In normal circumstances, the government is involved in regular tri-partite consultations. The Contact Committee (CC) – headed by the Prime Minister and consisting of senior cabinet members, representatives of farmers and fishermen, and the

leaders of the main employers and employees federations – has played an important role in incomes policy. The purpose of the CC is to exchange information about income objectives and the economic outlook prior to the wage negotiation round. The economic situation and prospects for the pending bargaining period are evaluated in detail by the Technical Reporting Committee (TRC) on the Income Settlement. The TRC consists of representatives from both the LO and NHO, as well as delegates from ministries and the official research institute. Both committees have served to narrow down disagreements stemming from different perceptions about the future. As such, Norwegian incomes policy has aimed at facilitating discussion among the social partners, by including a broad range of measures, such as increased public spending and tax benefits, in exchange for moderate wage settlements, in particular during the second half of the 1970s, late 1980s and early 1990s. In recent years, by contrast, the government has made no ''tax concessions'' of this kind. In fact, the Solidarity Alternative programme of wage moderation, introduced in 1992/93, incorporated plans for public expenditure restraint.

A considerable part of wage increases – the so-called wage drift (the difference between actual wage growth and increases negotiated at the central level) – tends to be determined outside the centralised wage bargaining process, and varies significantly from industry to industry. Since the 1960s, wage drift in manufacturing has on average accounted for more than half the total annual wage increases, but in the 1990s this share has declined to around one-third. The centrally negotiated contracts are used to set a basic wage growth, but firms are expected to grant increases beyond this contractual minimum. With local bargaining being introduced in more and more sectors since the 1960s, all central wage agreements have tended to become minimum contracts. Industry-level bargaining is not completely uncoordinated, as LO and NHO are usually acting behind the scenes, which has had a significant effect on the ability of the social partners to achieve wage moderation. Since the 1960s, the contract period of central wage settlements has, without exception, been two years. Agreements have always had some provision for inflation adjustment over the first year, although there are no formal indexation clauses.

Consistent with the strong tradition in Norway to emphasise equity considerations, such a wage formation system appears to provide little scope for relative wage adjustment and to have a bias towards wage compression.[49] Central wage

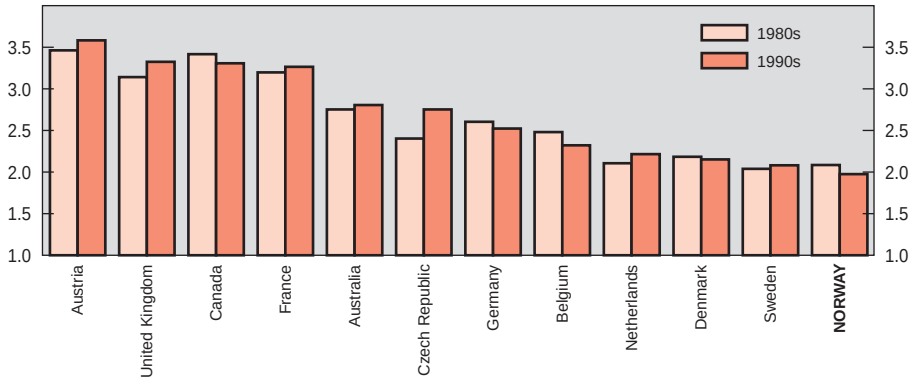
contracts typically embody solidaristic goals of the union movement. In practice, since the 1960s, no major central agreement between LO and NHO has been concluded without special provisions for low-wage workers. Moreover, the fact that the initial central ‘‘tariff‘‘ agreement sets relatively generous binding increases for the lower pay scales in absolute nominal terms – pre-empting the outcomes of higher income earners at the branch level – has tended to narrow the wage structure. Indeed, the rigidity of relative wages is indicated by the fact that wages have been found to be weakly influenced by local labour market conditions, while strongly affected by aggregate unemployment developments.[50] Such inflexibility is further compounded by the traditional practice of paying labour market entrants a wage equivalent to that paid to workers with some job tenure.[51] Moreover, in the rare case where minimum wage provisions are included in wage agreements (mostly for non-unionised labour market entrants for a period not exceeding two years), the effective wage ‘‘floor’’ normally exceeds the minimum wage, due to the payment of wage supplements based on skill, experience and productivity.[52] As a result, earnings dispersion – as measured by the ratio of earnings of full-time employees in the highest earnings decile to those in the lowest – is among the smallest in the OECD area and has tended to narrow further since 1980 (Figure 23), with most of the change explained by a rise in earnings of the lowest decile.[53]

Such institutionally driven wage compression may weaken the allocative role of the wage system. On the supply side of the labour market, students may lack incentive signals to select studies leading to better career opportunities. On the demand side, firms may have insufficient ability to adapt wages to attract high-skilled workers in emerging professions. With wages of high-skilled workers constrained – not reflecting the true cost of their human capital endowment – firms are receiving an indirect wage subsidy, which could be acting to crowd out lower-skilled workers. Low-productivity jobs may, at the same time, be lost due to the relatively high level of wages at the lower end of the pay scale.

It is well known that ‘‘static’’ distribution measures conceal differences in ‘‘earnings mobility’’ (the ability of employees over time to move up the ‘‘earnings ladder’’). Earnings mobility in Norway, is, in fact, higher than in many European countries, with about 78 per cent of the workers, the majority of whom were women, moving up or down deciles of the income distribution from 1980 to 1990.[54] For the 80 per cent of employed men and 50 per cent of employed women

Figure 23. **DEVELOPMENT OF EARNINGS DISTRIBUTION**

Ratios[1]



1. Ratios of highest (9th) to lowest (1st) decile.
Source: OECD, *Employment Outlook*, 1996.

working in the business sector, income was found to rise as workers gained more experience with age or where their skill level rose through additional education. Moreover, the upward earnings mobility of men was largely associated with moving from public to private sector employment, while women tended to prefer to stay in the public sector, due to the greater flexibility of working-time arrangements. Moreover, after the government, in the early 1990s, introduced a pay policy aimed at reducing gender earnings differences, pay dispersion has decreased in the public sector. Elderly persons and those of low educational attainment, however, tend to remain stuck at lower levels of income.[55] This feature has been exacerbated by the concentration of unemployment among such workers during the recession.[56] Such lack of earnings mobility and high unemployment probabilities were most likely mutually reinforcing, due to ‘‘stigmatisation’’ and poverty traps facing such workers.

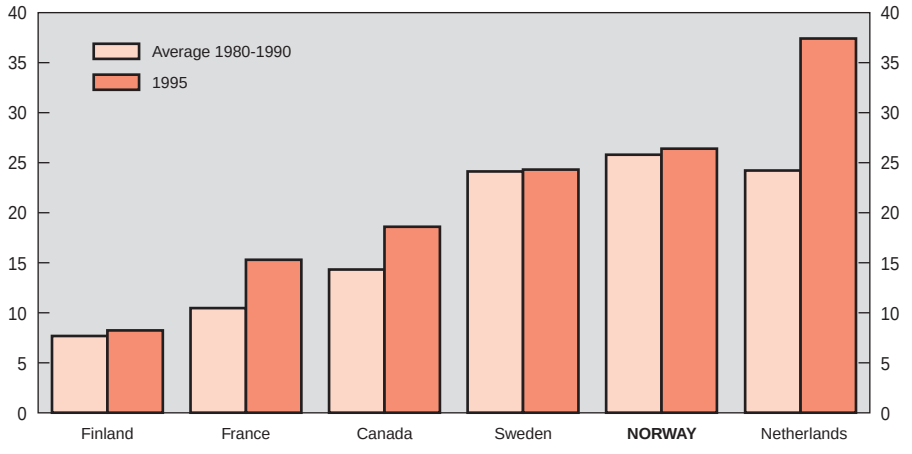
Flexibility of employment and working time

About nine-tenths of all workers are in regular employment, including full-time, part-time, or shift work, with around a quarter of the labour force in part-

time employment[57] (Figure 24). The relatively high share of part-time employment in Norway reflects *i)* the high and rising share, currently at 45 per cent, of females in total employment, largely in the public sector, and *ii)* the granting of a part-time unemployment benefit. In 1996, 44 per cent of all unemployment benefit recipients in Norway received a part-time benefit.[58] With only 16 per cent of part-time workers considering themselves as being under-employed, the majority are in such employment by choice. Norway has a relatively low share – 13.6 per cent in 1996 – of its labour force classified as being engaged in atypical jobs, with almost 8 per cent in temporary jobs – mostly in the public sector – and less than 4 per cent employed through fixed-term contracts in the private sector.[59]

As stipulated in the 1994/95 Act on Workers Protection, there appears to be no discrimination involving different kinds of work classified as regular employment (see above).[60] However, many firms have reported significant restrictions on the use of atypical jobs related to new legislation effectively restricting the use by firms of fixed-term contracts, by limiting such work to cases where the ''nature''

Figure 24. **PART-TIME WORK**
Percentage in total employment



Source: OECD Secretariat.

of the work requires it, as well as that it be different from the work ordinarily performed in the enterprise.[61] Generally, restrictions on the use of labour on a temporary basis reduce firms' flexibility when faced with fluctuations in demand and raise the costs of hiring labour of uncertain quality – such as new entrants, who, by default, face a greater obstacle in gaining valuable on-the-job experience. However, there has been some scope for substitution, as firms have reduced the use of fixed-term contracts while increasingly hiring people through temporary work agencies, with the latter type of employment increasing rapidly in recent years.[62] Consequent on Norway's entry into the European Economic Area (EEA) on 1 January 1994, some changes in labour laws are to be implemented in early 1997, including lengthening the time spent for ''rest'' from 10 to 11 hours in a 24-hour period.[63]

The *OECD Jobs Study* underlines that Employment Protection Legislation (EPL) tends to have opposite effects on the labour market. On the one hand, EPL imposes transactions costs on firms and reduces their ability to quickly adjust employment, thus contributing to lower overall employment levels. On the other hand, such legislation may strengthen links between workers and firms, thereby raising productivity growth through human capital investment and increased sense of job security, leading to higher employment levels. Ideally, EPL should aim to balance these two opposing forces. As Table 12 shows, the restrictiveness of EPL in Norway was in 1991 assessed as no more on average than in most other European countries.[64] Only in the case of provisions related to ''unfair dismissal'' was the legislation found to be among the most restrictive in the OECD area. Although it is not the intent of Norwegian EPL to prevent firms from making adjustments which are economically necessary, it may, by sufficiently raising the cost of dismissal, discourage new hiring. Though relatively few dismissals are challenged, the actual number almost certainly understates the economic influence of the legislation.[65]

Taxes and transfers

The general social insurance scheme in Norway – consisting primarily of a National Insurance Scheme (NIS) which has granted benefits since its inception in 1966, and the Family Allowance Scheme, established in 1946 – was designed to provide a basic income to individuals and families in need through, for

Table 12. **Employment protection legislation in Europe**

Summary rankings by main area[1]

	Regular procedural inconveniences	Notice and severance pay for no-fault dismissals	Unfair dismissal	Overall ranking for strictness of protection against dismissals
Portugal	12	15	16	16
Spain	15	14	13	15
Italy	3	16	15	14
Austria	10	10	11	13
Greece	8	12	10	12
Sweden	11	8	9	11
Finland	14	9	4	9-10
Germany	13	2	12	9-10
Norway	**6-7**	**6**	**14**	**8**
Netherlands	16	1	8	7
France	9	7	6-7	6
Belgium	4-5	13	3	5
Denmark	1	11	5	4
Ireland	6-7	3	6-7	3
United Kingdom	4-5	5	1	2
Switzerland	2	4	2	1

1. All rankings increase with the strictness of employment protection.
Source: OECD *Jobs Study.*

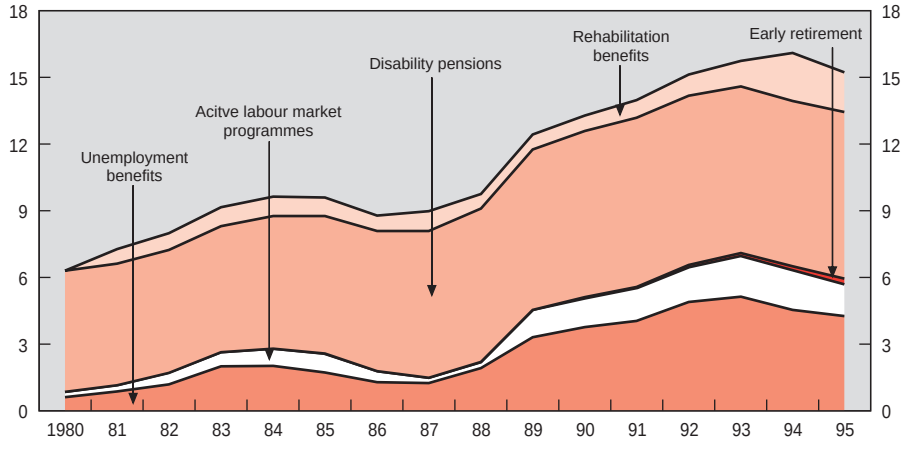
example, disability, rehabilitation, sickness, maternity and unemployment benefits. The NIS was established at a time when there was limited concern with adverse incentive effects associated with the structure of the benefit system. In recent years, a review of the NIS has been underway, with the aim of developing a ''Working Approach'' – emphasising better screening, a redesign of programmes to minimise disincentives and increased use of targeted rehabilitation and training programmes to enhance human capital – and to ''increase public saving''.[66] In 1994, with the NIS providing income support for 28 per cent of the population (approximately 900 000 persons in permanent income or pension schemes and 300 000 on temporary income maintenance),[67] it has become more apparent that the generosity and practically unlimited duration of functionally overlapping benefit schemes act to constrain the potential labour supply. Not only do those involved cease to exercise an influence on wage bargaining and thus weaken the adjustment of wages following cyclical and structural shocks, but when the interaction between the tax and benefit systems are taken into account,

severe disincentives may arise – such that many unemployed persons, notably the young, elderly, low-skill or long-term unemployed – get caught in ‘‘unemployment traps’’ (an incentive not to take paid work), while employed persons at the low end of the income distribution find themselves in ‘‘poverty traps’’ (an incentive not to move up the job ladder). Figure 25 shows that the number of people of working age receiving social benefits from the social insurance schemes rose considerably in the recession, although this was partly reversed in the ensuing years. Nevertheless, government spending (including social transfers) in Norway remains among the highest in the OECD area (Figure 26), and as such contributes to a high tax burden.

In the early 1990s, eligibility rules *for disability pensions* were tightened and enforced more strictly (including requiring second opinions from neutral physicians and a more exact interpretation of the medical requirements) while the commitment to occupational rehabilitation was strengthened (see section below on Active Labour Market Policies). Nevertheless, the overall number of disability pensioners rose from 232 000 in 1990 to 236 000 persons in 1995 (about 7 per

Figure 25. **RECIPIENTS OF LABOUR MARKET INCOME SUPPORT**

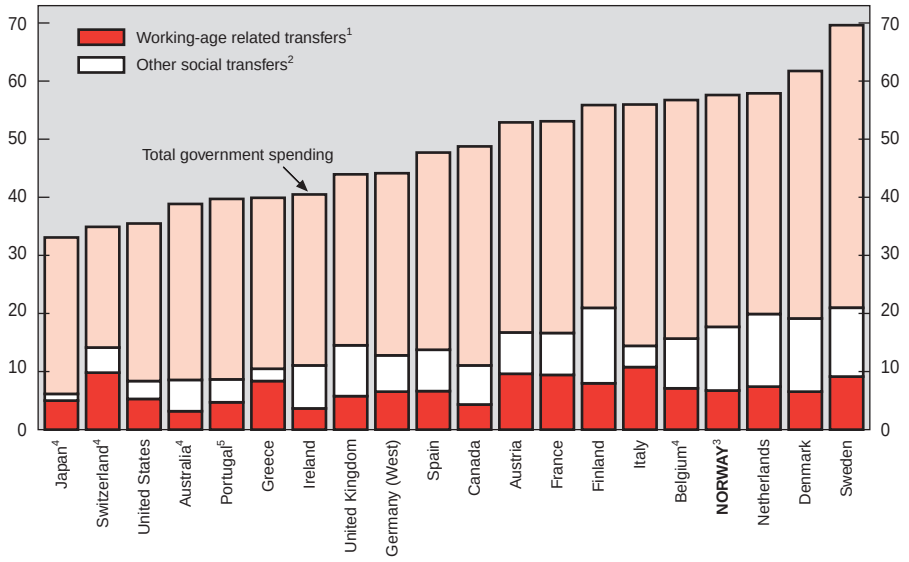
Percentage of working-age population



Source: Ministry of Local Government and Labour.

Figure 26. **GOVERNMENT SPENDING AND SOCIAL TRANSFERS**

Per cent of trend GDP (1993 or nearest year)



1. Includes disability pensions, sickness compensation, survivors benefits, family cash benefits, unemployment benefits and other contingencies.
2. Includes basically old age transfers.
3. Per cent of mainland GDP, 1991.
4. 1992.
5. 1990.

Source: OECD, Social Expenditure Database.

cent of the working age population), with around one-third of these estimated to be there for non-medical reasons.[68] This increase – which kept Norway close to the top of the OECD countries in terms of the high level of disability pensions – may partly be explained by the fact that more than 40 per cent of people over 55 use disability benefits as a bridge to retirement – well before they become entitled to a retirement pension at the age of 67.[69] At the same time, an increased use of sickness and rehabilitation benefits also occurred in the early 1990s, related also to the fact that Norway exceeds other OECD countries in terms of a

generosity index of sickness insurance.[70] When ''non-medical'' disability pension and rehabilitation benefit recipients, early retirees and those ALMP participants not counted as being in the labour force are combined with persons receiving unemployment benefits – as much as 13 per cent of the ''potential'' labour force were actually economically inactive and on public income support in 1995.

Along with the overall unemployment rate, the average number of beneficiaries of unemployment insurance rose from less than 2 per cent of the working-age population in 1988 to over 5 per cent in 1993, and declined to just over 4 per cent in 1995. As a result, average annual outlays on unemployment benefits, which rose from around ½ per cent of GDP (NKr 3.1 billion) in 1988 to above 1½ per cent (NKr 11.7) in 1993, declined to around 1 per cent of GDP (NKr 9.9 billion) in 1995. At the same time, the average duration of unemployment rose from just over three months in 1988 to six months in 1993, but had declined to around five months in 1995. Although, as noted above, the average duration of unemployment benefit spells tends to be artificially interrupted with labour market programmes, the design of the benefit system clearly encourages a long duration of unemployment spells. In Norway, as in other Nordic countries, the maximum allowable benefit duration is practically indefinite. Moreover, despite some tightening, job-search requirements remain relatively relaxed and the benefit level – which is calculated relative to the previous annual income – is generous (as measured by the replacement rate). However, a new legislation, limiting the combined duration of unemployment benefit and active labour market programmes, entered into force at the beginning of 1997 (see below).[71] The fact that unemployed job seekers have been permitted to refuse up to three suitable job offers in every twelve-month period, with only temporary suspension of benefits,[72] has contributed to reduce the rate of job acceptances. This is corroborated by the low exit rates from unemployment support, which have tended to rise, albeit slightly, towards the very end of the benefit period.[73] More generally, the introduction of waiting periods and experience rating for unemployment insurance contributions along with setting a time limit on unemployment benefits has been shown to decrease the disincentive effects associated with unemployment compensation.[74] Table 13 describes the main features of the unemployment and related cash benefit system in 1996.

Table 13. **Income replacement for people at working age**

1996

	Daily cash benefits during unemployment[1]	Disability and rehabilitation benefits	Daily cash benefits during sickness or maternity
Eligibility	All insured persons, registered at an unemployment office, able and willing to work, with an annual income of at least 75 per cent of the basic amount (NKr 29 423) in the last calendar year or as an average of the past three years.	All insured persons with entitlement to pension benefits whose working capacity has been reduced by 50 per cent or more due to illness, injury of defect, also taking into account job opportunities.	All insured persons with an annual income of at least half of the basic amount (NKr 19 423). Any woman not entitled to daily cash benefits receives a maternity grant of NKr 32 138.
Waiting period	None. Three days of unemployment in any ten day period qualifies for benefit.	None.	None. Four weeks notice to employer for reduced work under the maternity time account system.
Benefits	A daily rate of 0.2 per cent of the previous annual income not exceeding NKr 235 380.	Disability pension carries a basic amount of NKr 39 230. A supplement for dependent spouse of up to 50 per cent of basic amount.	100 per cent of pensionable income, up to six times the basic amount (NKr 235 380).
Means Testing	No.	Means test applied to disability pension supplement.	No.
Maximum duration	80 weeks. A renewal of benefits at 90 per cent of the previous benefit level is possible after a 13-week period. However, benefits are also paid during the interim period if no job or labour market measure has been offered.	52 weeks for rehabilitation benefits.	52 weeks for medical benefits. Up to 42 weeks of maternity benefits, and access to longer period if lower compensation.
Funding	Contributions from employees (7.8 per cent), employers (0-14.1 per cent, depending on zone) and the State (effective contribution of 30.7 per cent).	Contributions from employees (7.8 per cent), employers (0-14.1 per cent, depending on zone) and the State (effective contribution of 30.7 per cent).	Contributions from employees (7.8 per cent), employers (0-14.1 per cent, depending on zone) and the State (effective contribution of 30.7 per cent).
Administration	National Insurance Scheme (NIS).	NIS.	NIS.
Number of Beneficiaries (1995 average)	133 700[2]	290 900	408 700
Per cent of working-age population	4.3	9.3	13.0

1. New laws governing unemployment benefits took effect on 1 January 1997, raising the income requirement for benefit eligibility to 125 per cent of the basic amount (b.a.). Moreover, for incomes exceeding 200 per cent of the b.a., the maximum duration of benefits has been extended to three years, but without possibility of extension.
2. Full-time units. On average, there were 102 150 persons registered as full-time and 70 030 persons as part-time unemployed in 1995.

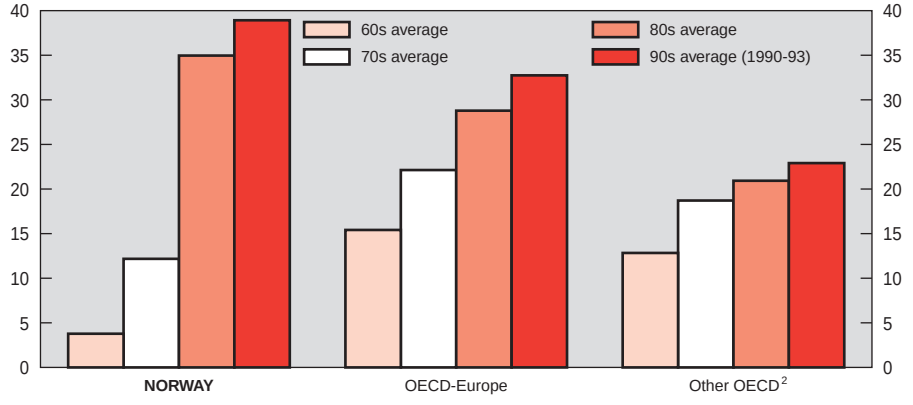
Source: Ministry of Finance

The ability of unemployed persons to renew unemployment benefits by leap-frogging to ALMPs, has resulted in practically indefinite duration of benefits.[75] During the past two decades, the unemployment insurance scheme has been repeatedly extended to the point that, in conjunction with various ALMPs, it is possible for an unemployed worker to receive between 40 and 60 per cent of his working salary for as much as six years without ever getting a non-subsidised job. Moreover, the ''summary measure of income replacement rates'', which captures both the duration and level of the benefit, has increased rapidly in Norway, and was in the early 1990s higher than the OECD Europe average, and higher still when compared with the rest of the OECD area (Figure 27).[76]

Unemployment traps may be especially high for those taking a job at average or below average wages, who may retain as little as 20 to 25 per cent of their salary once benefit withdrawals and taxes are accounted for. This is demonstrated by the *net* replacement rate – taking account of taxation and social welfare benefits – which gives a clear indication of the disincentives created by the interaction of the tax and benefit systems. As shown in Table 14, the net replace-

Figure 27. **UNEMPLOYMENT BENEFITS**

The summary measure of income replacement rates[1]



1. Benefits before tax as a percentage of previous earnings before tax.
2. Australia, Canada, Japan, New Zealand, United States.

Source: OECD Database on Unemployment Benefit Entitlements and Replacement Rates.

Table 14. **Net replacement rates**[1]

1994

	First month of unemployment				60th month of unemployment[2]	
	Couple, no children		Couple, 2 children		Couple, 2 children	
	2/3 APW[3]	APW	2/3 APW	APW	2/3 APW	APW
Australia	66	49	78	71	78	71
Belgium	75	64	76	66	91	70
Canada	64	63	67	67	61	47
Denmark	92	69	95	83	95	83
Finland	67	63	89	88	100	98
France	79	69	88	80	83	65
Germany	60	60	77	78	80	71
Ireland	67	49	70	64	70	64
Italy	36	37	45	47	14	11
Japan	49	43	48	42	86	68
Netherlands	79	77	84	84	95	80
New Zealand	70	48	86	70	86	70
Norway	**66**	**67**	**75**	**73**	**100**	**83**
Spain	74	75	77	74	66	46
Sweden	82	81	89	89	121	99
Switzerland	75	77	87	89	97	89
United Kingdom	52	35	90	77	90	77
United States	66	60	60	68	19	17
Average (unweighted)	68	60	77	73	80	67

1. For single-earner households, taking account of the effects of taxation and social security contributions.
2. Including social assistance.
3. The average production worker wage.

Source: OECD, *Employment Outlook*, 1996, Chapter 2.

ment rate for a family with two children, previously earning two-thirds of the average production worker wage, rises over time, from around 75 per cent in the first month of unemployment to 100 per cent after 60 months of unemployment, due to means-tested supplements.[77] Such net replacement rates are among the highest in the OECD area, matched only by those in Finland, Sweden and Switzerland. Clearly, there is not much financial incentive for members of a family in such a situation to switch to paid employment.

The *marginal tax wedge* – the difference between the cost to a firm of employing someone and the amount of after-tax consumption an employee can finance out of his take-home pay – is another gauge of the effect of the tax and benefit system on labour market outcomes. For Norway, it is calculated on the

basis of *i*) a marginal income tax rate (including social security contributions of employees) – ranging from 30.2 per cent for the lowest income level to 49.5 per cent for the highest incomes – and *ii*) a regionally differentiated social security contribution from employers, ranging from 0 to 14.1 per cent.[78] Thus, the marginal tax wedge in 1996 varied between 30.2 and 55.7 per cent,[79] down from the range of 37.3 to 81.3 per cent in 1980, due mostly to the tax reform in 1992.[80] Nonetheless, the level of marginal taxation in Norway is still high relative to most other OECD countries, tending to adversely affect work motivation for low-income workers in the presence of income replacement programmes[81] (Figure 28). In particular, there are pronounced disincentives for low-income single parents receiving income support, including child care, education and transitional benefits. In 1997, their effective marginal tax rate – including the reduction in benefits as incomes rise – ranges from 44 to 66.4 per cent for incomes up to approximately NKr 195 000.[82] Moreover, although the low social security contribution rates in regions with relatively high unemployment has acted to reduce the local tax wedge, it has also worked as an implicit regional subsidy reducing the cost of labour to firms. This has had an impact in regions with above-average concentrations of seasonal employment, such as fisheries and tourism.

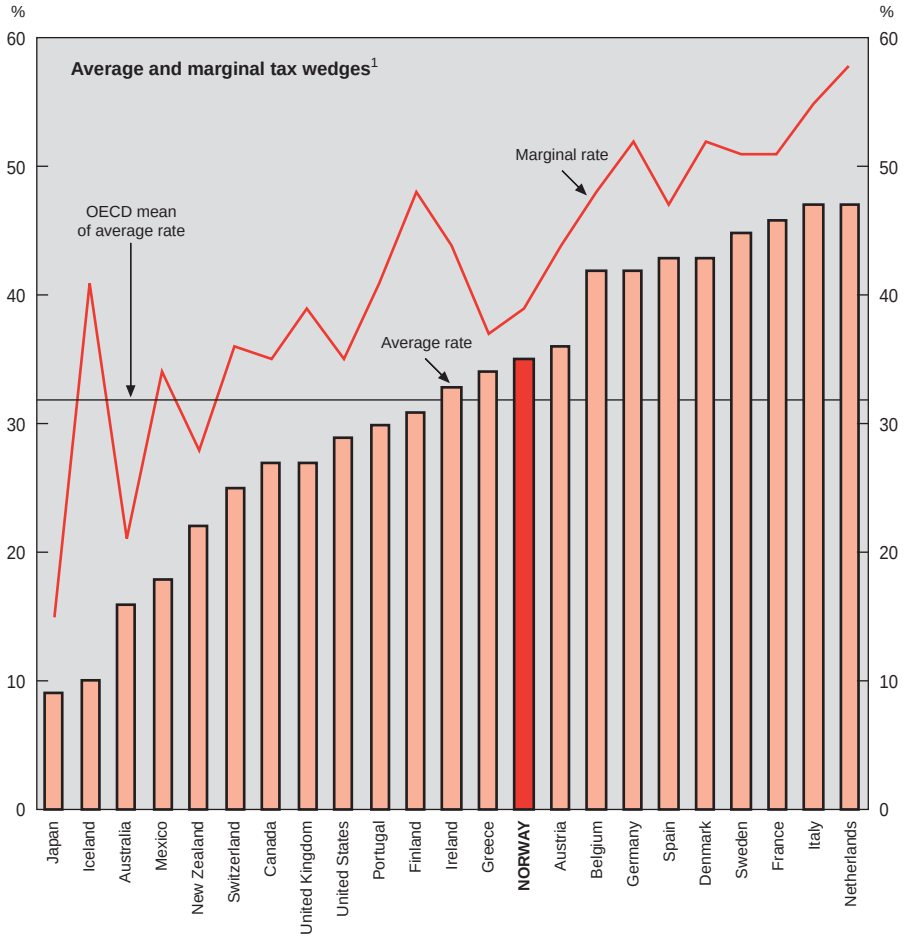
Active labour market policies

Although Norway has one of the lowest unemployment rates in the OECD area, it registered the fifth highest expenditure relative to GDP on Active Labour Market Programmes (ALMPs) in 1995, at 1.35 per cent. Of the 63 400 ALMP participants on average in 1995, roughly one-third were involved in labour market training (LMT) courses (including trainee places and substitute training), less than one-third were in other active programmes targeted on job-seekers (including public-sector job creation and wage subsidies) and the remaining third were ''occupationally handicapped'' persons under a variety of measures to improve their vocational skills – as opposed to placing them on a disability pension.[83]

While ALMPs, in particular LMTs, are in principle available for all persons who are unemployed or at risk of losing their jobs, most ordinary participants are persons with upper-secondary education or less, as the programmes tend to be targeted on people with low education and weak employability (Table 15). Participation in ALMPs for the ordinary unemployed rose steadily in the

Figure 28. **TAX WEDGES**

For a single person receiving 66% of an average production workers's income



1. The average wedge is the sum of employees' and employers' social security contributions and personal income tax as a percentage of gross labour costs – gross wages plus employers' social security contributions. The marginal tax wedge is the percentage of any marginal increase in gross labour costs arising as a result of an increase in gross wages which is paid in employees' and employers' social security contributions and personal income tax.
   Tax and wage data refer to the following years:
   1991: Austria, France, Germany, Iceland, Ireland, Japan and the Netherlands.
   1992: Belgium, Finland, Mexico, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland and the United States.
   1993: Australia, Canada, Greece, Italy and the United Kingdom.
   1994: Denmark.

Source: OECD, *The tax benefit position of production workers;* Secretariat estimates.

Table 15. **Active Labour Market Programmes for Ordinary Unemployed**[1]

1995

	Labour market training (LMT)	Public sector job creation	Job subsidies	Trainee places	Substitute training
Programme description	Vocational training to qualify job-seekers for demands of the labour market.	Temporary employment, with at least 15 per cent of the time spent in training.	Temporary placement in the public or private sector.	Work experience, preferably with training. Can be arranged in combination with secondary school.	On-the-job training for unemployed persons, while substituting for employees taking a temporary leave to study, train, or act as managers in programmes for unemployed.
Eligibility	Unemployed > 19 years of age. Groups with particular training needs.	1) Unemployed who have received unemployment benefits for > 70 weeks. 2) Unemployed ineligible for unemployment benefits.	Unemployed who need special assistance in addition to ordinary labour market services.	Persons without previous work experience.	Unemployed persons. Enterprises with need to upgrade skills.
Maximum Duration	Ten months.	Ten months. Two years for participants who hold managerial positions within the programme.	One year.	Six months, extendible to ten months in special cases.	Ten months.
Monthly Compensation 1996	Unemployment benefits, or grants NKr 149-199 per day. Allowance for traveling expenses and dependents.	Standard wages. No compensation for training period. Employer receives NKr 13 000 a month per participant.	Standard wages. Employer receives 25 per cent of total wages.	Depending on type of training, a grant NKr 149-199 a day and a subsidy to employer NKr 2 500-5 000.	Standard wages. Employer receives NKr 13 000 a month per participant.
Participants 1995 (average)	18 320	8 340	3 940	9 820	3 330

1. There were 19 560 persons participating in a variety of vocational training measures for the occupationally handicapped in 1995.

Source: Ministry of Labour

recession and peaked at 2.7 per cent (57 300 persons) of the labour force in 1993, before declining to 2.0 per cent (44 800 persons) of the labour force in 1995, due to the strong employment creation since 1993. Despite a fall in LMT measures in 1995, overall ALMP participation did not decline, as active measures targeted at the occupationally handicapped kept growing, to reach almost 20 000, double their 1990 level.[84]

Compared with passive income support schemes, such as unemployment insurance, ALMPs have the advantage of potentially strengthening the employability of participating workers. ALMP measures are designed to upgrade unemployed workers' skills and to interrupt long spells of unemployment to prevent the erosion of labour market skills. As individuals participating in certain ALMPs are not registered as unemployed, they directly reduce registered unemployment, especially long-term unemployment. However, the efficacy of the programmes should be judged in terms of how far they subsequently reduce mismatches, as well as the extent to which they improve the future job and earnings prospects of participants. ALMPs meeting these criteria are likely to reduce structural unemployment and wage pressure. Conversely, if they simply withdraw persons from the labour force, they may place upward pressure on wages. The effectiveness of these measures is therefore critically dependent upon the extent to which they actually upgrade or maintain relevant skills.

Cross-sectional macroeconomic estimates suggest that, in Norway, the beneficial effects of ALMP do outweigh their negative impact.[85] The effectiveness of LMT measures to increase employment has, however, been found to be mixed.[86] While programmes which offer formal training and are demanded by the labour market participants are most effective, especially in raising the employment probability of the participants, joining such ''popular'' programmes is rationed, perhaps due to their high cost.[87]

With the remaining unemployment in Norway mostly of a structural nature – including skill mismatches – the mechanisms for matching people with jobs, based on regional, sectoral and occupational mobility become more important. The government's Public Employment Service (PES) still maintains a monopoly on the placement of unemployed persons. A recent study found that the dissemination of vacancy information could be improved.[88] The PES also monitors job search efforts, and this activity, as noted above, is not very demanding of the job seeker.

Increasing the knowledge base, efficiency and innovative activity

Product market competition

The rationale for a strong competition policy may be especially compelling for a country like Norway, where a persistent current-account surplus, resulting from growing petroleum exports, and an associated real appreciation of the exchange rate has tended to raise the share of the ''non-tradeables'' sector in the economy, including the public sector. In these sectors, competitive pressures are generally weak due to a lack of effective competition from abroad and the presence of state monopolies. Indeed, productivity and/or efficiency in the distribution, construction, railways and the postal service has been found to be lower than in most other OECD countries.[89] By reducing prices of intermediate goods and services, enhanced competition in the non-tradeables sector would also tend to improve the international competitiveness of the tradeables sector.[90] In addition, there appears to be scope for tighter competition policies directed at the tradeables sector itself, where average mark-up ratios have been found to be high, in particular in non-metal and glass products, industrial chemicals, office appliances and transport equipment.[91] This is indicative of high barriers to entry, with both trade restrictions and economies of scale playing a role.[92]

Competition policy in Norway was reformed significantly in January 1994 – upon entry into the EEA – when the new competition law replaced the Price Act of 1953. The new competition law represents a shift from the ''abuse'' principle, to the now commonly used ''prohibition'' principle, making various anti-competitive practices illegal, as opposed to giving the government a mandate to act against firms, after establishing proof of abusive (anti-competitive) practices.[93] In particular, the competition authorities can act against public enterprises who financially support subsidiaries that are in competition with private suppliers. While they cannot impose administrative fines or other sanctions, they are charged with examining whether there are grounds for prosecution or caution, and pass on their deliberations to the competent body (such as the ministry supervising the activity of a specific public enterprise).

The new competition policy also emphasises enhancing effective competition in goods markets, by allowing private service providers to enter into public sector industries. In particular, the new competition policy has the aim of *i*) deregulating domestic markets, including the airline industry,[94] *ii*) abolishing state or

local government monopolies, including the import and distribution in the grain market and the wholesale monopoly in the drugs distribution, *iii*) transforming the administrative structure of several state-owned stock companies, by empowering management to frame and implement new business strategies, in particular in the telecommunications industry, and *iv*) separating service provision from regulatory activity. There has been significant progress in improving the market orientation of state enterprises since the 1980s and the deregulation of several markets has already occurred, including those for (hydro) electricity production and distribution,[95] as well as medicine production and supply.[96] On the other hand, as discussed in the previous chapter, re-privatisation of two of the three major commercial banks is proceeding slowly, with over 50 per cent of shares in *Den norske Bank* and *Kreditkassen* still in public hands.

Despite the progress noted above, direct support to industry and trade protection still imply considerable distortions in the Norwegian economy. The agriculture sector, in particular, benefits from a high degree of protection against imports, with effective tariff rates for meat and dairy products exceeding 300 per cent.[97] Measured by the Producer Subsidy Equivalent (PSE), agricultural support amounted to 74 per cent of the production value in 1995, almost twice the OECD and EU averages.[98] In 1995, the average direct support paid to each farmer, NKr 140 000, exceeded the NKr 130 000 calculated average return to own work and capital of farmers.[99] Budgetary support to the manufacturing sector also remains high by international standards, with *e.g.* energy-intensive industries recently receiving an exemption from electricity consumption tax and the shipping industry being granted a deferral of corporate income taxes (which are henceforth based on dividends rather than on corporate profits).

Promoting entrepreneurship

In 1995, small and medium-sized firms (SMEs) – firms with fewer than 100 employees – accounted for 99 per cent of private enterprises in Norway and almost 75 per cent of private business sector employment. This predominance of SMEs has been growing in recent years, despite the fact that some branches in manufacturing and the oil industry are dominated by large companies, while state monopolies are still prevalent in *e.g.* communication, transportation and postal services. Nevertheless, many existing company laws and regulations are considered complex and to favour large companies.[100] On the other hand, with some

exceptions, there appears to be sufficient amount of financing available for SMEs. The Savings Banks, which comprise half the banking sector in Norway, tend to be locally oriented and an important source of funding for SMEs. Moreover, the Industry and Regional Development Fund (SND) had in 1995 combined assets of NKr 4 billion (½ per cent of total business credit), a significant portion of which was available for the funding of high-risk loans to business start-ups and SMEs. Nonetheless, information and transaction costs create scale economies which make stock market financing more difficult for SMEs, given the low volumes of each transaction. Although the Oslo Stock Exchange has set up a separate listing for SMEs on the exchange (with less stringent requirements for being listed) to counter this, the volume traded is still quite small.

Technology, innovation and industrial policy

The OECD *Jobs Study* emphasised that technological research, innovation and diffusion are the main forces determining growth in productivity, and thus living standards in the medium and long-term, by stimulating new product development and increasing the efficiency of production processes.[101] For trend employment to grow in the presence of technological advances, product and labour markets need to be competitive to ensure that cost savings are passed on to consumers, thus stimulating demand, and should have the requisite flexibility to encourage firms to innovate in efficient production technologies, while at the same time encouraging the formation of an adequate labour supply with the right set of skills, to reduce mis-match bottlenecks (see also Chapter IV for a more detailed discussion).

At an estimated 1.8 per cent in 1995, total expenditure on research and development (R&D) in Norway remains below the OECD average of 2.3 per cent of GDP, partly due to the high number of SMEs with a relatively small share of R&D. In fact, private business R&D expenditure as a share of GDP has fallen since the 1980s, reflecting the declining output share of Norwegian Mainland industry. Moreover, the latter is based on food, wood and paper products, sectors which are traditionally not large performers in the R&D area. By contrast, although R&D spending in the information technology, electrical appliances, chemicals and metals sectors is close to the OECD average, these sectors are relatively small in Norway. Although the government accounts for about half of R&D funding, there appears to be a lack of interaction between the public and

private research sectors. Moreover, the small size of the domestic market may be an obstacle for the growth of innovative or technology-based firms. The largest five companies in Norway account for about 60 per cent of total industrial R&D. Despite the sluggish growth of domestic R&D funding (both industry and government) in recent years, this has partly been made up by increases in foreign R&D spending in Norway.

The government is taking an activist stance to technology policy. The Research Council of Norway (NFR) implements three technology-based programmes for small firms: Innovation and Entrepreneurship (FORNY), Regional Development of SMEs (RUSH) and Technology Diffusion from Research Institutes (TEFT). The FORNY programme promotes small start-up firms based on university research, providing financial support and legal advice to researchers. The RUSH programme is a regional scheme to develop links between regional industries and universities. The TEFT programme promotes collaboration between small firms and research institutes. Evaluations of the FORNY and TEFT programmes are now being conducted. There is some concern that these programmes may overlap with each other as well as the schemes of the Industrial and Regional Development Fund (SND), which generally promotes regional development through support to SMEs.

Many of these programmes are also intended to improve the links among industry, universities and public research institutes, which do not have a tradition of collaborating in research and technology development. Universities perform almost a third of the country's research but maintain weak ties to the business sector, resulting in little commercialisation of R&D. The SND has implemented two schemes to increase joint public/private research in the development of advanced technology: the Public R&D Contracts Scheme (OFU) and the Industrial R&D Contracts Scheme (IFU). The latter is now being evaluated, while an assessment of the former for the period 1988-92 showed positive results in terms of sales, exports and employment for participating companies.

Reforming education and training

As discussed in detail in Chapter IV, the system of non-compulsory education has in the past decade been subject to severe strains, mostly due to student enrolments exceeding the official projections by large amounts in the wake of the 1988-92 recession. These developments have generated a reservoir of university

graduates which is now starting to flood the labour market, in particular in the areas of law, economics, business administration, social sciences and construction engineering. This is a significant factor in explaining why unemployment is currently levelling-off despite strong employment growth (see Chapter I).

In the vocational branch of upper-secondary education, which in Norway is included in the comprehensive upper-secondary school system, the expansion in student enrolment has given rise to rationing of vocational student places and apprenticeship contracts, prompting students to remain in the system while queuing for a place in the field of study of their first choice. As a result, the effective study duration has tended to become excessively long, while the labour market is increasingly being confronted with a shortage of graduates with vocational skills, especially in urban areas. Recent projections, moreover, indicate that these shortages are likely to increase in the medium run. The lack of absorption capacity in the (vocational) upper-secondary school system became evident to the authorities already in the course of the 1980s, which led the government to initiate a far-reaching expansion and streamlining of the system as part of the ''Reform 94''. The key objective of that reform was to facilitate the transition of pupils within the system of upper secondary education from the general ''foundation classes'' to vocational courses. With the implementation of the reform, the municipalities running the schools were legally committed to provide adequate numbers of student places and to seek apprenticeship contracts with private enterprises. Given that the first cohort of apprentices under the new system has started only at the beginning of the current school year (1996/97), an assessment of the reform is not possible yet. However, preliminary results indicate that the shortage of (adequate) apprenticeship places has not yet been resolved.

The reform process in higher education has focused so far on: *i)* reducing the number of establishments in the non-university sector through mergers; *ii)* improving the system of budgeting and cost control; and *iii)* revising the planning of student intake per discipline. Concerning the latter issue, labour market considerations have been introduced in the planning for a number of disciplines where shortages persist, in particular teacher training and health-care professions (physicians, dentists and nurses). However, in other disciplines the actual number of student applications remains the overruling determinant of decisions regarding admission capacity, which, moreover, stays firmly in the hands of the central education authorities.

Norway has also an extensive network of adult education and training programmes, many of which have evolved from private initiatives, but are mostly managed by the local authorities. In 1995 some 25 per cent of the working-age population participated in adult courses of various durations. Roughly the same number of people were participating in work-related courses provided by employers. There is some evidence to suggest that such adult training and education in Norway is rather general in nature, and hence less conducive to enhancing the earnings potential of the participants. However, with a view to improving the impact of human capital formation on earnings and labour market prospects of individuals at large, the government's Vocational Training Act makes it possible for lower-skilled workers to obtain official craft and trade certificates upon demonstration of work experience.

Recent policy initiatives

In recent years, the government has focused on ways to improve the functioning of the labour market, in order to increase labour force participation rates and reduce structural unemployment, primarily through active labour market measures and by making adjustments to the unemployment and related social benefit system. The authorities have also implemented policies aimed at improving the functioning in product markets. These policies appear to cover only a limited part of the areas for structural reform identified by the OECD (see Box 2 above). This section discusses the specific policies initiated to this end in the past two years.

The *unemployment benefit* scheme has been changed, with the Storting recently passing legislation which entered into force on 1 January 1997. The reform has the stated aim of limiting unemployment benefit payments to those who have an employment history, to improve income security for the long-term unemployed and to establish a finite period for the receipt of such benefits. Three main changes to the unemployment benefit system were announced. First, access to unemployment benefits will be tightened, with the lowest yearly income level insured against unemployment to be raised from NKr 30 750 to 51 250, affecting possibly 12 000 – mostly young – unemployed benefit recipients. Second, the maximum legal duration of unemployment benefits is to be shortened from 3.7 to 3 years for persons previously earning more than NKr 82 000, and to 1.5 years

for those earning between NKr 51 250 and NKr 82 000. After the benefit period has expired, it will be necessary to re-qualify for unemployment benefits through regular employment. In the past, long-term benefit recipients have been able to move from one benefit programme to another in order to obtain continued (and almost indefinite) income support. The government intends to end this practice, although the first cohort of beneficiaries concerned will be affected only as of January of the year 2000. Finally, the earlier practice of reducing the benefit rate by 10 per cent after 93 weeks of unemployment has been discontinued, affecting about 25 000 people. In early 1996, the Storting passed legislation to reduce the maximum benefit period for *Social Assistance income support for single parents* from ten to three years, and to five years for single parents enrolled in education.

Recent measures to improve *competition in product markets* have aimed at reducing over-staffing in state monopolies, in particular the railway company. Moreover, the 1997 Government Budget proposal stipulates that the state railway and postal system will be transformed into state owned enterprises in 1997, which will remove them from the budget and make their operation dependent on their profitability. The process of deregulating the telecommunications sector has started and is to be completed in 1998. In the air transportation sector, capacity has increased in recent years, but foreign competition will be allowed on domestic routes after 1997/98, as in other EEA countries. Direct support of industries is being re-evaluated, to make Norwegian industries more competitive and, where necessary, to comply with the EEA and WTO agreements. As a result, support for the agricultural sector will be gradually reduced. While the above reforms represent moves in the direction of establishing a level playing field for competition, the government was persuaded to reduce considerably the taxation of shipping companies in June 1996, in order to prevent a significant loss of jobs associated with the migration of the merchant fleet to countries offering better tax treatment.

The government has implemented in recent years a number of programmes to ease access to financing, technology and skills for *small- and medium-sized enterprises* (SMEs). The FORNY, RUSH and TEFT schemes, focused on promoting innovative start-ups, supplement the existing programmes of the SND, which has long provided financial support to smaller firms and regions in difficulty. In addition, the National Institute of Technology (TI) and the North Norwegian Institute of Technology and Innovation (VINN) bring technology

consulting services to SMEs, while the Norwegian Business Advisors group gives guidance to small firms on questions ranging from general management to product development and the Office for Guidance to Inventors offers grants to smaller firms for development of inventions. The Industry and Trade Legislation Committee has put forward a proposal to simplify laws and regulations, for example the law on non-public limited companies, to give SMEs more equal footing *vis-à-vis* larger enterprises.

Concerning *education* policies, in addition to ''Reform 94'', the government is currently implementing an extension of the compulsory school age by one year, beginning (see Chapter IV) with pupils at six years, to bring it more in line with the practice in other OECD countries. Further, in higher education financial incentives have been created for reducing study durations. As of the academic year 1996/97, universities and other institutions for higher education have been offered the possibility to compete for ''result-oriented grants'' from the State, based upon the number of graduations. In the same spirit, students are being awarded extra grants when completing three rather than the usual two terms per academic year.

Assessment and scope for further action

The rapid growth of the Norwegian economy has resulted in significant employment creation and a continued reduction in the unemployment rate, close to the levels in Japan and the United States and well below the European average. Set against this background, and in view of the substantial fiscal and external surpluses, the problems confronting the Norwegian labour market appear to be minor compared with most other OECD countries. It should be noted, however, that the current favourable situation represents an opportunity to reduce welfare dependency. Further structural measures would, moreover, lessen the risk that economic shocks (*e.g.* volatility in the oil price and international economic activity), as well as the inevitable decline in oil revenues and the rising cost of the welfare system associated with an ageing of the population in coming decades (see Chapter II), may result in higher unemployment and an associated deterioration of the budgetary situation. Additionally, high and equitably distributed employment opportunities may more successfully ensure the attainment of Norwegian long-run objectives for a fairer income distribution than the welfare

system as currently designed. Clearly, the government has begun to change the current labour market practices, by *e.g.* reducing access to, and the duration of unemployment and related social benefits, maintaining a high level of active labour market policies and continuing to encourage moderate overall wage growth – through consensus building by the social partners – to maintain external competitiveness and to improve employment prospects. The impression, however, remains that more could be done, especially in view of the tightening labour market conditions at present.

In particular, with wage pressure building up, policies directed towards further increasing the *effective* labour supply and a more pronounced reduction in the generosity of the social benefit system are called for along with more flexibility in relative wages to allow young and low-skill workers to price themselves into employment, and more generally to foster continued private sector employment growth. A more flexible wage policy in the public sector could contribute to make the public sector more efficient. In order to enhance labour force participation it is important to ensure that labour market programmes are well designed and targeted. Moreover, reforms in product market competition and the education system should strengthen the ability of the economy to adapt to future shocks and ensure the prospects for sustainable growth.

The reduction in the duration of unemployment benefits which comes into operation in 1997 aims at increasing the incentive for effective job search by long-term unemployed persons. However, the rise in the benefit level beyond 93 weeks is bound to make the unemployment trap facing these persons more pronounced during the allowable benefit period. The reduction in the benefit period for income replacement for unemployed single parents, likewise, should help to prevent single-parent families from falling into long-term income dependence. The rise in the benefit rate, however, is likely to work in the opposite direction, by making unemployment traps more pronounced during the benefit period. Moreover, future action could focus on introducing waiting periods before benefits are paid.

At this juncture in the business cycle, certain labour market institutions and policies – the unemployment insurance system, some features of ALMPs, employment protection legislation (EPL) and the disability pension system – may be contributing to bottlenecks in the labour market. To counteract these developments, the PES placement service could be liberalised to improve job-skill

matching (*inter alia* through improved dissemination of vacancy announcements), tighter job search provisions, reasonable waiting periods and effective enforcement should be aimed for, especially since the generosity and duration of unemployment benefits have not been cut back sufficiently, so as to remove work disincentives. Active labour market programmes, especially those found to raise human capital, should be continued and targeted more closely on low-skilled workers, especially the young. The fact that access to the unemployment insurance system has been tightened and the maximum duration of benefits reduced is an important step in the right direction, but further effort would be needed along these lines.

Although approximately in line with European standards, Norway's EPL may to some extent have contributed to reduce the labour market's margin of flexibility. A number of job protection provisions should therefore be re-evaluated, especially if they impede the successful insertion of new entrants' into the labour force. More liberal regulations, especially with respect to short-term contracts, should be introduced. Experience in other OECD countries suggests indeed that successful moves from unemployment to employment often involve fixed-term contracts. Based upon market needs, such contracts may contribute to reduce youth unemployment and prepare the ground for full-time employment opportunities.

Policy towards SMEs could be changed to better meet the need of smaller firms to form clusters and networks for co-operation and so more effectively compete with larger and established firms in the global marketplace. The government could also realise economies of scale in dealing with the SMEs as groups rather than individually, while passing co-ordination costs to the firms themselves. Direct support for industry, among the highest in the OECD area, should be reduced more substantially; in particular, support for the agricultural sector. Competition policy is moving on the right track, in terms of privatising state enterprises and removing regulations that restrict competition in sectors dominated by state enterprises.

As noted, education policy in recent years has mainly focused on upper secondary education. With ''Reform 94'', upper secondary education has practically completed the transition to universal participation, while reducing the differences in curricula between general and vocational education and limiting the number of vocational programmes. Moreover, facilities for apprenticeship train-

ing have improved. However, the question arises to what extent this policy will actually succeed in matching demand for and supply of workers with specific vocational skills. Medium-term projections suggest that imbalances may persist, unless more labour market participants or entrants convert to occupations in high demand. Moreover, with the allocation of funds and student places across disciplines and institutions likely to remain based on student applications, only more pronounced ‘‘price signals’’, inherent in flexible relative wages, will encourage a shift in applications towards courses of study relevant to the labour market needs.

In sum, with the remaining unemployment at a very low level, reflecting essentially skill mismatches as well as the rise in the labour force participation rate to its pre-recession peak, the scope for macroeconomic policies to reduce it further is now close to being exhausted. In order to generate more employment and keep the economy on a sustainable path further improvements in the structural features of the labour and product markets are thus required, including *i)* enhancing the scope for relative wage adjustments to facilitate a more efficient allocation of labour resources, in particular the entry of young and low-skilled workers into employment, *ii)* reducing the generosity of the social transfer system, which currently provides income replacement for 15 per cent of the working-age population, *iii)* a better targeting of the education and training system to meet changing skill requirements in the labour market, and *iv)* the lowering of entry barriers in product markets to facilitate sustained favourable labour market outcomes. In a longer-term perspective, balancing the size and scope of the public sector in the economy should be considered in order to avoid long-term negative effects on the international competitiveness of Norwegian producers, and hence prepare the economy for the next century.

IV. Education and training

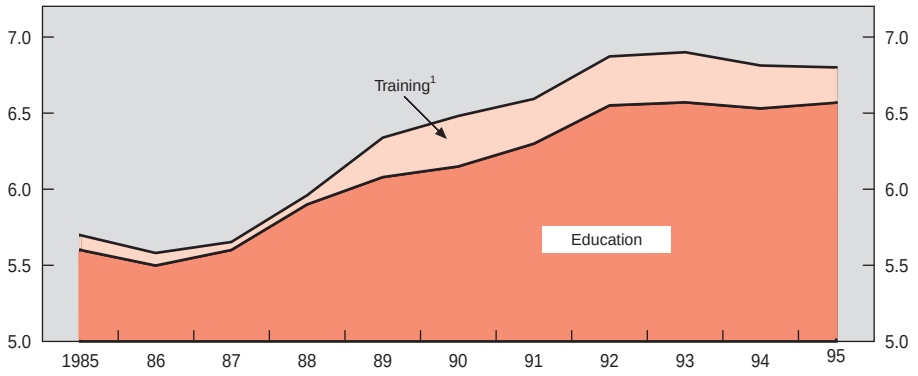
As discussed in detail in the 1995 *Economic Survey*, the exploitation of oil and gas from the North Sea has considerably benefited the Norwegian economy, allowing much higher growth rates of real earnings *per capita* and employment than could have been attained otherwise. Such a favourable outcome could not have been achieved had domestic firms not been in a position to rely on the specialised human capital needed to extract the available energy resources and to perform related activities such as the construction and maintenance of platforms, vessels and landing facilities, as well as logistics and catering services. However, human capital proves to be of a much wider interest to Norway, which – as other OECD countries – is in the process of adapting to the rapid technological change and associated industrial restructuring, implying a growing demand for more educated labour.

The Norwegian labour market is already now characterised by shortages of labour with vocational skills, which could act as a brake on economic growth and become a potential source of inflation. At the same time, there is excess supply of workers with university education in some disciplines, due to a considerable extension of Norway's education and training system since the mid-1980s to raise the educational standards of the population. This has been reflected in rising government expenditure in this field from 5¾ per cent of GDP in 1985 to almost 7 per cent in 1995 (Figure 29), with real *per capita* expenditure increasing by over 40 per cent in the same period.

These issues are addressed in the present chapter. The first section briefly illustrates the importance of human capital formation for the development of Norway's economy in the future. This is followed by a review of the Norwegian education system, with an emphasis on recent reforms having affected the structure, performance and cost of the system. The chapter then examines the process of continuing education and training for adults, including labour market training.

Figure 29. **GOVERNMENT EXPENDITURE ON EDUCATION AND TRAINING**

Per cent of GDP



1. Labour market training only.

Source: Statistics Norway, Ministry of Education, Research and Church Affairs; OECD, *Employment Outlook,* 1996.

It concludes with an overall assessment of the reform process and suggestions for further policy actions.

Why human capital is essential for Norway

Increasing mainland labour productivity

Recent analysis of economic growth has shown that human skills and competencies are an important determinant of a country's performance in raising aggregate labour productivity.[102] Increased human capital not only raises the productivity of labour inputs in a narrow sense, it also enhances the potential of other production factors – such as capital, technology, economies of scale and the socio-political environment – to contribute to a higher level of labour productivity. Moreover, there is evidence to suggest that a higher level of human capital contributes to a higher growth rate of productivity, reflecting the ability of appropriately educated workers to implement new technologies and to adjust to the institutional and organisational changes that go with it. Hence countries that

are better endowed with skilled labour reap a growth premium.[103] Indeed, the presence of adequate human capital influences investment decisions through the scope for innovation and the diffusion of new technologies, and a more and broader educated work force may become a seed-bed for new technologies, adjusting more smoothly to the structural, institutional and organisational changes that are typical for modern economies. It is in this context that the OECD Jobs Study concluded that investment in, and the proper orientation of, education and training confers benefits for the economy and society in the form of higher output and productivity growth.

Cross-country comparisons invariably rank Norway high with respect to aggregate labour productivity – in line with developments in a number of other ''productivity leaders'' in Europe, such as France, the Netherlands and Belgium (Table 16). This results in part from the tremendous expansion of petroleum production on the Norwegian continental shelf. Indeed, after correction for such production, output per man hour in Norway appears to have reached only 83 per cent of the United States level by 1993 (rather than 99 per cent) – still above the average for the European OECD as a whole, but significantly below the score of the above mentioned leading countries. In order to offset the loss in *per capita* income associated with the expected decline in petroleum production after the turn of the century, increases in mainland labour productivity are clearly needed, which would require continued efforts in the field of education and training. Given that educational qualifications of the population in Norway are already among the highest in the world, such efforts would need to focus on optimising the composition rather than increasing the overall level of skills and competencies.

One factor which constrains labour productivity in Norway's mainland economy is the large size of the public sector, even though national accounts statistics usually underestimate public sector labour productivity (in the absence of a market valuation, public sector production is essentially gauged by the value of its labour inputs only, which may be less than the true value added). Indeed, there is evidence of lower efficiency in public sector activities due to a lack of market scrutiny.[104] Moreover, the rate of return to human capital in the public sector – which measures average annual real earnings per worker as a percentage of expenses and earnings foregone associated with school enrolment – appears to

Table 16. **Comparative levels of labour productivity**

GDP per man-hour relative to the United States[1]

	1913	1929	1938	1950	1960	1973	1987	1993[1]
United States	100	100	100	100	100	100	100	100
Norway	**43**	**45**	**50**	**43**	**52**	**64**	**90**	**99**
of which: **Mainland economy**[2]	**43**	**45**	**50**	**43**	**51**	**60**	**82**	**83**
Netherlands	69	74	64	46	54	77	92	97
France	48	48	54	40	49	70	94	96
Belgium	61	55	53	42	45	64	86	90
Canada	75	66	58	75	79	83	92	83
Italy	37	35	40	31	38	64	79	82
United Kingdom	78	67	64	57	56	67	80	79
Austria	48	37	33	27	38	59	74	78
Australia	93	77	75	67	69	70	78	78
Sweden	44	38	43	49	54	76	82	75
Germany[3]	50	42	46	30	46	64	80	74
Japan	18	22	23	15	20	46	61	69
Denmark	58	59	54	43	46	63	68	68
Switzerland	51	57	55	56	59	67	68	66
Finland	33	32	33	31	36	57	67	60

1. Ranked according to 1993 level.
2. OECD Secretariat estimates.
3. In 1993 including eastern Länder.

Source: A. S. Englander and A. Gurney, ''Medium-term determinants of OECD productivity'', *OECD Economic Studies*, No. 22. Spring 1994; OECD Secretariat.

fall short of that in the private sector, which is also indicative of comparatively low labour productivity of public sector activities (Table 17).[105]

Raising the level of mainland labour productivity and *per capita* earnings may thus call for a leaner public sector and a shift of human capital resources towards mainland business activities. Such a transition would be facilitated if the private sector succeeded in attracting highly-educated women, who currently tend to opt for positions in the public sector where working time arrangements are relatively more flexible. Indeed, as the earnings differentials between the private and public sectors for female workers are relatively narrow, women have only weak incentives for a business career. By contrast, highly-educated male workers typically aim for jobs in the private sector, although they often use the public sector as a means to benefit from training facilities offered by the public administration. This also explains the greater tendency for men to see an advanced level of education to be associated with higher private sector earnings compared to

Table 17. **Rates of return to formal education in the Nordic countries**

Per cent[1]

	Norway	Denmark	Finland	Sweden
Men	**5**	5¼	7¾	5
Private sector	**6½**	6	7¾	6¼
Public sector	**4½**	5¼	8¼	5
Women	**5**	3½	6½	4½
Private sector	**5½**	4	5	5½
Public sector	**5**	3½	7½	4

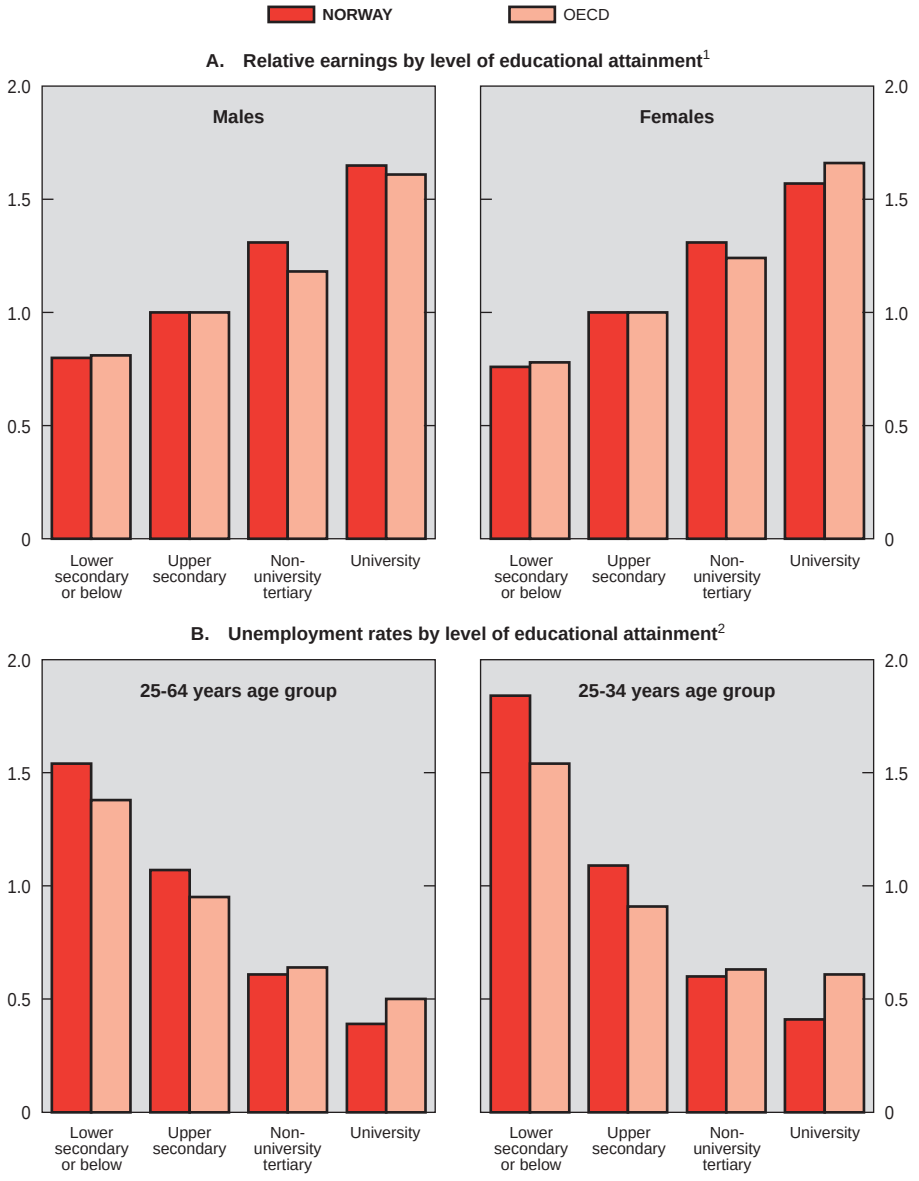
1. Pre-tax rates of return of one additional year in postcompulsory education.

Source: R. Asplund *et al*., ''Human capital and earnings in the Nordic countries'', in R. Asplund (ed.), *Human capital creation in an economic perspective*, Physica-Verlag, 1994, pp. 68-94.

women (Figure 30, Panel A).[106] A larger degree of flexibility of working time arrangements in the private sector could change these outcomes, with more higher educated women finding employment in the private sector and their rates of return on human capital becoming more akin to those of men (Table 17).

The future evolution of the mainland economy will be constrained by the ageing of the population which, on current demographic trends, should occur in the first half of the next century. Part of the required increase in mainland productivity should therefore take the form of enhancing life-time production and earnings per worker in the mainland economy. One way to achieve this would be to extend the effective retirement age of the Norwegian labour force. At present, the effective retirement age amounts to 61 years, six years below the official one, largely on the account of workers opting for early retirement through the disability pension scheme. While a stricter enforcement of eligibility rules regarding disability pensions would certainly help contain the number of beneficiaries (see Chapter III), evidence abounds that raising the human capital embodied in the work force would also contribute to extending the effective retirement age through better job tenure and prospects for upward earnings mobility. However, due to the smaller size of cohorts entering the labour market, such expansion of human capital cannot be achieved only through the renewal of the work force by the arrival of better educated youth; retraining adult workers is also necessary to a greater extent than in the past. In fact, this development is one of the main motivations behind the OECD-wide emphasis on continued education or

Figure 30. **EDUCATION AND LABOUR MARKET PERFORMANCE**



1. Ratio of mean annual earnings by level of educational attainment to mean annual earnings at upper secondary level (1992).
2. Ratio to overall unemployment rate (1992).

Source: OECD, *Education at a Glance,* 1995.

‘‘life-long learning’’.[107] This principle has also been embraced by the Norwegian authorities, and will need to be translated into policy action in the years to come.[108]

Meeting skill needs

Persons with a higher level of education and training tend to receive better pay, have a lower unemployment risk, are more likely to receive formal job-related training and retraining, and hence also experience more secure job tenure. Recent empirical analysis for several OECD countries clearly indicates that human capital characteristics such as skills, educational level and experience raises the degree of ‘‘upward earnings mobility’’ – *e.g.* the predisposition of workers to move up the job ladder.[109] The importance of formal training for upward earnings mobility is also confirmed for Norway, especially if such training is received at an early stage of one’s career,[110] even though the relationship between human capital and earnings at the individual level is bound to be weaker than in most other OECD countries, due to the rather compressed wage structure. The earnings distribution, moreover, has continued to narrow during the 1980s, both across branches of industry and levels of educational attainment.[111]

As in other OECD countries, the employment chances for higher educated workers are generally better than for those with lower levels of education, with the unemployment rate of the latter exceeding that of the former by a wide margin (Figure 30, Panel B). Although the number of unemployed at any level of education in Norway is typically lower than in other European countries, the unemployment rate of workers having completed only primary education more than doubled during the recent recession from a 3.6 per cent low in 1985 to 8.9 per cent in 1992, and has broadly stayed at that level since (Table 18). This occurred despite a marked decline in the labour force participation rate of primary educated people, which suggests that it is entirely due to falling job openings for this category of workers. At the other extreme, the unemployment rate for higher educated workers remained very low even during the recession in the early 1990s, climbing from 1 per cent in 1985 to a peak of 3.7 per cent in 1993. It has, however, not fallen much since then, due to excess supply of labour in a number of professions (see below). By contrast, the unemployment rate for workers with upper-secondary education, although also hit by the recession, dropped significantly from a 6.4 per cent peak in 1992 to 5.1 per cent in 1995,

Table 18. **Labour market situation by level of education**

Per cent

	1985	1990	1992	1993	1994	1995
Unemployment rate	**2.5**	**5.2**	**5.9**	**6.0**	**5.4**	**4.9**
of which:						
Primary school	3.6	8.4	8.9	8.9	8.4	8.5
Lower-secondary	2.9	5.6	6.4	6.3	6.0	5.1
Upper-secondary	2.2	5.4	6.4	6.4	6.0	5.1
Tertiary	1.0	2.0	2.8	3.7	3.3	3.3
Participation rate	**68.5**	**69.2**	**68.4**	**68.2**	**68.6**	**69.6**
of which:						
Primary school	52.3	49.3	46.9	46.3	46.3	47.1
Lower-secondary	72.4	71.6	69.8	69.4	69.2	69.2
Upper-secondary	77.9	78.6	77.0	76.7	77.1	78.1
Tertiary	87.1	86.3	84.3	82.6	82.0	82.4
Employment rate	**66.8**	**65.6**	**64.3**	**64.1**	**64.9**	**66.2**
of which:						
Primary school	50.4	45.1	42.7	42.2	42.4	43.1
Lower-secondary	70.3	67.6	65.4	65.0	65.1	65.6
Upper-secondary	76.1	74.3	72.1	71.8	72.5	74.1
Tertiary	86.2	84.6	82.0	79.6	79.3	79.7
Employment (× 1 000) [1]	**2012**	**2030**	**2004**	**2004**	**2035**	**2079**
of which:						
Primary school	476	372	329	317	306	302
Lower-secondary	613	544	501	502	489	466
Upper-secondary	482	610	630	627	643	673
Tertiary	395	483	527	448	471	501

1. Including employment not classified by educational attainment.
Source: Statistics Norway.

reflecting a substantial recovery in employment. Conversely, a similar fall in the unemployment rate for people who have completed lower-secondary education has been due to a decline in the participation rate, with the employment level remaining flat.

Such broad tendencies mask important differences between the labour market developments for individual occupations and professions. Among higher educated people, the recruitment of nurses, physicians and dentists has markedly increased with the expansion of the health care system in recent years, to a point where the education system had insufficient capacity to deliver the number of graduates required. As can be inferred from Table 19, almost 90 per cent of health care workers easily found a job within six months after completing educa-

Table 19. **Labour market situation of individuals six months after completing education, by level and field of study**

Per cent, 1993 figures

	Employed	ALMP	Unemployed	Further education	Withdrawn pension	Withdrawn others	Total number (× 1 000)
Upper-secondary education							
General studies	28	8	3	37	1	22	33.2
Humanities and arts	28	6	4	46	1	14	5.2
Teaching	40	13	4	26	2	15	1.6
Business, economics	29	7	3	49	1	11	38.7
Science, technical	30	8	4	44	1	12	37.9
Transportation	36	13	8	24	1	17	4.0
Health care	62	8	6	7	1	16	4.6
Agriculture, fishery	36	4	2	45	1	13	4.4
Services	34	9	4	38	1	13	9.6
Total	31	8	4	41	1	15	139.4
Higher education I (bachelor's equivalent)							
Humanities and arts	42	1	2	48	0	7	8.7
Teaching	72	1	1	18	0	7	9.6
Business, economics	57	3	3	28	0	8	13.8
Science, technical	44	5	5	28	0	17	2.1
Transportation	56	8	7	5	1	22	1.2
Health care	86	1	0	2	0	11	2.8
Agriculture, fishery	51	8	8	11	3	20	0.2
Services	69	1	2	27	0	0	1.3
Total	58	2	2	28	0	9	39.7
Higher education II (master's equivalent)							
Humanities and arts	62	0	4	15	1	16	0.0
Teaching	86	0	2	2	0	8	0.2
Business, economics	70	2	7	8	1	12	0.8
Science, technical	65	5	5	7	0	17	1.9
Health care	87	0	1	2	0	10	0.4
Agriculture, fishery	63	3	3	4	0	25	0.2
Services	99	0	0	0	0	1	0.2
Total	70	3	4	7	0	15	3.9

Source: Statistics Norway.

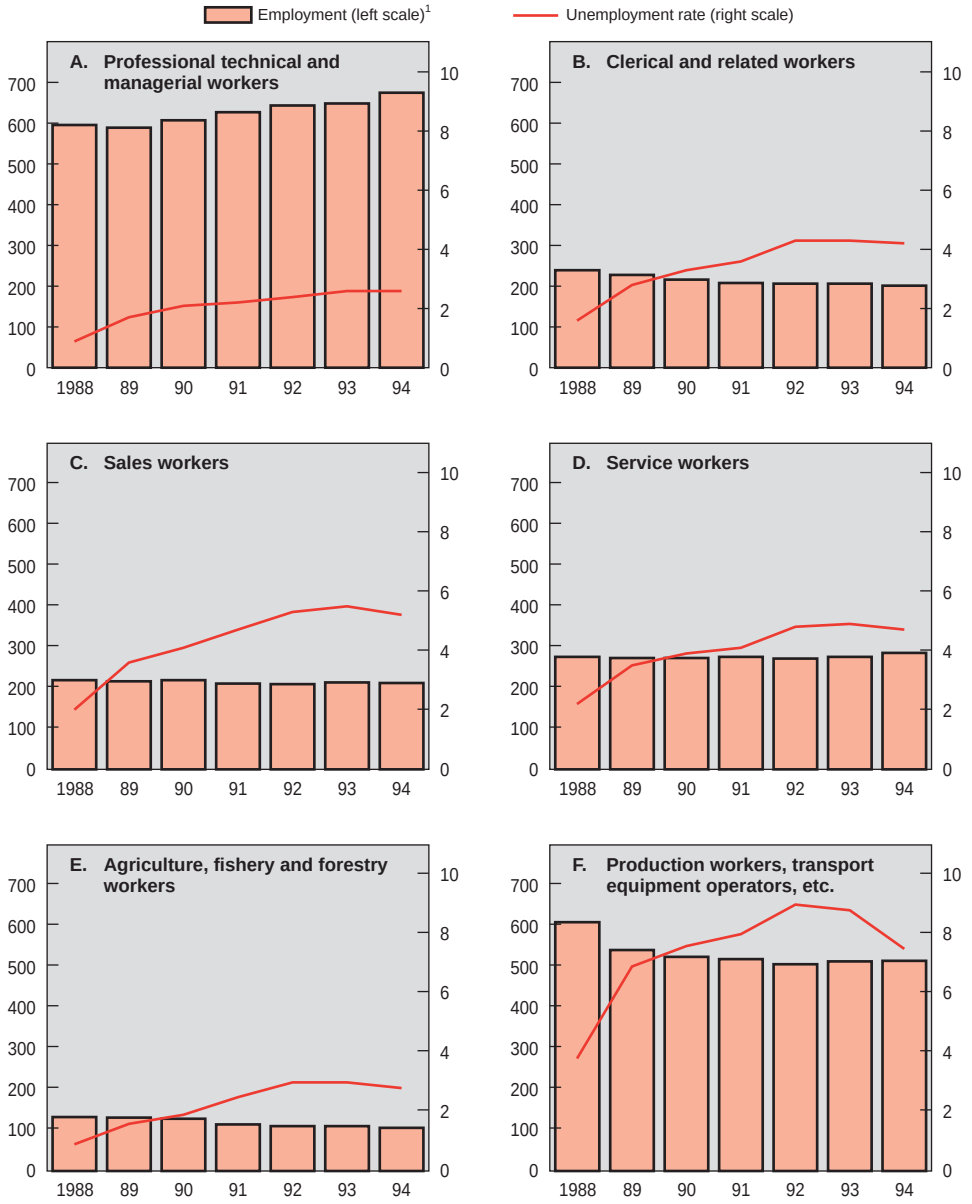
tion even when the number of unemployed peaked in 1993. Labour market conditions in this area have tended to tighten further since. The demand for teaching staff also increased in recent years, which is explained by the expansion of the education system since the late-1980s (see below). Indeed, at the peak of unemployment in 1993, around 80 per cent of new teachers found a job within

six months after graduation. At the same time, the labour market position of engineers (chemicals, construction) has been weakening with the unwinding of the investment activities on the Norwegian continental shelf (see Chapter I). The labour surplus for most other professions requiring higher education has also been widening, due to sharp increases in the enrolment of students in tertiary education since the late 1980s, in particular in the fields of law, economics, business administration and social sciences.

By contrast, unlike developments in other European countries, the unemployment rate for manual workers, although still significant, has been falling rapidly since the onset of the economic recovery (Figure 31). There are currently also rising employment opportunities in the hotels, restaurants and retail business (especially in the Oslo area), which is reflected in declining unemployment rates for sales and services workers.[112] According to recent projections for employment by occupation carried out by Statistics Norway, such tendencies should become even more pronounced in the medium run, with demand for workers having completed vocational education at the upper-secondary level expected to exceed supply by the year 2000 in several occupations, which may give rise to quantity rationing and wage increases.[113] At the same time, the excess supply of workers in occupations requiring higher education (other than health care workers and teachers) is projected to increase further.

In the long run, the decline in the government's petroleum revenues will certainly lead to further changes in the composition of labour demand by occupation. At present, Norway has one of the highest shares of professional, technical and managerial (PT&M) workers in total employment in the OECD area, only surpassed by the United Kingdom and Finland (Table 20). This phenomenon reflects the strong employment growth in the public sector in past decades, which has absorbed the bulk of high skill labour supply (in particular health workers and teachers – see above). With fiscal constraints tightening once the petroleum revenues start to decline in the long run, a shift in emphasis from public sector to mainland business job creation may become inevitable, and the share of PT&M workers employed in the public sector could fall accordingly. This raises concern that many higher educated people will have to accept a reduction in relative pay, or convert to other professions. New opportunities for higher educated people are likely to emerge in the ''high-tech'' segment of manufacturing, though, as noted in the OECD Jobs Study, these are still relatively small in Norway.[114] The next

Figure 31. **EMPLOYMENT AND UNEMPLOYMENT BY OCCUPATION**



1. Thousand persons.
Source: International Labour Office, Yearbook of Labour Statistics 1995.

Table 20. **Employment by occupation**

Percentage shares of total, 1994[1]

	Professional, technical and related workers	Administrative and Managerial workers	Clerical and related workers	Sales workers	Service workers	Agricultural workers, fishermen, etc.	Production workers, transport equipment operators, etc.
Norway	**26.2**	**7.0**	**9.9**	**10.4**	**14.1**	**5.2**	**25.4**
United States	17.4	13.3	15.1	12.0	13.7	2.9	25.5
Europe	14.4	7.7	16.2	10.2	12.2	6.6	30.5
of which:							
Austria	16.1	5.6	16.6	9.0	11.6	7.0	33.5
Belgium	14.2	7.7	9.6	6.2	5.5	6.2	49.8
Denmark	25.4	4.6	17.7	8.0	12.4	4.8	25.6
Spain	12.4	2.0	13.7	11.6	14.5	9.9	35.1
Finland	34.8	2.4	22.4	11.2	15.2	5.7	7.8
Germany	17.0	3.3	20.9	9.1	11.2	3.6	31.9
Greece	13.0	1.9	11.3	12.2	9.9	22.0	28.9
Ireland	17.0	3.7	15.1	10.3	11.5	14.0	27.4
Italy	1.1	9.4	12.8	14.0	15.4	4.6	39.4
Netherlands	24.0	4.3	17.8	10.2	12.3	3.9	21.6
Portugal	12.7	11.3	13.7	7.1	13.6	8.5	32.5
Switzerland	6.3	14.6	19.5	15.9	13.1	4.4	26.2
United Kingdom	19.1	15.5	15.4	7.9	10.0	8.8	22.6

1. Or latest year available.

Source: International Labour Office, *Yearbook of Labour Statistics 1995*.

two sections examine the extent to which the education and training systems, as currently designed, are able to cope with such concern.

The education system in the light of past and planned reforms

Main features of the system

Similar to the other Nordic countries, Norway has an education system which is highly regulated by the public authorities. At the central government level, the Royal Ministry of Education, Research and Church Affairs has full control over the core curricula for primary and secondary education, as well as for various disciplines in tertiary education.[115] The system is almost entirely public, although a small (but growing) private education sector exists – affecting around 2 per cent of all pupils in the 7 to 19 age group and mostly focusing on a

specific religion, philosophy of life or alternative tuition method. The State directly owns and manages the higher education system, although the institutions concerned (4 universities, 6 specialised university-level colleges and 26 non-university state colleges) have academic discretion with respect to teaching, research and strategic decision making. Conversely, the municipalities are in charge of the primary and lower-secondary schools, combined in so-called comprehensive schools (3 300), while the counties manage the upper-secondary schools (700). However, the municipal and county schools are closely supervised by 18 State education offices (one in each county) and are, for the main part, financed through state grants.[116] Public education is available for every citizen free of charge, while approved private schools receive a grant funding 85 per cent of the running costs at the comprehensive level and between 75 and 85 per cent on the upper-secondary level, with the remainder covered by tuition fees.

The municipal comprehensive schools admit pupils at compulsory school age (6 to 16 years),[117] of which seven years in primary education and three years in lower secondary education. Class units remain unchanged up to and including the second year of lower secondary education (except for participation in optional subjects). The average class size is about 18 children at the primary level and 22 at the lower secondary level, against legal maximum numbers of 28 and 30 respectively. Children with special learning needs are mostly enrolled in the mainstream education system; but there are 20 resource centres to provide assistance to the local schools, thirteen of which are specialised in sight, hearing, speech, reading and writing difficulties and seven dealing with behavioural and emotional disorders, including mental disability. There is also a facility for children from the Sami language minority, who can opt for it as either the first or the second language of tuition (there are currently about 1 400 Sami pupils). The around 26 000 immigrant children also receive education adapted to their special needs. Apart from the regular schools, most municipalities have set up after school day-care centres for children aged 7 to 10 years.

A key element of the Norwegian system is the emphasis on comprehensive education up to the upper-secondary level. After completing compulsory school, almost all pupils (96 per cent) are enrolled in full-time upper secondary education at the age of 16 up to (mostly) 19 years, which offers both general and vocational tuition. As a result, a highly structured vocational pathway into the labour force exists for all qualification levels and there is a well-defined framework for

certifying the level of qualifications required for specific trades. This feature has been further intensified after a far-reaching reform in 1994 (''Reform 94''). Under the new system, pupils, after leaving the compulsory school system, are enrolled in one of the thirteen new ''foundation courses'' (of which three with a general and ten with a vocational orientation; prior to the reform the number of foundation courses from which to choose was more than 100). Subsequently they enter an advanced course ''Level I'' which provides a further specialisation. In these first two years all pupils receive tuition in Norwegian, English, social studies, maths and natural sciences/environmental studies, irrespective of their choice as between general or vocational subjects. Pupils choosing a general option enter Advanced course ''Level II'' to prepare them for higher education, while those choosing a vocational option will usually spend another one or two years as an apprentice, at the end of which they will receive a craft or journeyman's certificate or other occupational qualification.

Since the implementation of the reform in August 1994, upper-secondary and associated apprenticeship training has become a statutory right for all students in the 16-19 years age group, with the counties carrying responsibility for seeking contracts with enterprises to supply the required number of apprenticeship places. Enterprises receive state grants to cover the cost of instruction, amounting to NKr 53 000 per apprentice for two years, which is roughly the equivalent of the cost of in-school education. Apprentices receive a salary according to a progressive scale to reflect the accumulation of skills, amounting on average to 50 per cent of the entry wage of a fully qualified worker.[118] Students who have completed vocationally-oriented upper secondary education (foundation course, advanced courses and apprenticeships) with pass marks or a craft or journeyman's certificate, and having also attained a specified level in the subjects of Norwegian, English, social studies, maths and natural sciences/environmental studies, are normally admitted into tertiary education if they wish to pursue this option.

As in all other OECD countries, the tertiary education system in Norway comprises a university and a non-university sector. The four universities and six specialised colleges at university level offer bachelors and masters degrees. Issues concerning degrees, exams, limits on student numbers and normal times of study are decided by the government. Graduate degree studies may be expanded into doctoral study programmes over a three to four-year study period. In addi-

tion to completing a doctoral thesis under contract bound supervision, the candidate has to undergo obligatory training in scientific theory and methodology. Apart from the university sector there are 26 state colleges, offering programmes running from one to three years, although at some institutions longer courses and graduate programmes of four to six years have also been introduced. As part of the reform in higher education in 1994, these colleges have been formed through mergers between 98 regional and/or vocational institutions. Most programmes at colleges are oriented towards specific professions, their graduates becoming professional or para-professional in areas such as teaching in pre and compulsory schools, engineering, social work, administration, economics, electronic based data, health professions, libraries, journalism, etc. Many of the undergraduate programmes offered by the colleges are equivalent to undergraduate degree programmes at universities.

All students in higher education are entitled to financial support via the State Educational Loan Fund. The monthly allowances include a grant and a loan component, with the former being considerably smaller than the latter (Table 21).

Table 21. **Financial support for students**
1995

	Upper-secondary education[1, 2]	Higher education[3]
Basic allowance, per month	NKr 3 010	NKr 3 650
Supplement if not living with parents, per month	NKr 1 900	NKr 1 900
Book allowance, per month	NKr 235	NKr 470
Total allowance, per month	NKr 5 145	NKr 6 020
In US dollars	*US$ 830*	*US$ 970*
of which: **grant**	Dependent upon parental income	24 per cent
Means testing (parental income)	Yes	No
Allowance for home leave, per year	Actual cost of three journeys, minus NKr 810	
Interest rate on loans[4]	6.0 per cent for loans interest bearing during seven years or less, 6.5 per cent for loans interest bearing for more than seven years, 0 per cent in specific cases (see text), no interest accrues until studies are completed	

1. Tested against parental income.
2. The conditions for higher education apply to students aged 19 or older.
3. The conditions for upper secondary education apply to students aged 18 or younger.
4. Situation in 1996.

Source: Royal Ministry of Education, Research and Church Affairs.

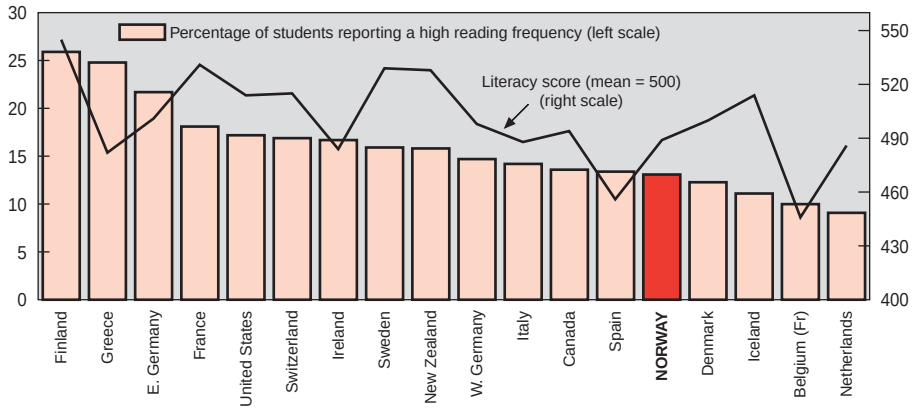
The maximum monthly allowance (excluding book and home leave allowances) amounts to an equivalent of around US$900, which is very generous by international standards, in particular since no tuition fees are levied and no test on parents' means is applied. Students in upper-secondary education are entitled to financial support if 19 years or older or if their parents' incomes are low. Official proposals in 1994 to reduce the grants for students above a certain age limit, in order to contain study durations, were rejected in Parliament. Instead, the Draft 1995 Budget in the autumn of 1994 introduced bonus grants for students completing more than the standard number of terms per year (see below). The maximum repayment period for student loans is 20 years, although the average is around 15 years, with advanced redemption growing. The interest rate on student loans have been reduced in steps from 11.5 per cent in 1989 to 6.0 per cent for loans that are interest bearing for less than seven years and 6.5 per cent for other longer loans, and are income tax deductible. Exemption for interest may be given for economic or social reasons (unemployment, illness, military service, pregnancy, further studies, low income). Moreover, no interest accrues until studies are completed.

The performance of the system

The performance of Norwegian students in international test scores is about average. The Second International Science Study, for example, places Norway slightly above the average when it comes to proficiency in chemistry, biology and physics.[119] At the same time, relatively few students in Norway report a high reading frequency, while the literacy score of 14 year-old Norwegian pupils is also about average – though clearly lower than in the other Nordic countries (Figure 32). Given that resource inputs, as gauged by student/teacher ratios, are higher in Norway than in most other OECD countries, these results are a reason for concern (Table 22).

The increase in participation in non-compulsory education since the late 1980s has implied that Norway ranks high among OECD countries in terms of the numbers graduating from upper-secondary and higher education (Table 23). Indeed, in no OECD country do so many young people complete upper-secondary education as in Norway – almost 96 per cent of the population at theoretical graduation age, compared with 78 per cent on average in the OECD area as a whole. With regard to the number of people acquiring the equivalent of a

Figure 32. **READING FREQUENCY AND LITERACY**



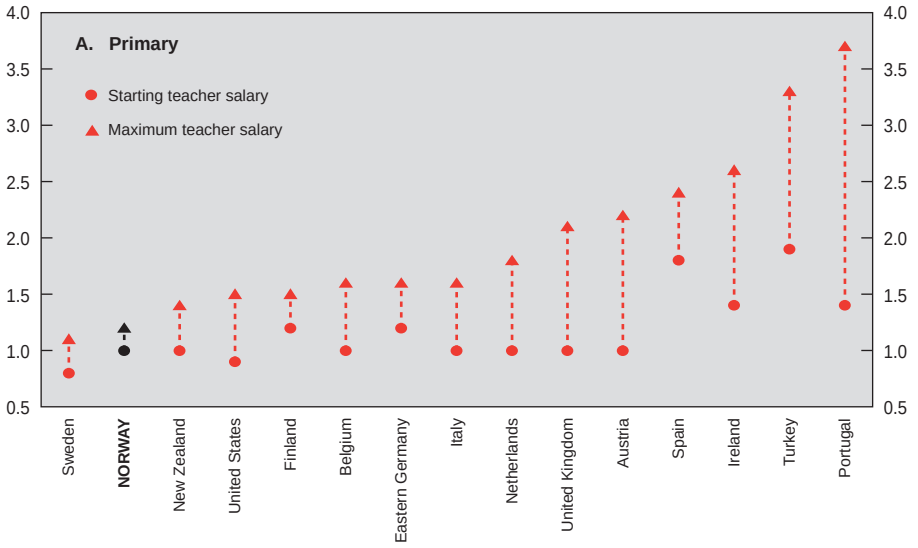
Source: OECD, *Education at a Glance*, 1995.

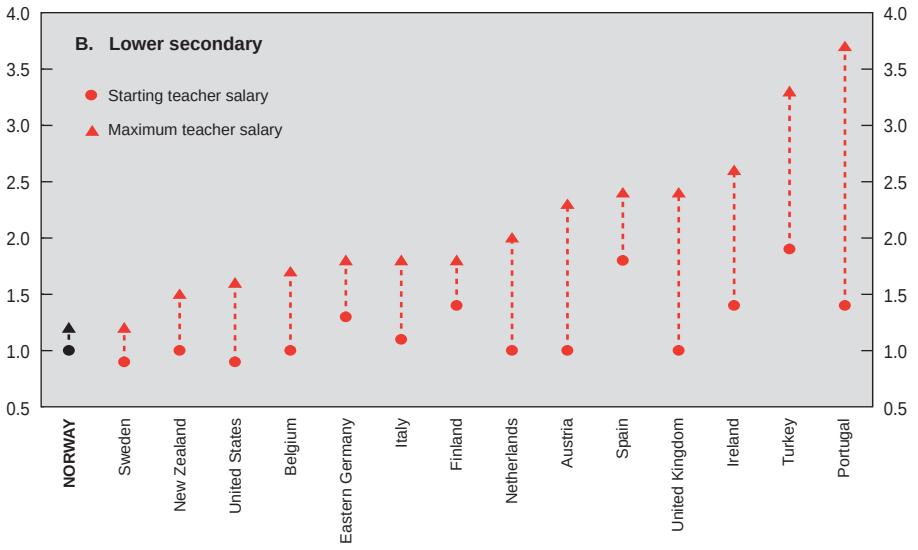
bachelors degree, almost 20 per cent of the relevant age group, Norway ranks third after Denmark and the United Kingdom. Graduation rates at the masters level, in contrast, are somewhat below the OECD average. In Norway university graduation rates are higher for women than those for men.

As noted, a key feature of graduation in upper-secondary education in Norway is the emphasis on school-based vocational education. This is reflected in the comparatively high share of vocational studies, with over 55 per cent of the relevant age group taking a vocational degree – the country ranking fifth in this regard after Germany, Switzerland, Sweden and the Netherlands. In fact, in recent years there has been a substantial excess demand for upper-secondary vocational education, reinforcing the shortage of graduates in this field (see above). This has led to rationing of vocational student places and apprenticeship contracts, prompting students to remain in the system while queuing for a place in the field of study of their first choice. As a result, the effective study duration tended to become excessively long.

Figure 33. **TEACHER COMPENSATION**

Ratio to per capita GDP, 1992





Source: OECD, *Education at a Glance,* 1995.

Table 22. **Resource inputs in education**[1]

1992 figures

	Norway	Other Nordics	OECD Europe	OECD
Education expenditure				
As a percentage of GDP	**6.6**	8.0	5.8	5.7
As a percentage of total public expenditure	**14.0**	12.7	11.4	12.2
Education expenditure per student[2]				
Early childhood education	**41.6**	38.6	22.6	16.0
Primary education	**25.3**	26.6	20.3	20.8
Secondary education	**35.1**	31.7	28.9	26.7
Tertiary education	**49.4**	46.9	45.4	49.0
Student/teaching staff ratios				
Primary education	**10.6**	14.0	17.6	17.4
Secondary education	**8.3**	11.2	13.6	13.8
Teacher salaries[3]				
Primary education	**1.2**	1.3	1.9	1.8
Lower secondary education	**1.2**	1.5	2.0	1.9

1. Public and subsidised private education.
2. As a per cent of per capita GDP.
3. As a ratio to per capita GDP.

Source: OECD, *Education at a Glance*, 1995.

In the course of the 1980s it was already becoming evident to the authorities that the system of vocational upper secondary education would increasingly suffer from the kind of bottlenecks signalled above.[120] As the number of students increased, the absorption capacity of the system clearly became too small, with, as noted, a persisting lack of apprenticeship places. Moreover, students had to specialise too early, giving rise to wrong study choices and high drop-out rates. These considerations led the government to embark upon a far-reaching expansion and streamlining of the upper secondary school system in 1994, as part of the ‘‘Reform 94’’. A key objective of that reform was to facilitate the moving of pupils within the upper secondary education system from the ‘‘foundation courses’’ to advanced vocational courses and to reduce the practice whereby pupils tended to use the system as a ‘‘waiting room’’ for entering such courses. To accomplish this the government established a statutory right for all people aged 16 to 19 years to three years’ upper secondary education, be it general or vocational. The regional authorities (counties) were legally obliged to provide the adequate number of pupil places to honour this right, establish a follow-up

Table 23. **Graduation in upper secondary and university education**

As a per cent of the population at theoretical graduation age, 1992

	Upper secondary education			University education	
	General	Vocational	Total	Bachelor	Masters
Norway	**40.3**	**55.4**	**95.7**	**19.4**	**6.4**
Denmark	45.1	37.1	82.2	22.1	7.9
Finland	43.3	49.5	92.8	6.5	11.8
Sweden	21.8	61.1	82.9	11.4	..
Belgium	34.9	41.2	76.1	..	13.6
France	32.3	45.9	78.2	..	14.5
Germany	24.3	68.3	92.6	..	13
Greece	60.1	24.2	84.3	11.8	0.1
Ireland	76.5	9.1	85.6	17.4	3.5
Italy	21.8	37.1	58.9	0.7	9.8
Netherlands	33.6	56.1	89.7	17.8	8.6
Spain	43.6	19.5	63.1	8	12.1
United Kingdom	63.6	16.6	80.2	20.4	7.2
Switzerland	16.1	66.6	82.7	..	8
Turkey	16.8	12.5	29.3	6	0.4
OECD Europe [1]	38.3	40.0	78.3	12.9	8.3
United States	..	..	75.7	27.4	9.1
Japan	67.4	24.8	92.2	23.4	9.6
OECD [1]	40.0	38.4	78.4	13.4	7.4

1. Unweighted average.

Source: OECD, *Education at a Glance*, 1995.

service for young people who have dropped out and ensure that adults also have an opportunity to obtain upper secondary education.

The progress in implementing the reform and its results are closely monitored by the authorities, with most of the field research out-sourced to a network of seven research institutes. The findings will be available by 1998 or 1999. Preliminary results indicate that the twin aim of full participation of the relevant age group in upper secondary education, with half of them enrolled in the vocational branch, has been practically achieved (96 per cent of the age group currently participates, of which 52 per cent follow vocational courses). However, the first cohort of apprentices under the new system has started only last Autumn, given that ''Reform 94'' was implemented at the beginning of the school year 1994/95. Although the number of apprentice places initially promised by

employers (17 000, or one-third of the relevant age group) seemed to be largely sufficient, of the total of 15 000 applicants only 9 500 were actually placed. Of the 5 500 applicants who have been refused admittance to an apprenticeship, 2 500 have left the system and 3 000 will spend an extra year in school in accordance with their statutory right to upper secondary education. It should be noted, however, that in addition to the apprenticeship placements realised under the new regime, around 7 000 apprenticeship contracts were granted to older applicants who were still in the queue.

The higher education system is also facing bottlenecks in certain disciplines. Although Norway already stands out by its very high graduation rates in medical science (Table 24), reflecting the large share of health-care employment in total employment,[121] further increases in student capacity are needed to accommodate rising demands triggered by recent initiatives to expand the health care system, including the increased number of hospitals beds. Excess demand for student

Table 24. **University degrees by disciplines**

As a per cent of total degrees, 1992

	Medical science	Natural and physical science	Engineering and architecture	Law and business	Human sciences
Norway	**17.2**	**8.7**	**25.9**	**18.3**	**31.3**
Denmark	14.2	8.2	16.6	20.0	40.9
Finland	9.5	15.6	23.9	11.0	40.0
Sweden	17.1	12.0	16.6	23.4	31.0
Belgium	12.8	12.1	23.9	29.3	21.9
Germany	11.7	17.6	22.2	24.7	23.7
Greece	10.5	16.8	13.7	12.4	46.6
Ireland	5.8	19.2	12.4	16.8	45.8
Italy	25.1	12.2	11.4	27.1	22.8
Netherlands	15.6	9.7	16.0	20.5	38.1
Spain	13.4	10.4	9.7	25.3	41.2
United Kingdom	6.8	17.1	15.2	21.8	39.1
Switzerland	15.2	19.1	10.3	29.8	24.8
Turkey	10.4	5.5	17.2	5.6	51.3
OECD Europe [1]	13.2	13.2	26.4	20.4	35.6
United States	7.1	10.3	8.1	27.3	47.3
Japan	5.3	7.3	21.6	39.4	26.3
OECD [1]	11.7	12.7	14.7	22.3	37.7

1. Unweighted average.

Source: OECD, *Education at a Glance*, 1995.

places in humanities, social sciences and law, led to the introduction of student quotas in 1990. Engineering and architecture faculties also apply strict admission regulations (except for the three-year non-university engineering programme), although the former are operating below full capacity due to a lack of applicants qualifying on mathematics and natural sciences. A problem concerning all disciplines to some degree is the very long study durations, which significantly exceed legal norms. This phenomenon may reflect weak labour market conditions for some groups of academics, the relatively generous system of student loans and grants, as well as the large number of students having part-time jobs.

The reform process in higher education has focused so far on three main objectives: *i)* streamlining the institutional structure; *ii)* improving the system of budgeting and cost control; and *iii)* revising the planning of student intake. Concerning the institutional structure, the number of establishments has been drastically reduced in the non-university sector from 98 regional and/or vocational colleges to the present 26 State Colleges, effective from August 1994. In addition, the administrative structure in both the university and non-university sector has been consolidated, with the university or college board empowered with authority regarding budgeting, financial reporting, quality control, legal affairs, staff recruitment, strategic choices and co-ordination of activities with other higher education institutions (''Network Norway''). An infrastructure was set up to monitor the results of this new approach. This was complemented by a pilot project, running from 1992 to 1997, to evaluate the quality of tuition in five selected disciplines nationwide.

As noted, the planning of student admission numbers per institution and discipline remains firmly in the hands of the central education authorities. Moreover, the supply of students has been and will remain the overriding determinant of decisions regarding admission capacity. This has been reiterated in a Bill of 10 May 1996, which stipulates that the largest possible number of qualified applicants (*i.e.* applicants that satisfy the minimum admission requirements) should be admitted, in accordance with ''individual wishes and the country's need for a highly educated work force''. In the same spirit, the Norwegian Council of Universities recommended in November 1994 to lift the limits on student numbers in humanities, social sciences and natural sciences which had been introduced in 1990. This recommendation was adopted by the parliament (Storting) in the Revised National Budget of June 1995. It was also decided that

the authority to reintroduce such limits at the universities should be transferred from the government to the Parliament.

However, labour-market considerations have been introduced in the planning for a number of disciplines. The admission capacity for teacher education has been increased recently in order to meet the extra demand for teaching staff emanating from the reforms in the school sector (*Reform 94* and *Reform 97*; see below). Similarly, the admission capacity for students in the field of health care will be raised to accommodate the extension of hospital capacity in the Oslo and Bergen areas. The admission capacity for future engineers, as well, has been raised in recent years for labour-market reasons, although this has so far led to over-capacity due to, as noted, a lack of qualified applicants. The education authorities do acknowledge the risk of future excess supply of higher educated people in some disciplines (see above), but it remains uncertain to what degree this will affect the planning of admission capacity.

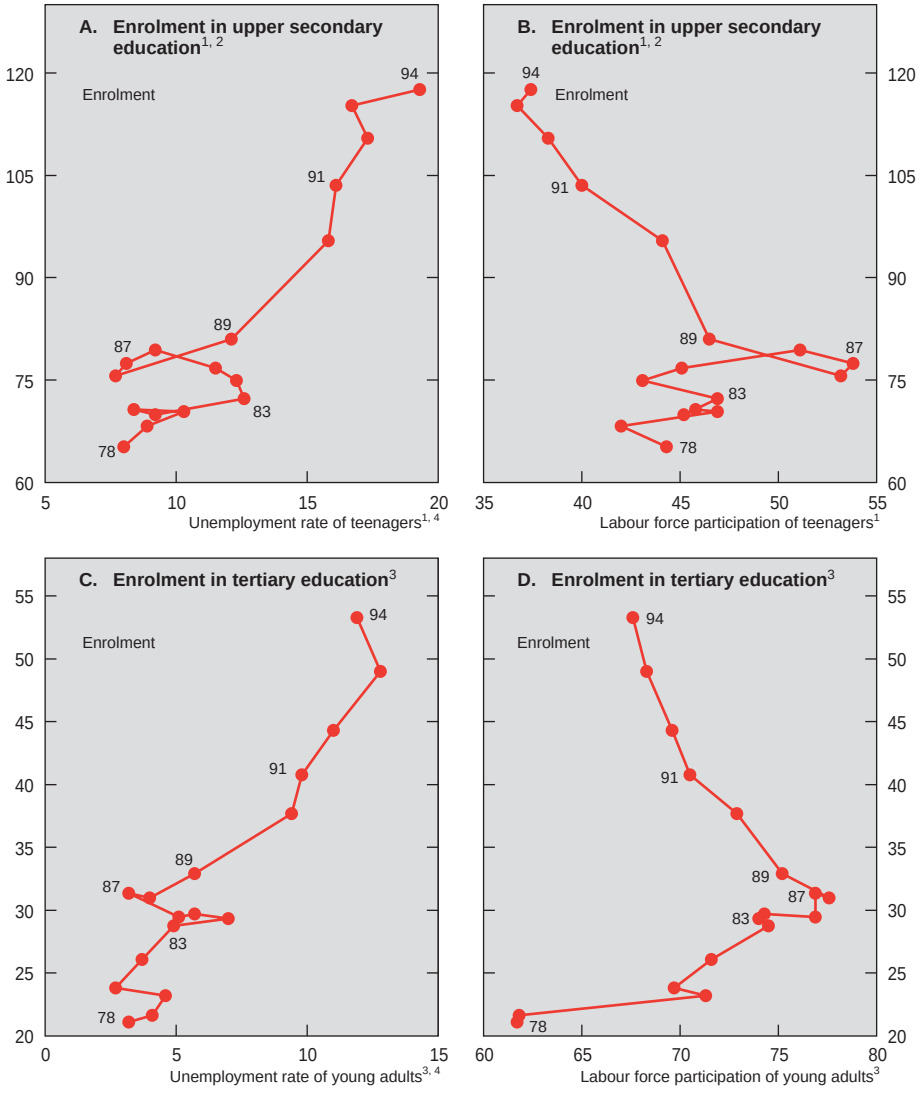
The cost of the system

The increase in youth unemployment during the prolonged recession in the 1988-93 period led to a surge in enrolment numbers in non-compulsory education. This, in turn, resulted in a noticeable decline in the youth labour participation rate (by around 10 percentage points), which implied a radical break with the earlier trend when school enrolment and labour participation rates of young people moved up in concert (Figure 34). The boost in upper-secondary and tertiary education enrolment in the late-1980s and early 1990s was made possible by a policy decision to expand the number of student places in university education, after large numbers of graduates from the local state colleges decided to continue their study in university. This development followed the more than tripling of the unemployment rate of young adults (20-24 years), from 3 per cent in 1987 to 10 per cent in 1989. The sharp increase in the enrolment rate in upper-secondary education occurred as increasing numbers of young adults chose to spend more years in school in order to avoid unemployment and improve their chances on the labour market, after the unemployment rate of teenagers (16-19 years) more than doubled in the years 1987-89.

Hence, while the number of pupils in compulsory education remained on the downward trend observed since the mid-1970s due to slower population growth, the number of students in upper-secondary education climbed from around

Figure 34. **THE TRADE-OFF BETWEEN SCHOOL AND WORK**

Per cent of corresponding age population



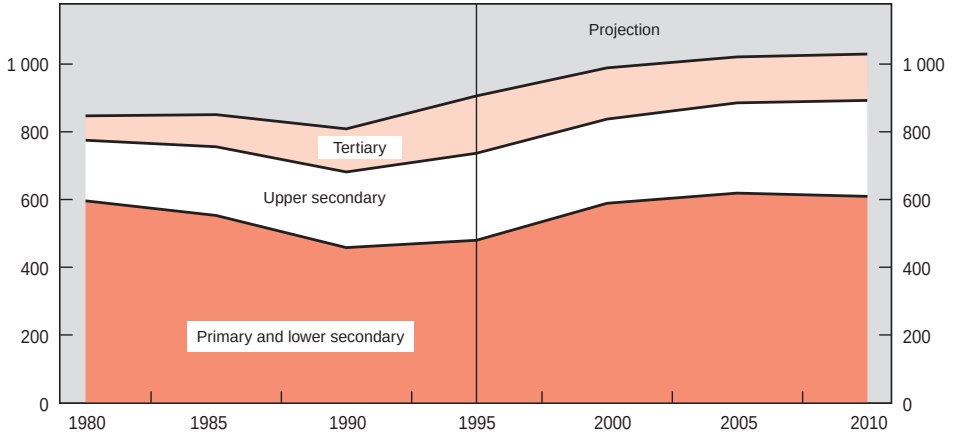
1. Per cent of 16-19 years age group.
2. Including apprenticeships.
3. Per cent of 20-24 years age group.
4. Preceeding year.

Source: Statistics Norway.

200 000 in 1988 to 240 000 in 1994 and in tertiary education from 100 000 to 170 000 (Figure 35). This development is due entirely to increased education participation rates of the relevant age cohorts. For 18 year-olds, for example, the enrolment rates increased from 60 per cent of the age group in 1987 to 85 per cent in 1994. Similarly, participation among the 24 year-olds increased from about 20 to 30 per cent in the same period. The growth in the number of students has been accompanied by a rising number of students receiving financial support from the government, with 159 000 loans and 199 300 grants having been awarded in 1993/94.

Due to the shift in the composition of the student population from compulsory to (more resource-intensive) non-compulsory education, real education expenditure *per student* grew by around 20 per cent in the period 1988-93. However, compulsory education has tended to become more costly over time as well. During the 1980s, the number of teacher full-time equivalents per class in compulsory education has increased from 1½ to 2, while the average class size declined from 19 to 17 pupils.[122] This trend has been particularly pronounced in

Figure 35. **GROWTH IN STUDENT NUMBERS**
Thousand students



Source: Statistics Norway; Ministry of Education, Research and Church Affairs.

remote areas, where falling enrolment has never been matched by an equivalent reduction in the number of classes per school, and the number of schools, due to local resistance and considerations regarding travelling time. In fact, the number of teachers in compulsory education has increased since 1985. The vocational branch of upper-secondary education has experienced similar developments, with the number of students per teacher falling from twelve in 1977 to nine in 1990.[123]

Such developments have resulted in a substantial increase in the overall cost of the system, with real *per capita* government expenditure on education rising by over 40 per cent between 1985 and 1995, almost twice the expansion in real GDP per head recorded over the same period. As a result, the expenditure/GDP ratio has risen from 5¾ per cent in 1985 to almost 7 per cent in 1995 – which is at the high end of the spectrum of OECD countries and significantly exceeding the area average of 5½ per cent.[124] These numbers exclude the increase in the stock of ‘‘bad loans’’ which burdens the Educational Loan Fund. In 1993 the total stock of outstanding loans was estimated at NKr 29 billion (3 per cent of GDP), of which NKr 1.7 billion was considered to be non-recoverable. Reflecting such trends, the Norwegian education system has become relatively expensive by international standards. Even though the overall education expenditure/GDP ratio in 1992 (the latest year for which international comparative data are available) was somewhat lower than the average for the other Nordic countries, expenditure per student appears very high in Norway, in particular in primary and secondary education (Table 22). This is primarily due to the extremely low number of pupils per teacher, as teacher salaries are typically lower than in other OECD countries (see above).

Official projections point to renewed increases in the number of pupils in compulsory education during the period 1995-2010, thus reversing the declining trend in the preceding fifteen years (Figure 35). Though accentuated by a rise in the number of births in the early 1990s, the bulk of the increase is due to the lowering of the compulsory age for entering primary school from seven to six years as of the school year 1997/98. This change, denoted as *Reform 97*, aims to extend the number of years in compulsory school from nine to ten years, which is more in line with the practice in other OECD countries. As a result, public expenditure on education as a share of GDP is likely to climb further in the foreseeable future. According to official calculations, ‘‘Reform 97’’ is estimated to raise current expenditure by NKr 1.7-2.1 billion per year, or the equivalent of

¼ per cent of GDP, while some NKr 4.8 billion worth of investment in equipment and building is required.

At the same time, in order to contain expenditure in higher education, tertiary institutions henceforth receive a ''framework budget'' from the central education authorities, authorising the boards to allocate funds across the various activities – provided the centrally determined number of student places per discipline is realised. The framework budget remains predominantly based upon historical costs, although increases in the annual allocations since 1990 have been determined by changes in the centrally-decided student admission capacity and standard unit cost per discipline. In addition, universities compete for additional ''result-oriented grants'' to be based upon the number of graduations, thus creating a financial incentive for institutions to reduce the excessively long study durations.[125] The central authorities have allocated NKr 200 million for this purpose in the academic year 1996/97. A similar arrangement will be introduced for colleges in the course of 1997. In order to collect the information needed to distribute these funds, the central authorities employ a budgetary reporting system which was created in 1991. The aim to stimulate quick graduation has also led the Ministry to decide to award extra grants to students completing three rather than two terms per year.

The contribution of continuing education and training

Adult education

Norway has an extensive network of adult education programmes, most of which have evolved from initiatives by humanitarian, political, occupational and other co-operatives in the nineteenth century. The broad aim of these programmes is to offer adults equal opportunities to raise their level of education as a vehicle for personal development and the transfer of cultural values, although more recently economic considerations, such as enhancing the productivity of the labour force, have also been put forward. In 1995 more than 750 000 adults attended courses at several levels (including higher education) and of various duration provided by the Adult Education Associations and the ''Folk High Schools'' (Table 25). The courses offered are very diverse and generally do not lead to formal evaluation and certification of skills. While such programmes have

Table 25. **Adult education and training programmes**

	Provider(s)	Funding	Number of participants in 1995	Per cent of working-age population
Adult courses	Adult Education Associations	Central authorities 25 per cent; clients 75 per cent	769 000	*24.5*
Folk High Schools	Folk High Schools	Religious organisations, foundations, counties, clients	6 100	*0.2*
Folk High Schools short courses	Folk High Schools	See above	20 400	*0.6*
Primary and lower-secondary education for adults	Primary and lower-secondary schools	Municipalities	2 900	*0.1*
Upper-secondary education for adults	Upper-secondary schools	Counties	30 400	*1.0*
Distance Education courses	Distance Education Institutions	Central authorities 25 per cent, clients 75 per cent	61 500	*2.0*

Source: Royal Ministry of Education, Research and Church Affairs; Ministry of Finance.

emerged from private initiatives, these are mostly co-financed by the local authorities.

Since the mid-1960s the local authorities have also become directly involved in providing adult education. In particular, the municipalities have a legal responsibility to offer primary and lower-secondary education for adults, in which 2 900 people were enrolled in 1995. Similarly, the counties provide upper-secondary education for adults, with 21 300 persons participating in 1995. Unlike most programmes managed by the Adult Education Associations and Folk High Schools, those implemented by the local authorities result in formal qualifications. The role of such formal adult education is likely to increase with the implementation of ‘‘Reform 94’’, which offers adults the possibility to complete their basic qualifications. Another important development is the establishment of ‘‘resource centres’’ attached to upper-secondary schools, state colleges and universities, which organise continuing education and training in specialised areas. They also offer education leading to formal qualifications. Most distance education (including education by correspondence and through electronic media) is

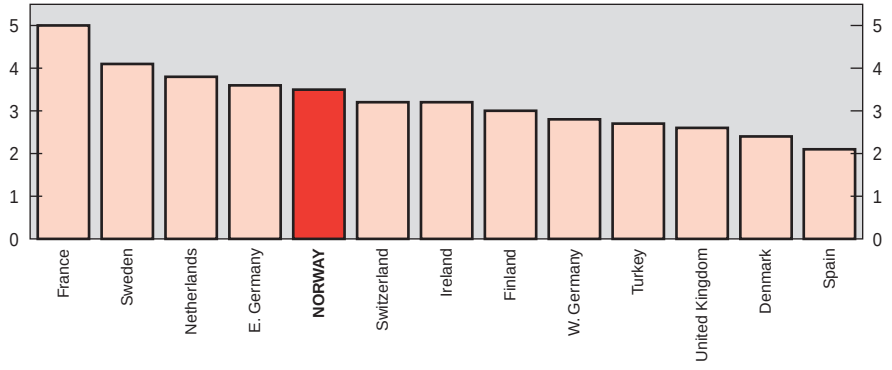
organised by independent distance-education institutions. In 1995 94 000 people participated in such education courses, the number of which is likely to grow with the wider application of new information technology.

Employer-provided training

Although the implementation of job-related training is left to the discretion of individual firms, the social partners play an important role in stimulating such training. In both the private and public sector, the collective wage agreements contain clauses which oblige the employers to document the need for continuing education and develop plans to meet those needs. Current estimates are that about one-third of Norwegian workers (600 000) per year participate in training courses provided by their employers. Norwegian enterprises spend annually an amount of around 3½ per cent of their wage bill on such courses, which is roughly in the middle of the 2 to 5 per cent range estimated for countries in the OECD area (Figure 36). With product-market competition increasing and technological change accelerating, this amount is widely expected to increase in the future, although official projections are not available.

Figure 36. **EXPENDITURE ON JOB-RELATED TRAINING AND EDUCATION IN THE PRIVATE SECTOR**

As a percentage of total wage costs



Source: SNF Bergen.

Such aggregate numbers mask the fact that training expenditure per worker tends to vary considerably across firms. Survey data for Norway suggest that companies employing mostly higher educated workers are likely to spend the largest amounts on training.[126] A *priori*, those firms whose staff is highly unionised are expected to show relatively high spending levels, since training (and associated certification of skills) is likely to raise wage levels. However, the unionisation rate appears to have had little impact on training expenditure in Norway to date. By contrast, the level of corporate profitability has been a significant factor raising training expenditure levels, indicating the importance of cash flow constraints for the amount of money spent on (re-)training in firms. Relieving the cash-flow constraint, for example through introducing a system of collatoralisation of human capital in the firm,[127] could be a useful policy objective. However, so far policies have primarily focused on certifying skills acquired through informal on-the-job training. As stipulated in Paragraph 20 of the Vocational Training Act, adults with sufficient work experience in a certain trade are allowed to register for the same certifying examinations as apprentices without having actually served as an apprentice.[128] In 1992/93 more than 7 000 adult workers obtained such craft and trade certificates, representing almost 45 per cent of the total number of certificates granted in that year.[129]

The impact of job turnover (defined as the sum of gross job losses and job gains) on training expenditure appears to be insignificant. This is an important finding, given that job turnover rates among Norwegian (manufacturing) firms are relatively high and increasing.[130] However, it also indicates that employer-provided training in Norway is not firm specific, which is corroborated by the fact that training received with a previous employer typically has a stronger impact on an individual's earnings profile than training offered by the current employer.[131] Such outcomes are at odds with findings for other countries, such as the United States where company-based training appears to lead to longer job tenure within the firm where the training is received and hence a lower job turnover.[132]

Labour market training

Like most other OECD countries, Norway has programmes for ''Labour Market Training'' (LMT), aimed at maintaining or enhancing the skills of unemployed workers, obtaining a better match between skills needed and offered and

motivating unemployed people to seek further education and training. In 1995, the Norwegian government spent around NKr 2.5 billion (some 0.25 per cent of GDP) on LMT, which is well above the OECD average, but substantially less than in the other Nordic countries – notwithstanding the higher rates of ''open'' unemployment in these countries.[133] All LMT is funded by the central government.

At present there are three programmes of this sort (see also Chapter III):

- The *main LMT programme,* in which 61 000 persons were enrolled in 1995, with on average around 18 000 people per month – or 0.6 per cent of the working-age population – attending courses. The courses mostly aim at providing unemployed workers with vocational qualifications, although the programme also includes general education courses for jobless persons with a weak educational background and job application courses. Most courses take the form of class room teaching in upper secondary schools, in separate centres attached to the schools or in enterprises. The duration of the courses varies between one and 40 weeks, with an average of nine weeks. Participants are selected by the labour market authorities on the basis of their labour market performance (the weakest have priority) or of their potential employability (those who are likely to benefit most from the courses are selected). Participants receive either a training allowance or keep their unemployment benefit.
- *Trainee places* aimed at providing a combination of work experience and training in firms to labour market entrants. The contracts are mostly of short duration (up to six months). The firm may receive a grant covering training expenses. The training allowance received by the trainees is the same as for those participating in the main LMT programmes. In 1995 almost 10 000 people, or 0.3 per cent of the working age population, were enrolled in this programme.
- A *training leave* programme, which allows companies to temporarily send existing staff on training leave, and replace them with unemployed workers who, if necessary, may receive training from the labour market authorities. Employers are granted a monthly subsidy per unemployed hired, while the unemployed receive the standard wage. The compensation level of the staff on training leave is agreed between the employer

and employee concerned. In 1995 on average more than 3 000 workers were participating in this programme.

The experience with LMT programmes is somewhat mixed in Norway, although they seem to be more effective than comparable programmes operated in *e.g.* Sweden. Norwegian studies mostly find a small positive impact of LMT on employment probabilities, of around 10 per cent, with the effects being strongest for training courses delivering qualifications which are in high demand.[134] Moreover, while labour market theory suggests that training programmes carry a risk of generating upward pressure on real wages as they reduce effective labour supply,[135] there is some evidence to suggest that active labour market measures in Norway, on balance, have reduced upward wage pressure.[136] However, now that the labour market is tightening, labour shortages and associated upward wage pressure could easily emerge. This would argue for a downscaling of LMT, except for those courses which are targeted on unemployed workers with low chances of being hired otherwise.

Looking ahead

Assessment of the reforms to date

With ''Reform 94'', upper secondary education has practically completed the transition to universal participation of teenagers, while, as in other OECD countries,[137] reducing the differences in curricula between general and vocational education and limiting the number of vocational programmes. In addition, the facilities for vocational education and apprenticeship training have clearly improved, and the most compelling bottlenecks relieved. However, the question arises as to what extent the progress in upgrading vocational education will be sufficient to achieve an improved matching of demand for and supply of workers with specific vocational skills. The experience since the reform was implemented suggests that imbalances between demand and supply across occupations and skill levels are still considerable and could persist, as relative wages appear not to be very responsive (see Chapter III). Such mismatches could eventually reduce the economic growth potential of the economy and contribute to upward pressure on the overall wage level – the more so since, as discussed earlier, the overall labour market situation is tightening.

The intensified scrutiny of the higher education system – with policies aimed at improving the quality of the system supported by a closer monitoring by the administration and a shift of public management from inputs to outputs, in line with the approach adopted in most OECD countries[138] – should be welcomed. However, the allocation of funds and student places across disciplines and institutions continues to be based primarily on student applications. As a result, there is a risk of the distribution of students across disciplines not corresponding to the needs of the labour market, the more so since, as noted, price signals inherent in flexible relative wages are limited. In fact, the labour market is currently unable to absorb the first wave of graduates in a wide range of disciplines following the sharp increases in student enrolment when the recession began in 1988. Apparently, the large numbers enrolled in higher education in the past six to seven years have masked the real degree of labour market slack in the economy, with the generous system of student grants and loans akin to income replacement for unemployment. As a result, the risk of abuse of such loans and grants by ''hidden'' unemployed workers is significant, thus unduly raising the cost of the system. Moreover, given that participation in higher education requires little or no financial sacrifices on their part, in particular when the labour market is weak, students may lose sight of the investment character of their efforts. Such an environment is not conducive to optimising study choices and could lead to inefficient learning.

Job-related training in Norway, as elsewhere in the OECD area, mostly attracts higher educated workers whose job and earnings prospects are usually relatively favourable to begin with. As a result, those most in need of skill-upgrading are typically deprived of adequate training. Active labour market policies, labour market training in particular, aims to offset this drawback by targeting training courses on unemployed workers with the lowest skills. However, the efficacy of such training is uncertain.

Scope for further action

Due to rapid changes in the industrial structure and technology and the (naturally) limited ability of the labour force to respond immediately to such changes, the match between supply and demand at the level of individual occupations and professions is never perfect. This feature tends to be heightened by the increasing prominence of knowledge-based modes of production and, hence, the

longer time required to endow workers with the appropriate skills and competencies. Norway, moreover, will be confronted with the challenge to convert the mainland economy to become less oil-dependent and revive manufacturing industries that have suffered in the past from the real appreciation of the exchange rate associated with the petroleum wealth. The Norwegian education and training system could certainly help to meet these challenges by providing people the appropriate skills and competences.

The widely-recognised importance of human capital formation as vehicle for ''transmission'' of petroleum wealth to future generations does not argue for increased public expenditure in this area *per se* – the Norwegian education and training system already being relatively costly by international standards – but needs to be reflected in a more rational allocation of resources across the various strands of the system. Indeed, in a situation with rather tight labour market conditions, the rationale for maintaining substantial labour market training programmes is weak. Moreover, given the emerging shortages of labour with vocational qualifications, a shift of resources from higher to vocational upper-secondary education could be justified. However, while necessary, a more rational education and training system is not sufficient to generate the changes that are needed in the labour force. The students themselves need to be motivated to choose the fields of study in line with their talents and ambitions, but taking into account also the employment possibilities and the associated pecuniary rewards. One measure which could serve as a catalyst for establishing a more rational attitude in this regard would be to develop in-house recruitment and counselling services in universities to provide assistance for placements in the private sector.

Finally, the incentives for job related training for lower skilled workers need to be improved in light of the expected changes in production technology and industrial structure and of the fact that investment in human competences is steadily becoming a more significant factor in the economy's competitive success. In line with recommendations by the OECD,[139] lifelong learning will figure high on the political agenda in the years to come, and a debate on how this could be shaped in Norway has started recently. The main federation of labour unions in Norway, the LO, has recently made the proposal, perhaps inspired by a Danish reform, to allow workers to spend one-tenth of their normal working time on education. This proposal has been motivated by the fact that, as noted, job-related

training in Norway mostly attracts higher educated workers; allowing *all* workers to participate in training programmes would thus make the system more equitable. This reform proposal, and any other proposals on this issue (including their funding) which may be put forward by relevant interest groups, will be subject to political decision making after the government has submitted to Parliament, by early 1998, a white paper which will pay specific attention to life-long learning. As yet it is unknown which direction such decision making will take. It should be noted, however, that the introduction of a large scale programme of paid education leave would have to be handled very carefully if the labour market continues to tighten further.

Notes

1. Another risk is implied by a possible dismantling of national monopolies for gas purchase and supply in EU countries and its potentially negative impact on gas prices from the point of view of gas producers.
2. It should be noted, moreover, that the measured rate of change in relative unit labour costs in Norway may be biased upward due to international differences in the method of calculating labour productivity.
3. Dependent upon the index used.
4. A new, wider definition of the quarterly survey of unemployment – including persons unemployed for up to two weeks prior to the survey date and sampled on a weekly basis – has boosted the unemployment rate by around one-half of a percentage point in 1996. For comparability with earlier numbers, a calculated unemployment rate – approximating the older definition – is used.
5. Youth unemployment in 1996 may be biased upwards due to the change in the sampling methodology of the labour market survey, as noted above.
6. Calculations by Statistics Norway indicate that the Solidarity Alternative has been successful in promoting wage moderation in the 1993-1995 period, as indicated by a significant estimated negative residual in the wage equation in their macroeconomic model. See Statistics Norway, *Economic Survey*, 3/96, pp.16-17.
7. *Long-term Programme 1994-97*, Report No. 4 to the Storting, Oslo, January 1993.
8. The No-vote in the Referendum on the EU-accession in October 1994 has not affected this process.
9. The initial budget proposal for 1997 released in October 1996 announced a 5 per cent surplus. This estimate was revised upward, however, reflecting a higher level of oil prices than initially projected (see Table 6).
10. Excluding *i)* a once-and-for-all drop in recorded State expenditure due to the conversion of the State Railways and the Postal Bank in state-owned companies; *ii)* the consolidation of the ''Tax Equalisation Fund'' into the State budget (see below); *iii)* a fall in subsidies to the State Housing and Agricultural Banks due to an increase in their interest margin; and *iv)* a one-off transfer to local governments of municipal power plants privatisation proceeds.
11. This implies as a side effect an increase in recorded State expenditure, which normally excludes the off-budget Tax Equalisation Fund.
12. *Stortingsmelding nr. 35* (1994-95).

13. The annual update of the generational accounts has become a standard procedure in the framework of the preparation of the Budget. See A.J. Auerbach, J. Gokhale, L.J. Kotlikoff and E. Steigum, *Generational Accounting in Norway: is Norway overconsuming its petroleum wealth?*, July 1993 (mimeo).

14. D. Roseveare *et al.*, *Ageing populations, pension systems and government budgets: simulations for 20 OECD countries*, OECD, Economics Department Working Papers, No. 168, 1996.

15. In this respect, too, Norway is unique among OECD economies, where public finances would normally be positively affected by a fall in real interest rates given their net liability position.

16. The starting capital could be determined such as to cover the valuation of accrued pension entitlements, usually denoted as the ''accrued to date'' or ''legal contract'' method. This would yield the smallest possible estimate of the required starting capital. Alternatively, the ''expected job duration'' approach could be adopted, which takes accruals of entitlements in the future into account, based on the survival probabilities underlying observed job durations. The ''implicit lifetime contract'' goes one step further, assuming workers to stay with their present employer and thus accruing full pension entitlements. See for further analysis and documentation: R. Disney and E. Whitehouse, ''What are the occupational pension plan entitlements worth in Britain?'', *Economica*, Vol. 63, No. 150, May 1996, pp. 213-238, and P. J. van den Noord and R. Herd, ''Estimating pension liabilities: a methodological framework'', *OECD Economic Studies*, Vol. 23, Winter 1994, pp. 131-166.

17. A. Boersch-Supan, ''Population aging, social security design, and early retirement'', *Journal of International and Theoretical Economics*, Vol. 148, 1993, pp. 533-557.

18. The operation of funded pension schemes appears indeed to be most effective in economies which would otherwise be susceptible to ''capital flight'', see for example R. Holzmann, *Pension Reform, Financial Market Development and Economic Growth: Preliminary Evidence from Chile*, IMF Working Paper 96/94, August 1996.

19. These additional guidelines were reiterated in the 1997 National Budget released in December 1996.

20. This included foreign currency purchases by the Bank of Norway in anticipation of an estimated allocation to the Petroleum Fund of NKr 45 billion in 1996; see Bank of Norway, *Economic Bulletin*, 1996/3, Oslo, pp. 220. It must be noted that once foreign currency is formally transferred to the Petroleum Fund at the end of the year, the official foreign currency reserves are reduced in the same proportion, as the Fund's assets are technically not part of the official reserves; see L. Eide, *Management of the Government Petroleum Fund and consequences for monetary policy*, Paper presented at the Paris Europlace Seminar, November 1996 (mimeo).

21. Fixed-rate loans by and fixed-rate deposits with the Bank of Norway essentially serve to counter seasonal fluctuations in net liquidity of banks.

22. OECD, *Agricultural policies, markets and trade in OECD countries, Main report*, 1996.

23. Proposition to the Storting No. 8 (1992-93).

24. These reductions were implemented in 1994 and 1995. In 1996, however, prices and budgetary support were slightly increased.

25. ''Agricultural policies, markets and trade in OECD countries, Summary and conclusions,'' OECD 1996.

26. In fact, commercial banks have succeeded to reverse earlier loss provisions and to recover loan losses that previously had been written off.

27. However, two of the three major commercial banks have absorbed companies in the life-insurance and mortgage sector.

28. In line with recommendations made in a White Paper presented in April 1994, the Banking Law Commission presented a Bill in late 1995, entailing the implementation of the EU's Deposit Guarantee Directive. However, the coverage of deposit insurance would be substantially higher than the EU suggests – up to NKr 2 million per depositor instead of ECU 20 000 (or NKr 160 000) according to the EU directive. The Banking Law Commission also proposes to continue the current practice whereby commercial banks and savings banks manage separate guarantee funds.

29. A reason given for this trend is that the scope for flexible working time – including part-time work – is considered greater in the public sector.

30. Direct employment in the offshore sector is small, at around 23 000 people in 1995, which is just over 1 per cent of total employment.

31. In terms of hours worked, public employment accounted for 26½ per cent of total employment, reflecting the large number of women in part-time employment in the public sector.

32. This excludes the approximately 2 per cent of the labour force in agricultural employment, an activity, although formally market based, now close to being fully publicly funded (direct subsidies and import protection).

33. The causality between labour force participation and employment conditions, however, is not obvious over the long run, in as much as employment conditions may also adapt to trend changes in labour supply.

34. The share of females in total employment rose from 31.7 in 1976 to 46 per cent in 1994.

35. The responsiveness of labour force participation in Norway, especially for youth and women, to variation in output has been found to be one of the most pronounced in the OECD area. The responsiveness of the real consumption wage to changes output was, however, found to be very low; see J. Elmeskov and K. Pichelmann, ''Interpreting unemployment: the role of labour force participation'', *OECD Economic Studies*, No. 21, Winter 1993.

36. Participants in some ALMPs, *e.g.* job creation measures, are, however, counted as employed in labour force surveys.

37. E. Hernæs and S. Strøm, ''Heterogeneity and Unemployment Duration'', *Labour*, Vol. 10, No. 2, 1996, pp.269-296. Moreover, another study indicates that the proportion of long-term unemployment also depends on an individuals unemployment history. K. A. Larsen and E. Devold, *Long Term Unemployment and Labour Market Programmes in Norway*, ECON, Center for Economic Analysis AS, 1996 (mimeo).

38. A study found that the introduction in 1992 of a maximum 80-week unemployment benefit period did not lead to an increase in job placements. Rather, when people reached the end of

the period, they dropped from the unemployment register. E. Bratberg and K. Vaage, *Spell Durations with Long Unemployment Insurance Periods*, Department of Economics, University of Bergen, (mimeo, 1996). A possible explanation for this is the fact that an unemployment spell of a long duration reduces the skill level of the persons concerned. The lack of scope for adjustment in the relative wage results in diminished employment prospects for these people. At the same time, they appear to be able to switch to alternative income replacement programmes.

39. Inadequately integrated immigrants, especially those who do not speak Norwegian, are likely to be caught in particularly difficult unemployment situations.

40. Another important reason behind the government's call for wage restraint was the necessity to relieve a profit squeeze experienced by manufacturers associated with weak output prices in the early 1990s.

41. With inflation subdued at a very low level, real wage growth in 1996 exceeded 3 per cent – five times the average annual real wage increase since 1975.

42. *OECD Survey of Japan*, 1996.

43. The existing structural unemployment in Norway is probably due, *inter alia*, to the skills of unemployed persons not matching the requirements of production (see Chapter IV), labour costs exceeding the market-clearing level for some types of labour and the generous unemployment and related welfare benefits system (see S. Scarpetta, ''Assessing the Role of Labour Market Policies and Institutional Settings on Unemployment: A Cross-Country Study'', *OECD Economic Studies*, No. 26, 1996/1).

44. See S. Holden, *The Unemployment Problem – a Norwegian Perspective*, August 1996 (mimeo).

45. In any event, the singular reliance on the use of demand management to prevent hysterises from occurring may be a mis-guided policy for a variety of reasons, but most significantly due to its ''alteration of the incentive structure and of the quality and the properties of the economic system''. W. Kösters and A. Belke, ''Slow adjustment to shocks or true hysteresis? – Lessons from the different labour market performances in western Europe and the USA'', *Intereconomics*, May/June 1996, pp. 107-121.

46. LO has become the principal partner of the government and employers in discussions on incomes policy, due to its ability to co-ordinate actions of its unions and their rank-and-file members, while this ability is less pronounced among the several different public employee unions.

47. The coverage of the central agreement extends to around 95 per cent of public sector, and 60 per cent of private sector employees.

48. In 1996, the wage negotiations at the branch level involved around 300 tariff agreements.

49. Income distribution considerations appear to have a large say in wage claims in Norway. See G. Høgnes and P. Veiden, ''Wage inequalities – a comparison between Austria and Norway'', *Sociologisk Tidssskrift*, 1992.

50. J.M. Dyrstad and K. Johansen, *Regional Wage Responses to Unemployment and Profitability, Empirical Evidence from Norwegian Manufacturing Industries*, 1996 (mimeo).

51. Interestingly, high union density – such as in Norway – may be linked to high youth unemployment, by making youth ''outsiders'' in the wage bargaining process. S. Scarpetta (1996) *op. cit*.

52. H. Torp, *Labour Market Institutions and Development*, Institute for Social Research, Oslo, 1995.

53. Although anecdotal evidence suggests a widening of wage dispersion in the private sector in recent years, it narrowed in the public sector, with the effect that the ratio of highest to lowest earnings decile (D9/D1) in Norway, not only fell after the 1980s, but was in 1993 the lowest in OECD Europe. *OECD Employment Outlook*, 1996.

54. Longva and Strøm: *Income Development 1980-90: growth, differences and mobility*, 1992 (mimeo). For information on earnings mobility in select European countries, see Chapter III in the *OECD Employment Outlook*, 1996.

55. Longva and Strøm (1992), *op. cit*.

56. The Norwegian Youth Unemployment Commission, *NOU* 1994:3 and the OECD.

57. Norwegian Ministry of Labour, *Labour Force Survey*, 1995.

58. *The OECD Jobs Study: Evidence and Explanations, Part II: The Adjustment Potential of the Labour Market*, OECD, 1994.

59. The Ministry of Local Government and Labour has initiated research on the use of fixed-term contracts, with the objective to find out who the main groups are on such contracts, what the working conditions are for such employment and to determine the share of out-sourcing of activity by firms in total fixed-term contracts.

60. The 1993 *Labour Force Survey* found that about 20 per cent of those classified in regular employment were engaged in ''atypical'' work forms, half of which was in shift work.

61. The Parliament has passed a law allowing researchers and artists to negotiate fixed-term contracts in collective agreements, while athletes are given general exemption from the law concerning fixed-term contracts.

62. According to the NHO, the change in hiring practices took place mostly in 1995 and 1996, after firm's legal counsel had completed a review of the new legislation and offered recommendations to employers concerning the proper use of such atypical work forms under the new legislation.

63. In addition, the entry into force of the EEA Agreement made it unnecessary for foreign nationals from other EEA countries to obtain the customary work permit prior to working in Norway. Moreover, new legislation gives a government committee the scope to extend conditions of nationwide collective agreements to foreign (including EEA) workers. However, no such claims have been put forward to the committee yet.

64. EPL in Europe is considered to be more restrictive than in the rest of the OECD area.

65. In practice, firms seek to avoid contentious dismissals, often providing generous dismissal packages in order to forestall court challenges.

66. A. E. Risa, *Objectives and Strategies in the Development of the Norwegian Welfare State*, Department of Economics, University of Bergen, Working Paper No. 0396, February 1996.

67. Moreover, 400 000 persons received health-related cost compensation from the Norwegian Insurance Scheme in 1994. Of the approximately 1 600 000 persons receiving some form of public income support in 1994, around one-tenth received the ''last resort'' social assistance benefit. T. Eardley, J. Bradshaw, J. Ditch, I. Gough and P. Whiteford: *Social Assistance in OECD Countries: Country Reports*, OECD, 1996.

68. Source: Ministry of Finance.

69. The main reasons for this are *i)* the replacement rate (the percentage of labour income replaced by benefits) for elderly disability benefits, which in Norway is one of the highest in the OECD area (64 per cent) and *ii)* the favourable tax treatment of disability benefits. Source: S. Blöndal and M. Pearson: Unemployment and other non-employment benefits, *Oxford Review of Economic Policy*, Vol. 11, No. 1, Spring 1995.

70. Source: *OECD Employment Outlook*, July 1995.

71. The new law governing unemployment benefits has limited the duration of benefits to up to three years. A decision has yet to be made, however, about what type of social assistance will replace this benefit after it is exhausted.

72. Sanctions have been increased for those who refuse suitable employment. Every time an individual rejects a suitable job offer, his access to benefits can be temporarily suspended (eight weeks after the first refusal and twelve weeks after the second); after three refusals within a period of twelve months, benefits can be suspended for six months. Previously benefits could be suspended for a maximum of twelve weeks.

73. *OECD Economic Survey of Norway*, 1994.

74. J. Kettunen, ''Duration-dependent features of unemployment insurance'', *Economic Letters* 51, 1966, pp. 115-121.

75. The maximum benefit duration for the partially unemployed (those working involuntarily reduced hours) is the same as for the fully unemployed and, in order to qualify for benefit, one must be at least 40 per cent unemployed.

76. J. P. Martin, ''Measures of Replacement Rates for the Purpose of International Comparisons: A Note'', *OECD Economic Studies*, No. 26, 1996/1.

77. Means-tested supplements are most pronounced in the ''last resort'' Social Assistance Programme. However, this programme may interact with other programmes, in cases where the benefits are deemed to be insufficient.

78. From 1993 the regionally differentiated social security contribution from employers has been reduced for all regions with non-zero rates (excluding Finnmark and a few other regions already with a zero rate).

79. It should be noted that if indirect taxes were included – as is standard in measures of the tax wedge – the marginal tax wedge at the high end rises to 63 per cent.

80. The reduction in the high-end marginal tax wedge from around 80 to about 55 per cent was accomplished through decreasing the progressivity of tax rates on earnings, while the reduction at the low-end marginal tax wedge, from around 37 to about 30 per cent, was obtained through lowered social security contributions.

81. The cost of a marginal public expenditure programme has been estimated at 150 per cent of the initial spending outlay due to the production and income losses associated with the

additional taxes required to finance such a programme (see OECD *Economic Survey of Norway*, 1995).

82. Above this income level, regular tax rules apply.

83. Depending on the nature of their vocational problem – whether it be physical, mental or social – occupationally handicapped persons participate in a variety of programmes suited to their special situation. See *Norwegian Labour Market Policy* 1996, Ministry of Local Government and Labour, Oslo, 1995.

84. Vocational training for the occupationally handicapped has risen steadily from around 10 000 in 1990 to over 20 000 in 1996. This is due to the integration of two such programmes in 1994, and increased effort in this area.

85. *OECD Employment Outlook*, (1993) and P.R.G. Layard, S.J. Nickell and R. Jackman, *Unemployment, Macroeconomic Performance and the Labour Market*, Oxford University Press, 1991, Oxford.

86. The experience with labour market training has been mixed, especially in the Oslo region, where 15 per cent of the population is of immigrant origin and characterised by high levels of unemployment. One reason is that many immigrant LMT participants are still mostly foreign speaking. Another reason is that Oslo appears to accumulate ''difficult groups'' who seem not to be very interested in finding work after training is completed.

87. O. Raaum, H. Thorp and H. Goldstein, *Effects of Labour Market Training – A multinomial Analysis*. Working Paper 95:9, Center for Research in Economics and Business Administration, Oslo 1995.

88. *Review of Public Employment Services in select OECD countries*, OECD, 1995.

89. D. Pilat, ''Labour productivity levels in OECD Countries: Estimates for Manufacturing and Selected Service Sectors'', *OECD Working Papers* No. 169, Paris, 1996.

90. A study indicates a ''pricing-to-market'' determination of import prices in Norway, such that improved competition in domestic markets could also lead to lower import prices. Bjorn Naugh and Ragnar Nymoen, ''Pricing to market in a small open economy'', *Scandinavian Journal of Economics*, 98(3), 329-350, 1996.

91. J. O. Martins, S. Scarpetta and D. Pilat, ''Markup-up Ratios in Manufacturing Industries'', *OECD Working Papers* No. 162, Paris, 1996.

92. P. Geroski, P. Gregg and J. Van Reenen, ''Market Imperfections and Employment'', *OECD Economic Studies*, No. 26, 1991/1.

93. *Competition Policy in the OECD Countries* 1992-1993, OECD, Paris, 1995.

94. From 1997, foreign airlines will be given access to the domestic market, complementing the EEA-agreement which requires that international flights within the EEA-area be opened to all airlines as of 1 July 1994.

95. Statenett SF (state owned) and local electricity distribution companies are natural monopolies with price regulation, while free competition has been promoted in the production of electricity.

96. The EEA agreement allows for competition in the wholesale market and parallel import of drugs. In addition, several measures have been implemented to promote competition in the market for the supply of drugs.

97. Moreover, the import-weighted average tariff rate in Norway remained constant around 4.3 per cent between 1988 and 1993. *Indicators of Tariff and Non-Tariff Trade Barriers*, OECD, Paris, 1996.

98. *Agricultural Policies, Markets and Trade in OECD countries*, OECD, 1996, Paris.

99. If price protection is included with direct support, the support per farmer averaged NKr 220 000 in 1995. (Source: Ministry of Agriculture.) Total transfers per farmer equivalent, including all transfers from taxpayers and consumers resulting from agricultural policies, less budget revenues from tariffs on imports, amounted to NKr 283 000 ($43 500) in 1995; see OECD, *Agricultural policies, markets and trade in OECD countries, Main Report*, 1990.

100. Source: Ministry of Finance, 1996.

101. The OECD Jobs Strategy: *Technology, Productivity and Job Creation*, Vol. 2, *Analytical Report*, Paris, 1996.

102. See for example N.G. Mankin, D. Romer and D.N. Weil, ''A contribution to the empirics of economic growth'', *The Quarterly Journal of Economics*, May 1992, pp.407-427.

103. D. O'Neill, ''Education and income growth: implications for cross-country inequality'', *Journal of Political Economy*, Vol. 103, No. 6, December 1995.

104. See 1994 *Economic Survey*.

105. R. Asplund *et al.*, ''Human capital and earnings in the Nordic countries'', in R. Asplund (Ed.), *Human capital creation in an economic perspective*, Physica-Verlag, 1994, pp.68-94.

106. G. Høgsnes, ''Wage determination and the male-female wage gap: the case of Norway'', *Comparative Social Research*, Vol. 15, 1995, pp.167-186.

107. OECD, *Education and training – Learning and working in a society in flux*, Paris, 1996.

108. Royal Norwegian Ministry of Finance, *Put Europe to work: elements in a strategy for increased growth, competitiveness and employment*, Oslo, 1994.

109. OECD, *Employment Outlook*, Paris, 1996.

110. P. Schøne, *On the job training and the effect on earnings for young workers in Norway, Institute for Social Research*, Oslo, 1995 (mimeo).

111. E. Barth and H. Mehlum, ''Lønnsforskjeller i Norge 1980-1991'', *Søkelys på arbeid-smarkedet*, Vol. 10, 1993, pp. 147-152.

112. K. A Larsen, *Rekrutteringsundersøkelsen 1995*, ECON, Oslo, 1996.

113. Å. Cappelen and N. M. Stølen, ''Forecasting labour market imbalances'', *Economic Survey 4/94*, Statistics Norway, 1994, pp.21-29 and K.A. Larsen, *Kommunenes arbeidskraft og kompetansebehov 1992-2007*, ECON report 109/94, Oslo, 1994.

114. The share of high-tech products in manufacturing value added and exports amounted in 1992 to 9.4 and 10.3 per cent, respectively, while such shares in most OECD countries are roughly

twice as high. See *The OECD Jobs Study: Evidence and Explanations*, Part I, Table 4.12, p.149.

115. In particular teacher education, nursing and engineering.

116. Since 1986 no-longer earmarked but part of the total lump sum. Moreover, since 1992 the administrative structures of municipal and county schools can be fixed by the local authorities.

117. The compulsory school age has been lowered from 7 to 6 years with effect from the school year 1997-98.

118. Average of 30, 40, 50 and 80 per cent in, respectively, the first, second, third and the fourth six-month period of the apprenticeship.

119. J. Bishop, ''Schooling, learning and worker productivity'', in R. Asplund (ed.), *Human capital creation in an economic perspective*, Physica Verlag, 1994, pp.13-67.

120. O. Briseid, ''Comprehensive reform in upper secondary education in Norway: a retrospective view'', *European Journal of Education*, Vol. 30, No. 3, 1995.

121. H.O. Oxley and M. MacFarlan, ''Health care reform: controlling spending and increasing efficiency'', OECD *Economic Studies* No. 24, 1995, pp. 7-55.

122. R. Sørensen, *Kommunes prioriteringer av grunnskolen. Laererkapasitet og klassestørrelse 1980-92*, Norwegian School of Management, Working Paper 1994/5.

123. Falch and Rattsø, *The politics of resource use in high schools: investigating the variation between Norwegian counties*, Notat 1995, University of Trondheim, Department of Economics.

124. Figure for 1992. Source: OECD, *Education at a Glance*, Paris, 1995.

125. Institutions receive an amount of NKr 8 000 for every male and NKr 12 000 for every female graduate.

126. T. Gulbrandsen, ''Etterutdanning i private bedrifter'', *Sociologisk tidsskrift*, No. 2, 1996.

127. OECD, *Measuring what people know – Human capital accounting for the knowledge economy*, Paris, 1996.

128. A worker is eligible if he has held a relevant job for a period 25 per cent longer than the amount of time usually served by apprentices.

129. T. Skinningsrud, *Coherence between compulsory education, initial and continuing training and adult education in Norway*, NORUT Social Research, Tromsø, 1995.

130. OECD, *Employment Outlook*, Paris, 1996 and K.G. Salvanes, *Job creation and job destruction in Norway, 1996-1992*, Oslo, March 1996.

131. P. Schøne, *op. cit.*

132. C. Chang and Y. Wang, ''A framework for understanding differences in labor turnover and human capital investment'', *Journal of Economic Behavior & Organization*, 28, 1995, 91-105.

133. For comparison, the share of LMT expenditure in GDP in 1995 amounted to 1 per cent in Denmark, 0.78 per cent in Sweden and 0.43 per cent in Finland, see OECD, *Employment Outlook*, Paris, July 1996.

134. O. Raaum, H. Torp and H. Goldstein, *Employment effects of labour market training in Norway*, Memo 95:8, Department of Economics, University of Oslo, 1995.

135. L. Calmfors, ''Active labour market policy and unemployment – A framework for the analysis of crucial design features'', *OECD Economic Studies*, No. 22, 1994, pp. 7-47.

136. O. Raaum and F. Wulfsberg, *Unemployment, labour market programmes and wages in Norway*, SNF-Oslo, (mimeo), 1996.

137. OECD, *Evaluating and reforming education systems*, Paris, 1996.

138. D. Kallen, ''New educational paradigms and new evaluation policies'', in OECD, *Evaluating and reforming education systems*, Paris, 1996, pp. 7-23

139. OECD, *Lifelong learning for all, Meeting of the Education Committee at Ministerial Level*, Paris, 16-17 January 1996.

Annex I

Calendar of main economic events

1995

January

The EFTA's surveillance body ESA declares the retail monopoly for wine and spirits in Norway as a violation of the EEA Agreement's prohibition of quantitative trade restrictions and provision in monopolies.

The Gas Negotiation Committee signs an agreement with Total and Gaz de France concerning the sale of natural gas to France. The agreement covers deliveries of 40 million tons of oil equivalents in 26 years, starting in 2001.

Parliament rejects government proposals on a change in the so-called split model for income taxation of self-employed.

February

The Ministry of Industry and Energy announces 56 blocks on the NCS to be open for exploration in the framework of the 15th licensing round.

The government decides to abolish the State import monopoly for wine and spirits, although it would maintain the retail monopoly of these products.

March

The government endorses the EU's licensing directive for oil activities, implying that oil and gas are henceforth included under the provisions of the EEA Agreement.

May

The Revised 1995 Budget is presented to Parliament. Based upon an assumed oil price of NKr 115 per barrel, the State budget deficit is expected to reach NKr 6 billion in 1995, compared with a projected NKr 11 billion in the December Budget Bill.

June

The Parliament approves the EU's licensing directive for oil activities.

July

Norway concludes an agreement with the EU ensuring duty-free import and export quotas between Norway and the EU area for processed agricultural goods, in line with the average imports and exports per product category between Norway and Sweden, Finland and Austria in the 1991-93 period.

October

The government presents the draft National Budget for 1996. Based upon an oil price assumption of NKr 105 per barrel, the State budget surplus for 1996 is projected at NKr 10.6 billion; excluding oil-related revenues and expenditures the budget balance shows a deficit which is expected to reach NKr 38.7 billion. The State surplus is to be transferred to the Government Petroleum Fund.

November

The Parliament endorses the privatisation of the railway company Ofotbanen, with the Norwegian and Swedish State Railways each participating in 24.5 per cent of the share capital of the company, renamed Malmtrafikk AS.

As an agreement on the OECD shipbuilding accord was not reached, the government decides to extend its shipyard support scheme into 1996.

The Parliament adopts a new Securities Trading Act, in force as of 1 January 1996, entailing changes in capital adequacy rules for securities-trade companies.

The Norwegian Competition Authority demands that school and text books be exempt from new market regulation for books.

December

The government presents the final National Budget Bill for 1996, with the State surplus and transfer to the Petroleum Fund revised upward from NKr 10.6 billion to NKr 12.6 billion.

The State sells shares in Christiania Bank (Kreditkassen), reducing its stake to 51 per cent of the bank's share capital.

1996

January

Allocations in the 15th licensing round for the Norwegian Continental Shelf are announced by the Ministry of Industry and Energy, with the Norwegian companies Statoil, Norsk Hydro and Saga Petroleum being offered operator responsibility for eight out of a total of eighteen joint licenses granted.

March

Norway and the EU reach an agreement on the tariff rates to be applied to processed agricultural products.

The Bank of Norway cuts its official deposit and overnight lending rates by 0.25 per cent to 4.5 and 6.5 per cent, respectively.

April

The central government accounts for 1995 show a surplus of NKr 4.1 billion, of which NKr 2.1 billion is transferred to the Government Petroleum Fund.

A breakdown of wage negotiations is followed by a strike in the hotel and restaurant industry.

May

The Revised 1996 Budget is presented to Parliament. Based upon an assumed oil price of NKr 115 per barrel, the State Budget surplus is expected to reach NKr 24.9 billion in 1996, compared with a projected NKr 12.6 billion in the December Budget Bill. The entire surplus is to be transferred to the Government Petroleum Fund.

The annual pay increase for public sector workers is fixed at 4 per cent. An agreement in the engineering industry, following industrial actions, entail a reduction in the retirement age to 62 and a pay increase of NKr 1.5 per hour. In the textile industry a pay increase of NKr 3 per hour is agreed.

June

A strike organised by the Norwegian Oil and Petrochemical Trade Union affects exploration activity and production drilling on the Norwegian Continental Shelf. The National Union of Electricians and Power Station Workers calls a strike on demands for paid leave for further education.

Iceland, Russian and Norway break off negotiations on uncontrolled fishing in the ‘‘loophole’’ area.

July

The Bank of Norway introduces a repo arrangement for government bonds and Treasury bills to supply liquidity to the money market.

The Gas Negotiation Committee and Ruhrgas sign a contract for natural gas deliveries for the 2000-25 period.

August

In a special meeting with the European Commission, the Norwegian government declines allegations by Scottish salmon farmers of subsidised dumping of farmed salmon.

October

The government presents the draft National Budget for 1997. Based upon an oil price assumption of NKr 115 per barrel, the State budget surplus for 1997 is projected at NKr 40.9 billion; excluding oil-related revenues and expenditures the budget balance shows a deficit which is expected to reach NKr 24.2 billion. The State surplus is to be transferred to the Government Petroleum Fund.

November

The central bank cuts its deposit and ''overnight'' lending rates by ½ percentage point to 4 and 6 per cent, respectively, to reduce appreciation pressure on the Norwegian krone.

The parliament approves the conversion of the Post Office and the Norwegian State Railways from public corporations to state-owned companies.

December

The National Budget for 1997 is adopted. Based upon an oil price of NKr 125 per barrel, the State budget surplus is expected to be NKr 54.6 per billion, which is to be transferred to the Government Petroleum Fund.

1997

January

The central bank cuts its deposit and ''overnight'' lending rates in two steps by ¾ percentage point to 3¼ and 5¼ per cent, respectively, to reduce appreciation pressure on the Norwegian krone.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	Average 1986-95	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
A. Percentage changes											
Private consumption[1]	1.4	5.6	–1.0	–2.8	–0.6	0.7	1.4	2.2	2.1	4.1	2.6
Government consumption[1]	2.4	2.2	4.0	0.5	2.2	4.1	3.9	5.5	1.0	0.7	0.2
Gross fixed capital formation [1]	0.7	23.9	–2.1	1.6	–8.8	–11.9	–1.3	–3.3	1.5	6.9	4.5
Residential[1]	–3.8	9.6	4.0	–3.8	–17.0	–17.1	–27.3	–12.6	–1.4	34.9	12.7
Oil sector[1]	2.4	62.4	–18.5	–4.2	25.0	–62.7	88.4	31.3	7.7	–7.3	–13.1
Ships and pipelines[1]	–14.6	67.8	–61.9	–448.9	48.3	–62.0	–45.0	–111.5	119.8	–30.5	–30.3
Other private business sector [1]	2.4	9.4	17.8	44.2	–7.0	–7.0	–18.3	–27.1	1.8	19.1	19.3
Public[1]	3.1	15.0	10.0	7.4	0.8	–5.5	9.6	3.8	–10.5	8.6	–0.5
Total domestic demand[1]	1.6	15.4	–5.6	–2.6	–2.4	–0.7	0.5	1.5	2.8	4.4	3.7
Exports of goods and services [1]	5.2	1.6	1.2	5.5	10.7	8.6	6.1	5.2	2.0	8.2	3.8
Imports of goods and services [1]	2.2	9.9	–7.3	–1.7	2.2	2.5	0.2	0.7	4.0	6.9	5.1
GDP[1]	2.4	4.2	2.0	–0.5	0.3	1.6	2.9	3.4	2.1	5.0	3.3
Mainland GDP[1]	1.3	3.4	1.2	–1.7	–2.3	0.4	1.1	2.2	1.7	4.3	2.7
GDP price deflator	3.0	–1.4	7.2	4.4	6.3	4.2	2.7	–0.5	2.6	0.6	3.1
Industrial production	1.0	1.1	1.0	–1.2	0.2	0.1	–1.6	1.2	2.1	5.6	3.6
Manufacturing	78.9	1.4	0.8	–2.0	0.4	0.4	–1.7	1.4	2.5	5.4	2.9
Employment	0.3	3.5	1.9	–0.6	–3.0	–0.9	–1.0	–0.3	0	1.5	2.0
Compensation of employees (current prices)	5.5	13.2	12.9	5.9	1.3	4.1	4.2	3.4	1.2	2.7	3.0
Productivity (real GDP/employment)	2.1	0.6	0.1	0.1	3.5	2.5	3.9	3.7	2.1	4.1	2.1
Unit labour costs (compensation/real GDP)	3.0	8.7	10.7	6.4	1.0	2.5	1.3	0	–0.8	–0.1	1.3
B. Percentage ratios											
Gross fixed capital formation as per cent of GDP at constant prices	23.4	28.7	27.6	28.2	25.6	22.2	21.3	19.9	19.8	20.7	21.0
Stockbuilding as per cent of GDP at constant prices	2.3	8.6	2.5	0.8	0.7	1.8	0.9	0.7	1.7	1.6	2.7
Foreign balance as per cent of GDP at constant prices	2.6	–6.4	–3.4	–1.2	1.4	3.4	5.4	6.9	6.3	7.2	6.8
Compensation of employees as per cent of GDP at current prices	47.1	48.1	49.7	50.6	48.1	47.3	46.7	46.9	45.3	44.4	43.7
Direct taxes as per cent of household income	16.7	15.2	16.1	17.5	17.7	17.3	16.8	15.7	16.3	13.5	13.7
Household saving as per cent of disposable income	1.3	–6.1	–6.2	–2.4	0.9	0.9	2.9	5.4	5.6	7.2	7.0
Unemployment rate	4.5	2.0	2.1	3.2	4.9	5.2	5.5	5.9	6.0	5.4	4.9
C. Other indicator											
Current balance (billion dollars)	1.0	–4.5	–4.0	–4.0	–0.1	2.9	3.9	4.5	3.5	3.0	4.5
Excluding shipping and oil platforms	1.6	–11.9	–8.9	4.8	19.2	9.7	1.7	–2.1	1.4	1.5	0.8

1. At constant prices.
Source: Central Bureau of Statistics; OECD estimates.

Table B. **Supply and use of ressources**

Kr million, current prices

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Private consumption	294 985	315 158	325 167	338 778	357 100	376 275	394 949	411 637	434 798	457 138
Government consumption	108 657	124 130	130 754	139 016	149 465	161 326	172 945	179 575	185 206	191 973
Gross fixed investment	155 203	166 586	181 428	175 057	156 210	157 426	156 340	168 202	183 558	197 663
Stockbuilding	21 860	20 062	5 616	4 297	11 903	6 068	5 774	9 634	13 445	23 401
Total domestic demand	580 705	625 935	642 965	657 148	674 678	701 095	730 008	769 048	817 007	870 175
Exports	195 021	200 433	213 858	262 658	293 752	308 046	300 094	315 960	334 837	355 041
Imports	212 470	210 746	217 232	237 459	246 359	246 367	245 806	261 669	282 104	299 352
GDP at market prices	563 256	615 622	639 591	682 347	722 071	762 774	784 296	823 339	869 740	925 864
Indirect taxes	99 922	107 493	106 704	107 049	110 849	116 724	122 212	130 332	. .	. .
Subsidies	29 569	31 515	28 035	30 001	32 565	34 176	34 914	36 322	. .	. .
GDP at factor costs	492 903	539 644	560 922	605 299	643 787	680 226	696 998	729 329	763 799	809 867
Depreciation and other operating provisions	. .	. .	. .	. .	. .	. .	. .	. .	. .	. .
Net domestic product at factor costs	. .	. .	. .	. .	. .	. .	. .	. .	. .	. .

Source: Norwegian National Accounts.

Table C. **Gross domestic product by origin**

Kr million, current prices

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Agriculture, forestry and fishing	17 760	19 616	19 609	19 475	22 537	22 229	20 386	21 544	22 199	22 028
Crude petroleum and natural gas production	51 373	52 172	44 737	70 312	87 606	90 386	90 416	95 761	96 282	100 494
Pipeline transport	5 678	6 003	5 982	7 155	7 983	10 020	10 939	11 017	12 591	13 463
Mining and quarrying	1 569	1 644	1 573	1 870	1 738	1 707	1 839	1 504	1 533	1 532
Manufacturing	70 133	78 421	82 978	84 609	83 777	85 838	88 328	94 667	102 509	116 609
Electricity, gas and water supply	16 442	18 006	19 747	20 966	22 329	22 741	22 079	21 642	20 450	23 939
Construction	30 981	38 220	40 463	34 120	30 751	27 501	27 971	25 764	28 605	33 897
Wholesale and retail trade	62 028	66 790	69 403	69 663	73 641	75 602	78 309	78 265	83 487	88 735
Hotels and restaurants	9 479	11 110	10 879	10 307	9 977	10 420	10 270	10 406	11 700	11 683
Ocean transport and drilling	11 694	8 604	11 316	16 430	18 266	23 404	17 815	21 053	18 882	18 528
Other transports and communications	32 411	36 010	40 443	42 634	45 034	45 525	47 453	51 867	54 415	58 266
Financing and insurance	29 063	35 546	35 406	36 862	35 741	35 574	36 905	38 640	38 622	36 717
Business services	29 004	32 799	35 667	35 619	35 810	37 417	39 275	43 159	46 239	47 866
Dwelling services	32 576	36 021	41 923	47 828	53 359	55 878	58 960	61 604	62 277	63 399
Other private services	27 712	30 573	32 931	33 195	35 944	38 205	40 771	41 906	43 642	45 442
Producers of government services	81 676	92 793	99 621	106 027	112 837	121 273	128 836	133 086	138 436	144 418
GDP at market prices	570 645	623 699	647 911	690 239	733 744	762 775	784 296	823 338	869 743	925 865

Source: Norwegian National Accounts.

Table D. **Gross domestic product by origin**

Volume, 1986 = 100

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Agriculture, forestry and fishing	100.0	104.7	105.8	108.9	121.4	122.5	114.9	133.0	135.1	145.3
Crude petroleum and natural gas production	100.0	113.3	127.0	161.0	165.8	189.2	213.4	224.2	251.2	269.2
Pipeline transport	100.0	112.6	109.6	130.2	128.6	141.4	159.2	153.8	172.3	187.1
Mining and quarrying	100.0	100.2	87.8	97.6	95.4	91.0	94.7	79.8	84.3	86.5
Manufacturing	100.0	101.9	97.0	95.5	96.1	94.1	97.0	99.4	104.7	107.8
Electricity, gas and water supply	100.0	106.5	113.3	124.8	125.3	113.0	119.5	122.3	115.5	125.5
Construction	100.0	103.7	102.7	94.4	90.6	85.7	85.9	79.8	84.7	93.0
Wholesale and retail trade	100.0	98.0	92.7	90.8	91.3	90.3	93.1	92.6	99.8	103.8
Hotels and restaurants	100.0	103.5	94.8	83.9	78.0	79.6	71.3	66.3	74.0	73.6
Water transport	100.0	79.0	75.9	90.4	105.1	109.1	103.5	102.4	95.3	95.6
Other transports and communications	100.0	103.4	110.6	110.7	117.4	122.7	125.6	139.5	153.8	164.7
Financing and insurance	100.0	106.7	109.9	101.8	97.8	91.6	89.7	87.4	93.2	87.8
Business services	100.0	104.2	103.7	98.8	92.1	91.0	93.8	100.5	105.1	105.3
Dwelling services	100.0	105.0	111.3	117.4	120.9	119.4	120.7	122.1	123.3	124.6
Other private services	100.0	100.4	101.3	97.8	100.5	101.4	102.7	102.7	102.1	102.0
Producers of government services	100.0	102.8	105.0	106.9	109.7	113.2	117.6	119.6	121.0	121.6
GDP at market prices	100.0	102.0	101.5	102.1	103.8	105.4	108.9	111.9	117.5	121.4

Source: Norwegian National Accounts.

Table E. **General government income and expenditure**

Kr million

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Current receipts	281 097	309 816	333 901	348 185	378 001	387 840	395 058	409 507	435 405	469 547
Indirect taxes	99 922	107 493	106 704	107 049	110 849	116 724	122 212	131 031	141 729	153 131
Social security contributions	67 460	79 718	79 456	76 147	79 434	83 340	87 334	83 134	87 372	90 794
Direct taxes	79 309	87 951	94 243	105 080	116 256	117 638	117 360	125 150	135 329	145 987
Capital income	30 855	33 463	44 887	47 898	51 226	53 135	52 676	50 832	46 314	47 800
Other current receipts	3 551	1 191	8 611	12 011	20 236	17 003	15 476	19 360	24 661	31 835
Current expenditure	239 946	268 976	298 830	317 377	342 190	366 355	387 214	404 279	415 220	423 989
Expenditure on goods and services	108 657	124 130	130 754	139 016	149 465	161 326	172 947	179 483	184 663	189 683
Subsidies	29 569	31 515	28 035	30 001	32 565	34 176	34 914	37 663	38 297	35 033
Interest paid	22 325	24 061	22 670	24 705	25 805	24 836	25 648	27 800	27 293	26 701
Current transfers	4 947	5 283	24 664	18 638	19 101	21 062	19 820	19 886	22 343	21 263
Net saving	41 151	40 840	35 071	30 808	35 811	21 485	7 844	5 228	20 185	45 558
Consumption of fixed capital	4 259	4 845	9 434	10 150	10 240	10 818	11 467	11 642	12 264	12 500
Gross saving	45 410	45 685	44 505	40 958	46 051	32 303	19 311	16 870	32 449	58 058
Gross fixed capital formation	16 458	20 087	26 141	27 151	25 740	28 885	29 996	27 368	27 701	27 778
Net lending	29 932	26 440	16 656	12 255	18 429	1 149	–13 415	–12 397	3 072	28 434
Memorandum item:										
Revenue from oil sector	18 268	15 402	9 940	24 573	40 684	46 269	38 248	44 569	43 628	50 200

Source: Norwegian National Accounts; OECD, *National Accounts.*

Table F. **Labour market**

Thousand persons

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Civilian employment	2 053	2 090	2 079	2 014	1 992	1 973	1 970	1 970	2 003	. .
Agriculture, forestry and fishing	151	139	134	132	129	116	110	111	107	. .
Oil production and mining	22	24	24	23	22	21	25	25	24	. .
Manufacturing	358	352	337	318	310	294	295	292	303	. .
Electricity, gas and water	21	23	21	22	23	21	20	22	22	. .
Construction	155	166	166	147	139	130	122	116	119	. .
Wholesale and retail trade	364	375	376	369	358	354	353	349	348	. .
Transports and communications	179	178	175	167	162	162	157	158	165	. .
Banking insurance real estate	142	155	166	154	150	153	153	153	160	. .
Community, social and personal services	658	673	674	675	696	716	729	741	752	. .
Registered unemployment	36.2	32.4	49.3	82.9	92.7	100.7	114.4	118.2	110.3	102.2
Unfilled vacancies	10.5	12.4	8.7	6.9	6.6	6.5	6.4	7.4	7.7	8.8
Unemployment rate (per cent of labour force)	2.0	2.1	3.2	4.9	5.2	5.5	5.9	6.0	5.4	4.9

Sources: OECD, *Labour Force Statistics, Main Economic Indicators.*

Table G. **Balance of payments**

Million US dollars

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Current account										
Merchandise exports	18 157	21 184	21 930	26 700	34 138	34 167	35 398	32 185	34 934	42 172
Crude petroleum and natural gas	7 173	8 603	7 434	11 117	14 988	16 190	14 031	13 843	15 741	17 919
Ships and oil platforms	1 550	1 626	872	975	1 780	2 357	1 899	1 450	1 558	1 659
Other	9 434	10 955	13 624	14 608	17 370	15 619	19 467	16 892	17 635	22 594
Merchandise imports	20 249	21 885	22 400	23 235	28 684	26 812	27 113	25 327	28 122	34 236
Ships and oil platforms	625	978	2 310	3 919	3 051	2 334	1 004	1 674	1 220	1 058
Other	19 624	20 907	20 090	19 316	25 633	24 478	26 109	23 653	26 902	33 178
Trade balance	–2 092	–720	–186	3 782	7 598	8 656	9 313	7 943	8 281	. .
Services net	–1 625	–2 360	–2 552	–2 416	–2 326	–2 068	–4 632	–2 826	–1 077	. .
Travel	–1 556	–1 853	–2 056	–1 642	–2 097	–1 770	–2 057	–1 838	–1 896	. .
Investment income	–1 129	–1 235	–2 515	–2 786	–3 437	–3 907	–2 812	–2 787	–2 879	–2 417
Other services	1 060	728	2 019	2 012	3 208	3 609	237	1 799	3 698	. .
Transfers net	–808	–978	–1 134	–1 127	–1 417	–1 514	–1 791	–1 399	–1 622	. .
Private	–140	–194	–168	–222	–237	–338	–491	–304	–448	. .
Official	–668	–784	–966	–905	–1 180	–1 176	–1 300	–1 095	–1 174	. .
Current balance	–4 524	–4 058	–3 872	239	3 855	5 075	2 889	3 719	5 582	. .
Capital account										
Long-term capital, net	2 817	19	4 599	3 028	–1 322	–2 885	2 540	1 188	–2 394	. .
Private, direct	–582	–706	–683	159	–474	–2 131	286	1 069	–985	. .
Private, portfolio	4 286	2 346	4 185	3 060	514	–2 966	949	–404	87	. .
Private, other	–734	–1 558	1 230	–188	–1 568	2 009	1 231	584	–1 317	. .
Public [1]	–152	–63	–132	–2	207	203	75	–61	–178	. .
Short-term capital net	116	5 501	107	–1 035	642	–4 604	–3 471	6 058	–228	. .
Private non monetary	1 287	317	155	–659	361	726	3 188	1 576	801	. .
Private monetary institutions	–1 171	5 184	–48	–376	281	–5 330	–6 659	4 482	–1 029	. .
Miscellaneous official accounts	25	–25	65	–58	–37	40	711	–4 146	1 476	. .
Errors and omissions	–1 650	–1 258	–1 063	–1 355	–2 803	–417	–3 372	–1 290	–758	. .
Change in reserves	–3 169	–84	–225	903	357	–2 767	–728	3 957	1 559	. .

1. Excludes special transactions.

Source: OECD Secretariat.

Table H. **Foreign trade by area**

Million US dollars

	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
Imports, total	15 554	20 298	22 578	23 211	23 668	27 200	25 523	26 064	24 003	27 388
OECD countries	13 935	18 521	20 220	19 275	18 222	22 385	21 528	22 764	20 928	23 435
EEC	7 643	10 163	11 196	10 638	10 085	12 335	12 211	12 607	11 609	13 364
of which:										
Germany	2 500	3 436	3 495	3 145	2 972	3 768	3 608	3 758	3 250	3 796
Belgium-Luxembourg	433	596	650	585	618	634	612	623	584	696
France	650	824	832	767	770	1 009	975	1 058	1 024	1 103
Netherlands	550	775	868	906	781	1 068	1 341	1 027	984	1 156
United Kingdom	1 553	1 783	2 020	1 724	1 681	2 325	2 128	2 315	2 126	2 758
USA	1 118	1 393	1 452	1 527	1 724	2 388	1 986	2 224	1 932	2 004
Sweden	2 776	3 644	4 265	4 071	3 586	4 234	3 927	4 011	3 378	4 077
Finland	632	811	981	808	711	843	804	921	786	995
Non OECD countries	1 619	1 768	2 347	3 792	5 290	4 687	3 880	3 176	2 964	3 847
COMECON	434	366	445	528	593	685	544	635	613	1 019
OPEC	128	82	65	112	117	144	127	70	63	61
Others	1 058	1 320	1 837	3 152	4 580	3 858	3 208	2 471	2 289	2 767
Exports, total	19 934	18 230	21 449	22 429	27 101	34 033	34 037	35 138	31 893	34 735
OECD countries	17 682	15 689	18 875	20 073	24 536	30 643	30 424	31 196	28 200	31 075
EEC	13 846	11 859	13 803	14 343	17 324	21 653	22 262	23 034	20 865	22 117
of which:										
Germany	3 104	3 485	3 193	2 699	2 934	3 680	3 703	4 548	4 077	4 128
Belgium-Luxembourg	193	215	288	557	672	723	821	1 003	721	1 003
France	1 048	627	1 086	1 596	2 380	2 606	2 537	2 686	2 459	2 694
Netherlands	1 214	1 086	1 562	1 497	1 753	2 636	2 668	2 480	2 686	3 268
United Kingdom	7 103	5 060	5 711	5 786	7 221	8 853	8 937	8 454	7 736	7 123
USA	1 021	988	1 222	1 334	1 775	2 168	1 609	1 825	1 934	2 280
Sweden	1 752	1 805	2 387	2 564	3 238	3 834	3 447	3 209	2 692	3 186
Finland	315	317	421	468	558	916	1 014	877	795	1 009
Non OECD countries	2 251	2 541	2 574	1 715	1 796	2 622	2 971	3 230	2 899	2 898
COMECON	175	174	223	270	287	382	406	488	497	601
OPEC	100	122	112	100	94	413	116	166	149	150
Others	1 977	2 246	2 240	1 345	1 415	1 826	2 449	2 576	2 252	2 147

Source: OECD, *Foreign Trade Statistics*, Series C.

Table I. **Prices and wages**

1990 = 100

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Consumer prices										
Total	79.2	86.1	91.9	96.0	100.0	103.4	105.8	108.2	109.8	112.5
Food	82.4	88.6	94.3	96.9	100.0	101.7	103.1	102.1	103.5	105.0
Rent heating and light	76.2	81.9	88.6	94.0	100.0	104.5	107.0	110.0	111.0	114.1
Wholesale prices										
Total	81.9	86.8	91.4	96.5	100.0	102.5	102.6	102.5	104.0	106.0
Consumer goods	81.4	86.9	91.9	95.7	100.0	103.7	105.6	105.7	107.1	108.7
Investment goods	85.1	90.5	94.6	97.9	100.0	102.1	103.0	106.2	108.2	110.8
Petroleum products	77.1	81.2	72.7	84.4	100.0	100.2	91.4	93.2	91.0	89.1
Hourly earnings										
Males	73.3	85.1	89.8	94.4	100.0	105.3	108.6	111.6	114.8	118.8
Females	71.4	82.9	88.1	93.5	100.0	106.0	109.1	112.3	115.6	119.7

Source: OECD *Main Economic Indicators*, Secretariat estimates.

Table J. **Money and credit**

Kr million

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Changes in money supply										
Central authorities [1]	–3 696	16 784	16 568	34 836	46 465	60 308	82 400	37 500	39 700	19 800
Commercial and saving banks [2]	50 651	83 298	11 290	1 303	1 076	12 520	–33 400	–26 300	20 900	39 200
Unspecified and statistical errors	14 695	–9 739	–6 200	–17 396	–17 498	–8 811	–9 200	–9 500	–14 400	2 000
Domestic liquidity supply	61 650	90 343	21 658	18 743	30 043	64 017	39 800	1 700	46 200	61 000
Foreign transactions	–47 176	–40 252	300	11 794	–4 147	–15 101	–2 400	1 300	–10 400	–30 900
Change in broad money (M2)	14 474	50 091	21 958	30 537	25 896	48 916	37 400	3 000	35 800	30 100
Domestic lending										
by financial institutions										
Total [3]	571 850	692 160	765 431	823 750	843 424	823 842	808 863	797 121	804 929	842 831
Commercial banks	173 610	217 871	229 780	256 034	261 184	244 981	251 237	249 566	256 826	281 687
Saving banks	129 822	159 072	173 158	184 100	191 950	191 652	192 312	202 474	216 931	236 352
State banks	129 559	137 446	145 777	154 725	163 044	174 949	185 586	180 794	176 625	175 439
Insurance companies	46 809	55 597	59 414	59 313	61 199	70 091	78 302	70 888	66 452	55 987
Mortgage credit institutions	67 467	99 870	126 488	142 782	144 659	121 606	83 132	73 144	63 219	62 289
Private finance institutions	18 751	21 898	30 347	25 963	20 742	19 986	17 636	18 981	24 264	30 487
Postal saving banks	5 460	6 047	7 101	8 394	10 196	13 089	15 246	20 873	25 623	. .

1. Government income surplus, loan transactions, Central and State banks. Excluding oil taxes.
2. Including tax-free allocations to funds and saving with tax productions.
3. Breakdown does not add up to total.

Sources: Bank of Norway, *Economic Bulletin;* Central Bureau of Statistics, *Monthly Bulletin of Statistics.*

Table K. **Production and employment structures**

	Per cent share of GDP at factor cost (current prices)				Per cent share of total employment			
	1962	1970	1980	1991	1962	1970	1980	1991
Agriculture, forestry and fishing	7.9	5.6	3.8	2.9	19.0	12.9	8.4	6.1
Mining crude petroleum and natural gas	0.8	0.7	14.8	13.3	0.6	0.6	0.8	1.1
Manufacturing	21.1	21.6	16.0	13.5	23.4	23.8	19.8	14.3
of which:								
Food, beverages and tobacco	3.9	3.3	1.7	3.0	3.7	3.6	3.1	2.5
Textiles, clothing, leather	2.3	1.5	0.7	0.3	3.2	2.4	1.2	0.5
Wood and wood products	1.6	1.8	1.4	0.8	2.1	2.2	1.9	1.2
Paper and paper products	2.6	2.8	2.0	2.1	3.4	3.5	2.8	2.4
Chemicals and products of petroleum, coal rubber, etc.	2.1	2.3	1.9	1.6	3.0	3.0	2.5	1.9
Fabricated metal products, machinery and equipment	5.7	6.4	5.6	4.3	6.4	7.4	6.8	4.9
Electricity, gas and water	2.8	2.7	2.9	4.0	0.9	0.9	0.9	1.0
Construction	8.0	7.0	5.9	3.9	8.1	8.2	7.9	6.9
Services	59.4	62.4	56.5	62.4	48.2	53.7	62.3	70.6
of which:								
Wholesale and retail trade, restaurants, and hotels	20.1	12.4	11.7	11.0	14.4	15.5	16.6	16.5
Transport, storage and communication								
Finance, insurance, real estate and business services	9.5	9.1	9.9	13.7	2.7	3.7	5.0	7.3
Producers of Government services	9.7	11.8	13.6	16.3	12.8	16.6	23.2	28.7

Source: OECD, *National Accounts.*

Table L. **Productivity and investment structure**

	Productivity growth (Sector GDP/sector employment)				Investment Per cent of total investment[1]			
	1963-69	1970-79	1980-91	1991	1962	1970	1980	1991
Agriculture, forestry and fishing	4.4	3.3	4.6	5.6	8.0	6.6	8.2	3.8
Mining, crude petroleum and natural gas	8.7	25.7	6.5	9.5	0.6	2.4	9.7	20.5
Manufacturing	4.6	1.7	2.3	1.4	17.1	15.3	12.8	10.3
of which:								
Food, beverages and tobacco	3.8	0.2	–0.1	4.5	3.2	2.9	2.3	2.0
Textiles, clothing, leather	3.1	0.6	4.2	8.0	0.7	0.5	0.4	0.1
Wood and wood products	6.4	1.6	0.5	–0.1	0.5	1.2	0.8	0.6
Paper and paper products	5.2	–0.1	2.0	0.4	2.8	1.8	3.0	1.7
Chemicals and products of petroleum, coal, rubber, etc.	8.1	4.0	4.8	–8.1	3.3	2.0	1.4	2.0
Fabricated metal products, machinery and equipment	2.9	1.3	2.4	3.5	3.4	3.4	2.6	2.1
Electricity, gas and water	5.2	0.7	0.7	–8.3	10.4	8.1	9.9	5.1
Construction	2.8	2.5	0.2	3.3	1.3	2.0	2.2	1.8
Services	2.4	1.7	0.5	0.2	62.6	65.5	57.3	58.5
of which:								
Wholesale and retail trade, restaurants, and hotels	1.3	–0.6	–0.2	1.7	4.7	4.6	4.6	4.2
Transport, storage and communication	4.5	3.3	3.7	5.1	25.8	20.5	10.9	16.7
Finance, insurance, real estate and business services	–0.2	–1.6	–1.8	–0.9	19.2	21.8	23.9	16.6
Producers of Government services	2.0	0.5	0.8	0.3	12.1	16.9	16.2	19.1

1. At current prices.
Source: OECD, *National Accounts.*

Table M. **Labour-market indicators**

A. LABOUR MARKET PERFORMANCE

		Cyclical Trough: 1983	Cyclical Peak: 1986	1988	1994
Standardised unemployment rate		3.4	2.0	3.2	5.4
Unemployment rate:	Total	3.4	2.0	3.1	5.4
	Male	3.2	1.6	3.0	5.9
	Women	3.8	2.4	3.3	4.9
	Youth [1]	8.9	5.0	7.9	12.6
Share of long-term unemployment in total unemployment [2]		6.3	7.5	6.4	28.0

B. STRUCTURAL OR INSTITUTIONAL CHARACTERISTICS

		1975	1980	1985	1994
Participation rate: [3]	Total	69.8	76.7	80.0	77.8
	Male	85.5	88.7	89.7	83.8
	Women	53.6	64.2	70.1	71.6
Employment/population (15-64 years)		69.1	74.0	75.5	73.6
Non-wage labour costs [4] (as a percentage of total compensation)		14.9	14.6	14.3	12.8
Unemployment insurance replacement ratio [5]		7.2	23.5	34.9	..

		Average percentage changes (annual rates)			
		1970/1960	1980/1970	1990/1980	1994/1991
Labour force		0.7	2.2	1.3	0.9
Employment:	Total	0.7	2.1	1.2	1.5
	Industries	1.2	0	–0.7	2.9
	Services	2.0	4.7	2.4	1.7

1. People between 16 and 24 years as a percentage of the labour force of the same age group.
2. Persons seeking a job for 12 months and over as a percentage of total unemployed.
3. Labour force as a percentage of relevant population group, aged between 15 and 64 years.
4. Employers' contributions to social security and pension funds.
5. Unemployment benefits per unemployed as a percentage of compensation per employee.

Source: OECD, *Labour Force Statistics;* OECD Secretariat.

Table N. **Public sector**
General government income and expenditure structures
As a percentage of GDP

	1962	1970	1980	1994
Current receipts	32.4	39.7	48.5	50.2
Indirect taxes	12.9	16.6	15.7	16.3
Social security contributions	5.7	8.8	10.9	10.0
Direct taxes	12.2	12.1	19.6	15.8
Capital income	1.5	2.1	2.3	5.3
Other current receipts	0	0	0	2.8
Current expenditure	25.5	33.3	40.4	47.6
Expenditure on goods and services	13.6	16.5	18.3	21.3
Defence	3.1	3.2	2.6	. .
Education	3.3	4.4	4.6	. .
Health	1.3	2.2	3.7	. .
Social security and welfare	0.7	0.8	1.5	. .
Economic services	2.0	2.2	2.1	. .
Other	3.2	3.7	3.8	. .
Subsidies	3.8	4.7	6.4	4.2
Interest paid	1.3	1.6	3.1	3.1
Current transfers	7.5	11.2	13.1	16.3
Net saving	6.9	6.4	8.1	2.6
Consumption of fixed capital	0.5	0.7	0.7	1.5
Gross saving	7.4	7.0	8.9	4.0
Gross fixed capital formation	3.2	4.1	3.7	3.3
Net lending	4.2	3.0	5.2	0.3
Memorandum item:				
Revenue from oil sector	0	0	7.5	3.0

Source: Norwegian National Accounts; OECD, *National Accounts.*

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Denmark	Finland	France	Germany	Greece
Population											
Total	Thousands	1994	17 840	8 031	10 124	29 251	5 206	5 088	57 960	81 407	10 430
Inhabitants per sq. km	Number	1994	2	96	332	3	121	15	106	228	79
Net average annual increase over previous 10 years	%	1994	1.4	0.6	0.3	1.6	0.2	0.4	0.5	2.9	0.5
Employment											
Total civilian employment (TCE)[2]	Thousands	1994	7 680 (93)	3 737	3 724 (92)	13 292	2 508	2 015	21 781 (93)	35 894	3 790
of which: Agriculture	% of TCE	1994	5.3 (93)	7.2	2.6 (92)	4.1	5.1	8.3	5.1 (93)	3.3	20.8
Industry	% of TCE	1994	23.7 (93)	33.2	27.7 (92)	22.6	26.8	26.8	27.7 (93)	37.6	23.6
Services	% of TCE	1994	71 (93)	59.6	69.7 (92)	73.3	68.1	64.9	67.2 (93)	59.1	55.5
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1994	331.6	198.1	227.9	544	146.7	97.2	1 328.5	1 832.3	73.1 (93)
Per capita	US$	1994	18 588	24 670	22 515	18 598	28 181	19 106	22 944	27 826	7 051 (93)
At current prices using current PPP's[3]	Bill. US$	1994	327.9	162.3	204.2	596.7	107	82.5	1 111.8	1 601.7	118
Per capita	US$	1994	18 382	20 210	20 166	20 401	20 546	16 208	19 201	24 325	1 450
Average annual volume growth over previous 5 years	%	1994	2.2	2.5	1.6	1.1	1.9	–1.6	1.1	2.6	1.4 (93)
Gross fixed capital formation (GFCF)	% of GDP	1994	21.4	24.8	17.4	18.7	14.8	14.3	18.1	18.5	17.4 (93)
of which: Machinery and equipment	% of GDP	1993	9.8	8.7	7.8	6.2	7.2	5.90	8.10	7.5	7.8
Residential construction	% of GDP	1993	5.2	6.3	4.6	6.1	2.9	3.7	4.9	6.5	3.6
Average annual volume growth over previous 5 years	%	1994	0.8	3.7	0.4	–0.1	–2.8	–12.9	–1	0.8	2.7 (93)
Gross saving ratio[4]	% of GDP	1994	16.8	25.3	22	16	17	16.6	19	21	15.5 (93)
General government											
Current expenditure on goods and services	% of GDP	1994	17.5	18.8	15	20.2	25.3	22.4	19.6	17.7	19.1 (93)
Current disbursements[5]	% of GDP	1993	36.9	48.4	55.3	49	61.1	58.9	51.5	45.6	51.2
Current receipts	% of GDP	1993	33.5	48.6	50.1	43	58.3	52.5	46.8	45.7	40.2
Net official development assistance	% of GNP	1993	0.34	0.31	0.41	0.46	1.03	0.76	0.66	0.44	..
Indicators of living standards											
Private consumption per capita using current PPP's[3]	US$	1993	10 803	10 546	12 090	11 863	10 042	8 814	11 395	10 733	6 367
Passenger cars, per 1 000 inhabitants	Number	1990	430	382	387	469	311	386	413	480[8]	169
Telephones, per 1 000 inhabitants	Number	1991	464	432	410	586	577	544	511	420[8]	413
Television sets, per 1 000 inhabitants	Number	1991	480	478	451	639	536	501	407	556[8]	197
Doctors, per 1 000 inhabitants	Number	1993	2.2 (91)	2.3	3.7	2.2	2.8 (92)	2.6 (92)	2.8	3.2 (92)	3.8 (92)
Infant mortality per 1 000 live births	Number		6.1	6.5	8	6.8	5.4	4.4	6.5	5.8	8.5
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1994	3	5.5	3.7	3.3	3.5	4.8	3.5	5.2	14.6
Consumer prices	%	1994	3	3.4	2.8	2.8	2.1	3.3	2.5	3.3	16.2
Foreign trade											
Exports of goods, fob*	Mill. US$	1994	47 363	44 881	137 259[7]	165 358	41 850	29 514	235 337	422 243	8 958
As % of GDP	%	1994	14.3	22.7	60.2	30.4	28.5	30.4	17.7	23	11.5 (93)
Average annual increase over previous 5 years	%	1994	5	6.7	6.5	7.1	8.3	4.9	5.6	4.4	3.4
Imports of goods, cif*	Mill. US$	1994	49 731	55 071	126 006[7]	148 297	35 932	23 091	220 508	376 566	21 111
As % of GDP	%	1994	15	27.8	55.3	27.3	24.5	23.8	16.6	20.6	30.1 (93)
Average annual increase over previous 5 years	%	1994	4	7.2	5	5.4	6.1	–1.3	3.5	6.9	5.4
Total official reserves[6]	Mill. SDRs	1994	7 730	11 523	9 505[7]	8 416	6 203	7 304	17 986	52 994	9 924
As ratio of average monthly imports of goods	Ratio	1994	1.9	2.5	0.9	0.7	2.1	3.8	1	1.7	5.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Iceland	Ireland	Italy	Japan	Luxembourg	Mexico	Netherlands	New Zealand
Population										
Total	Thousands	1994	267	3 571	57 190	124 960	398	93 010	15 382	3 526
Inhabitants per sq. km	Number	1994	3	51	190	331	153	47	377	13
Net average annual increase over previous 10 years	%	1994	1.1	0.1	0	0.4	0.8	2	0.6	0.8
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	138	1 168 (93)	20 152 (93)	64 530	162 (91)	32 439	6 631	1 560
of which: Agriculture	% of TCE	1994	9.4	12.7 (93)	7.5 (93)	5.8	3.7 (91)	25.8	4	10.4
Industry	% of TCE	1994	26.1	27.7 (93)	33 (93)	34	31.5 (91)	22.2	23	25
Services	% of TCE	1994	65.2	59.7 (93)	59.6 (93)	60.2	64.8 (91)	52.1	73	64.6
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	6.2	52	1 017.8	4 590	10.6 (92)	371.2	334.3	51.2
Per capita	US$	1994	23 199	14 550	17 796	36 732	27 073 (92)	3 991	21 733	14 513
At current prices using current PPP's[3]	Bill. US$	1994	5.1	54.3	1 068.4	2 593.7	11.7	673.3	285.9	57.3
Per capita	US$	1994	19 271	15 212	18 681	20 756	29 454	7 239	18 589	16 248
Average annual volume growth over previous 5 years	%	1994	0.6	4.7	1	2.1	4.1 (92)	3	2.3	2.5
Gross fixed capital formation (GFCF)	% of GDP	1994	15.2	15.1	16.4	28.6	20.4 (93)	20.7	19.3	20
of which: Machinery and equipment	% of GDP	1993	3.9	6.3	7.4	11.5	..	9.4	8.6	9.3
Residential construction	% of GDP	1993	4.4	4.1	5.3	5.4	..	4.9	5.1	4.9
Average annual volume growth over previous 5 years	%	1994	–4	1	–2.3	1.4	6.5 (92)	7.7	0.4	5.8
Gross saving ratio[4]	% of GDP	1994	16.9	19.5	18.8	31.2	60.2 (92)	15.1	24.4	20.7
General government										
Current expenditure on goods and services	% of GDP	1994	20.6	16	17.1	9.8	17.1 (92)	11.8[9]	14.2	14.7
Current disbursements[5]	% of GDP	1993	34.9	..	53.2	26.9	..	..	55.4	..
Current receipts	% of GDP	1993	35.9	..	47.1	32.9	..	..	54.5	..
Net official development assistance	% of GNP	1993	..	0.15	0.42	0.27	0.34 (92)	..	0.88	0.22
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	11 546	7 750	11 029	11 791	15 545	4 853	10 726	9 266
Passenger cars, per 1 000 inhabitants	Number	1990	464	228	478	282	470	85	356	440
Telephones, per 1 000 inhabitants	Number	1991	527	300	400	454	511	70	477	436
Television sets, per 1 000 inhabitants	Number	1991	319	276	421	613	267	148	485	443
Doctors, per 1 000 inhabitants	Number	1993	3	1.7 (92)	1.7 (91)	1.7 (92)	2.1 (92)	1	2.5 (90)	2
Infant mortality per 1 000 live births	Number		4.8	5.9	7.3	4.5 (92)	8.5 (92)	18	6.3	7.3
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	..	4.6	5.9	2.4	..	5.3	3.2	2.1
Consumer prices	%	1994	6.3	2.7	5.2	2	3.1	16.1	2.8	2.5
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	1 628	34 125	189 802	396 149	..	60 882	155 084	12 169
As % of GDP	%	1994	26.3	65.7	18.6	8.6	..	16.4	46.4	23.8
Average annual increase over previous 5 years	%	1994	2.7	10.5	6.2	7.6	..	21.7	7.6	6.5
Imports of goods, cif*	Mill. US$	1994	1 464	25 812	167 690	274 916	..	79 346	139 800	11 859
As % of GDP	%	1994	23.6	49.7	16.5	6	..	21.4	41.8	23.2
Average annual increase over previous 5 years	%	1994	0.7	8.1	1.9	5.5	..	25.5	6.1	6.1
Total official reserves[6]	Mill. SDRs	1994	201	4 189	22 102	86 214	..	4 301	23 655	2 540
As ratio of average monthly imports of goods	Ratio	1994	1.6	1.9	1.6	3.8	..	0.7	2	2.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population										
Total	Thousands	1994	4 337	9 900	39 150	8 781	6 994	60 573	58 375	260 651
Inhabitants per sq. km	Number	1994	13	107	78	20	169	78	238	28
Net average annual increase over previous 10 years	%	1994	0.5	–0.1	0.2	0.5	0.7	2.1	0.3	1
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	1970 (93)	4 372	11 760	3 926	3 772	19 664	25 044 (93)	123 060
of which: Agriculture	% of TCE	1994	5.6 (93)	11.5	9.8	3.4	4	44.8	2.2 (93)	2.9
Industry	% of TCE	1994	23.1 (93)	32.8	30.1	25	28.8	22.2	26.2 (93)	24
Services	% of TCE	1994	71.3 (93)	55.7	60.2	71.6	67.2	33	71.6 (93)	73.1
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	103.4 (93)	87	482.4	196.6	257.3	130.7	1 019.5	6 649.8
Per capita	US$	1994	23 984 (93)	8 792	12 321	22 389	36 790	2 157	17 468	25 512
At current prices using current PPP's[3]	Bill. US$	1994	95.3	122	531.7	153	167.4	319.3	1 030.2	6 649.8
Per capita	US$	1994	21 968	12 335	13 581	17 422	23 942	5 271	17 650	25 512
Average annual volume growth over previous 5 years	%	1994	2.1 (93)	1.4	1.5	–0.3	0.5	3.6	0.8	2.1
Gross fixed capital formation (GFCF)	% of GDP	1994	22 (93)	25.7	19.8	13.7	22.8	24.5	15	17.2
of which: Machinery and equipment	% of GDP	1993	..	..	5.7	5.7	7.5	10.3	7.3	7.7
Residential construction	% of GDP	1993	..	..	4.1	4	14.9[10]	9.1	3.1	4
Average annual volume growth over previous 5 years	%	1994	–3.93	2.7	–1.2	–7.6	–0.4	5.1	–2.1	4.6
Gross saving ratio[4]	% of GDP	1994	21.9 (93)	24.2	18.8	13.7	29.3	22.5	13.5	16.2
General government										
Current expenditure on goods and services	% of GDP	1994	22.1 (93)	17.2	16.9	27.3	14.1	11.7	21.6	16.4
Current disbursements[5]	% of GDP	1993	..	..	43.7	67.3	36.7	..	42.7	35.8
Current disbursements[5]	% of GDP	1993	..	..	40.1	59	36	..	36.8	31.7
Current receipts	% of GDP	1993	1.23	0.36	0.32	1.33	0.49	..	0.34	0.19
Net official development assistance	% of GNP	1993	9 826	7 780	8 412	9 240	13 730	3 617	10 942	16 444
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	378	260	307	418	441	29	361	568
Passenger cars, per 1 000 inhabitants	Number	1990	515	273	340	687	603	143	445	553
Telephones, per 1 000 inhabitants	Number	1991	423	187	400	468	406	175	434	814
Television sets, per 1 000 inhabitants	Number	1991	3.2 (92)	2.9	4.1	3	3	0.9	1.5 (92)	2.3 (92)
Doctors, per 1 000 inhabitants	Number	1993	5	8.7	7.6	4.8	5.6	52.6	6.6	8.5 (92)
Infant mortality per 1 000 live births	Number		4	..	7.2	5.4	..	..	6.7	2.8
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	2.7	9	5.6	5.7	3.9	73	4.6	3.6
Consumer prices	%	1994	34 645	17 072	73 129	61 122	70 467	18 456	205 170	512 627
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	30.9 (93)	19.6	15.2	31.1	27.4	14.1	20.1	7.7
As % of GDP	%	1994	5	6.1	10.5	3.4	6.4	9.5	6.1	7.1
Average annual increase over previous 5 years	%	1994	27 345	25 967	92 182	51 730	68 126	22 976	227 026	663 256
Imports of goods, cif*	Mill. US$	1994	23.3 (93)	29.9	19.1	26.3	26.5	17.6	22.3	10
As % of GDP	%	1994	2.9	6.6	5.2	1	3.2	37.9	2.8	7
Average annual increase over previous 5 years	%	1994	13 033	10 627	28 475	15 929	23 790	4 911	28 094	43 350
Total official reserves[6]	Mill. SDRs	1994	5.7	4.9	3.7	3.7	4.2	2.6	1.5	0.8

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

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